GreenHub RVO Pareto analysis

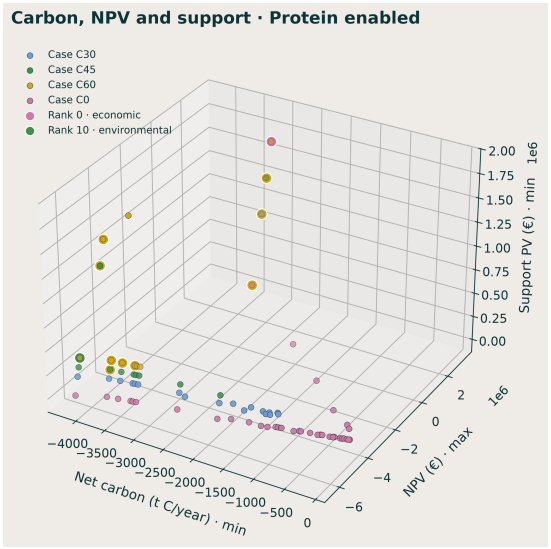
Current no-PEF BioSTEAM launch

Data assembled 2026-11-03T12:21:09Z UTC | Seed 20261102 | 128 sampled designs per protein group

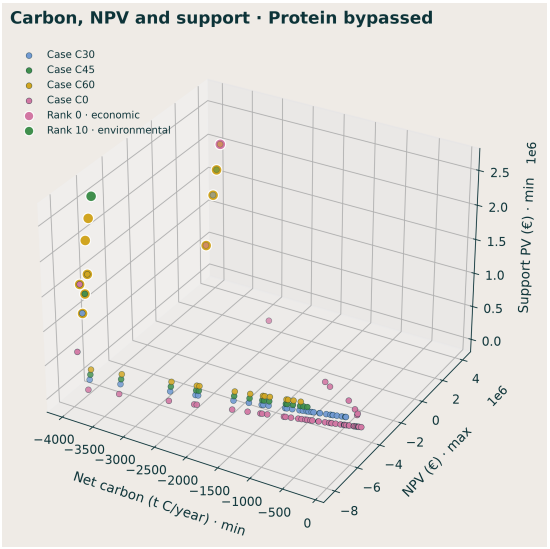
Separate protein-enabled and protein-bypassed process frontiers compare net carbon, screening NPV, and assumed public-support present value. The report includes selected candidates, baseline comparisons, indexed figures, model documentation, source batches, and selected model-run evidence.

RVO support amounts are user-defined what-if assumptions. They are not verified awards, eligibility determinations, or official SDE++ tariff calculations.

Three-objective front - protein enabled



Three-objective front - protein bypassed



Analysis summary

Each of 128 sampled process configurations in each protein group was simulated once, then paired with four financial support cases. Four explicit baseline references per group were also evaluated. PEF is disabled throughout (F1). Negative profits and dominated designs remain in the saved batches.

Protein group	Evaluated rows	Carbon / NPV	Support / NPV	Carbon / NPV / support
Protein enabled	516	9	10	78
Protein bypassed	516	2	12	120

Support cases

Case	Programme / named authority	CAPEX grant	Operating premium	Term
C30	C30 · RVO DEI+ CAPEX grant 30%; dynamic SDE++ value €1.87/Nm³ biomethane	30%	EUR 1.87/Nm3 gas	15 y
C45	C45 · RVO DEI+ CAPEX grant 45%; dynamic SDE++ value €1.87/Nm³ biomethane	45%	EUR 1.87/Nm3 gas	15 y
C60	C60 · RVO DEI+ CAPEX grant 60%; dynamic SDE++ value €1.87/Nm³ biomethane	60%	EUR 1.87/Nm3 gas	15 y
C0	C0 · no CAPEX grant; dynamic SDE++ value €1.87/Nm³ biomethane	0%	EUR 1.87/Nm3 gas	15 y

Assumptions and scope

NPV is a pre-tax, 20-year screening metric at an 8% discount rate. It uses high direct uninstalled equipment CAPEX, partial operating margin, upfront CAPEX grants, and a 15-year operating-support proxy. Installation, financing, escalation, replacements, working capital, residual value, and taxes are excluded.

Net carbon is operational gate-to-gate climate load converted to carbon equivalent minus assumed carbon retained in press-cake furniture and nutrient-rich solids. This mixed screening indicator is not a complete LCA or durable carbon-removal claim. Full methods, exclusions, and official-programme caveats are copied in the documentation folder.

Selected candidates - Protein enabled

Ranks 0 to 10 are selected from the three-objective front. Rank 0 maximizes NPV; rank 10 is the minimum-net-carbon endpoint among selected candidates. Switches use Protein-Biogas-N-P-K; 1 means pathway present and 0 means bypassed.

Rank	P-B-N-P-K	Routes	Case	Feed t/y	Net C t C/y	NPV EUR	Support PV EUR
0	P1-B1-N1-P1-K1	PR3/BG4/NR7	C60	24,204	-3,208.0	3,216,023	1,626,824
1	P1-B1-N1-P1-K1	PR3/BG4/NR7	C45	24,204	-3,208.0	2,873,926	1,284,727
2	P1-B1-N1-P1-K1	PR3/BG4/NR7	C30	24,204	-3,208.0	2,531,829	942,629
3	P1-B1-N1-P1-K1	PR3/BG4/NR7	C0	24,204	-3,208.0	1,847,634	258,435
4	P1-B1-N0-P0-K0	PR1/BG3/NR0	C45	27,345	-3,818.2	-5,829,115	1,351,015
5	P1-B1-N0-P0-K0	PR1/BG3/NR0	C60	27,345	-3,818.2	-5,591,803	1,588,327
6	P1-B0-N0-P0-K0	PR1/BG0/NR0	C60	25,672	-3,600.4	-5,429,473	334,864
7	P1-B0-N0-P0-K0	PR1/BG0/NR0	C60	26,896	-3,772.0	-5,508,895	339,508
8	P1-B0-N0-P0-K0	PR1/BG0/NR0	C60	24,314	-3,409.9	-5,341,248	329,628
9	P1-B0-N0-P0-K0	PR1/BG0/NR0	C45	26,896	-3,772.0	-5,593,772	254,631
10	P1-B0-N0-P0-K0	PR3/BG0/NR0	C60	29,975	-4,190.2	-5,972,643	350,912

Selected candidates - Protein bypassed

Ranks 0 to 10 are selected from the three-objective front. Rank 0 maximizes NPV; rank 10 is the minimum-net-carbon endpoint among selected candidates. Switches use Protein-Biogas-N-P-K; 1 means pathway present and 0 means bypassed.

Rank	P-B-N-P-K	Routes	Case	Feed t/y	Net C t C/y	NPV EUR	Support PV EUR
0	P0-B1-N1-P1-K1	PR0/BG2/NR7	C60	29,075	-3,879.8	3,940,186	2,048,684
1	P0-B1-N1-P1-K1	PR0/BG2/NR7	C45	29,075	-3,879.8	3,603,235	1,711,733
2	P0-B1-N1-P1-K1	PR0/BG2/NR7	C30	29,075	-3,879.8	3,266,284	1,374,782
3	P0-B1-N1-P1-K1	PR0/BG2/NR7	C0	29,075	-3,879.8	2,592,382	700,880
4	P0-B1-N0-P1-K0	PR0/BG3/NR2	C45	28,376	-3,972.2	-6,531,026	2,346,739
5	P0-B1-N0-P1-K0	PR0/BG3/NR2	C30	28,376	-3,972.2	-6,805,547	2,072,217
6	P0-B1-N0-P0-K0	PR0/BG2/NR1	C60	27,824	-3,894.3	-7,105,818	1,651,151
7	P0-B1-N0-P1-K0	PR0/BG3/NR2	C0	28,376	-3,972.2	-7,354,590	1,523,174
8	P0-B1-N0-P0-K0	PR0/BG2/NR1	C45	27,824	-3,894.3	-7,350,924	1,406,045
9	P0-B1-N0-P0-K0	PR0/BG2/NR1	C30	27,824	-3,894.3	-7,596,030	1,160,939
10	P0-B1-N0-P1-K0	PR0/BG3/NR2	C60	28,376	-3,972.2	-6,256,505	2,621,260

Baseline references

Exact current F1/no-PEF baseline references are simulated separately for each support case. The plotted baseline figures compare protein-enabled and protein-bypassed routes; detailed values and run IDs remain in the source batches.

Group	Case	P-B-N-P-K	Feed (t/y)	Net carbon (t C/y)	NPV (EUR)	Support PV (EUR)
Protein enabled	C0	P1-B1-N1-P1-K0	8,760	-1,220.0	-5,342,729	100,284
Protein enabled	C30	P1-B1-N1-P1-K0	8,760	-1,220.0	-5,006,263	436,750
Protein enabled	C45	P1-B1-N1-P1-K0	8,760	-1,220.0	-4,838,030	604,983
Protein enabled	C60	P1-B1-N1-P1-K0	8,760	-1,220.0	-4,669,797	773,216
Protein bypassed	C0	P0-B1-N1-P1-K0	8,760	-1,225.6	-5,552,765	211,170
Protein bypassed	C30	P0-B1-N1-P1-K0	8,760	-1,225.6	-5,230,716	533,219
Protein bypassed	C45	P0-B1-N1-P1-K0	8,760	-1,225.6	-5,069,691	694,244
Protein bypassed	C60	P0-B1-N1-P1-K0	8,760	-1,225.6	-4,908,666	855,269

Complete model warnings, limitations, and sample configurations are preserved in `evidence/current_run/`. The source PNGs are embedded below without edits to their visual styling.

pareto net carbon vs npv protein enabled

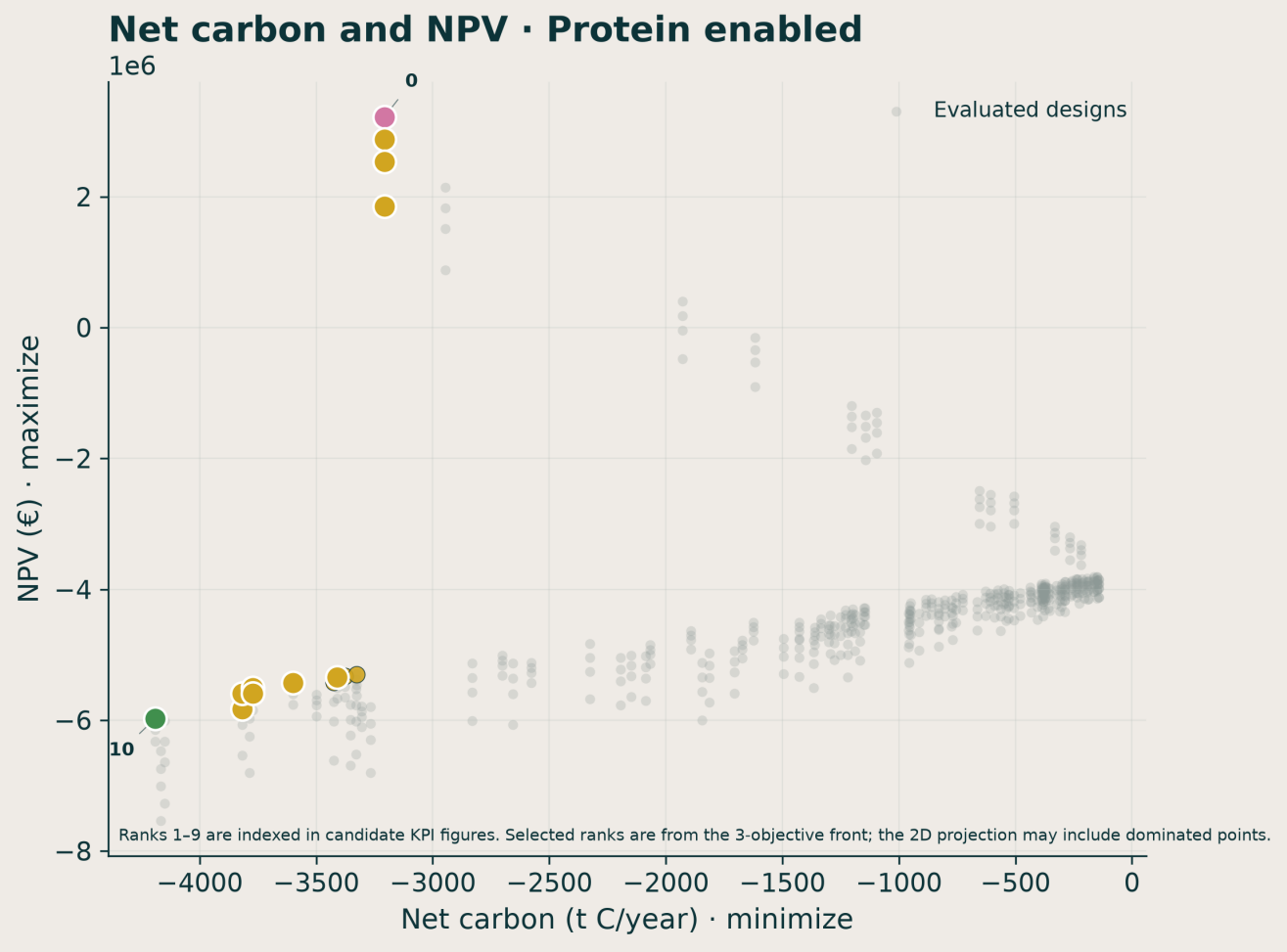


Figure 01. Net carbon and NPV · Protein enabled

protein_enabled / all support cases | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

pareto subsidy granted vs npv protein enabled

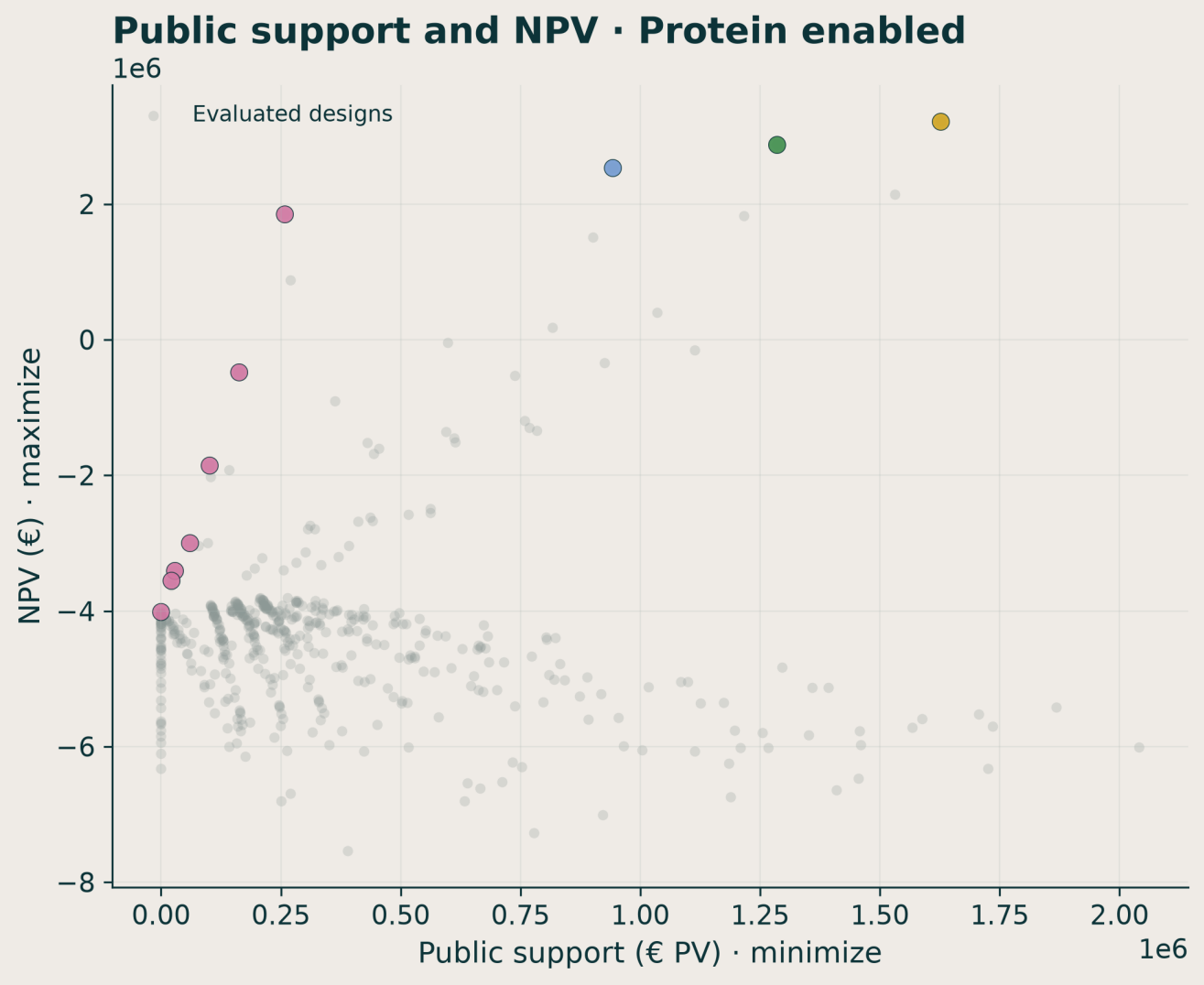


Figure 02. Public support and NPV · Protein enabled

protein_enabled / all support cases | Public support: € PV; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

pareto net carbon vs npv protein enabled case C30

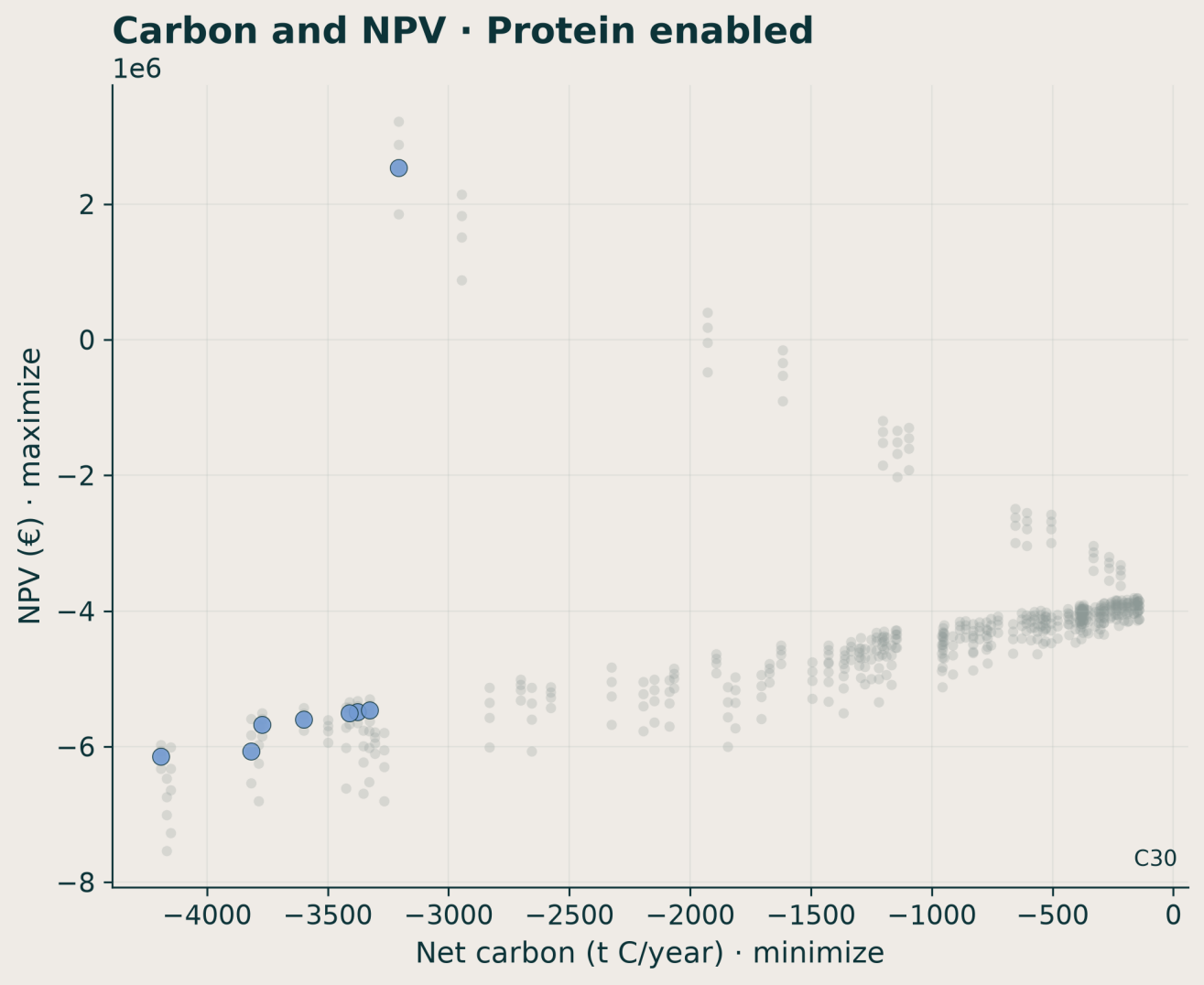


Figure 03. Carbon and NPV · Protein enabled

protein_enabled / case C30 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

pareto net carbon vs npv protein enabled case C45

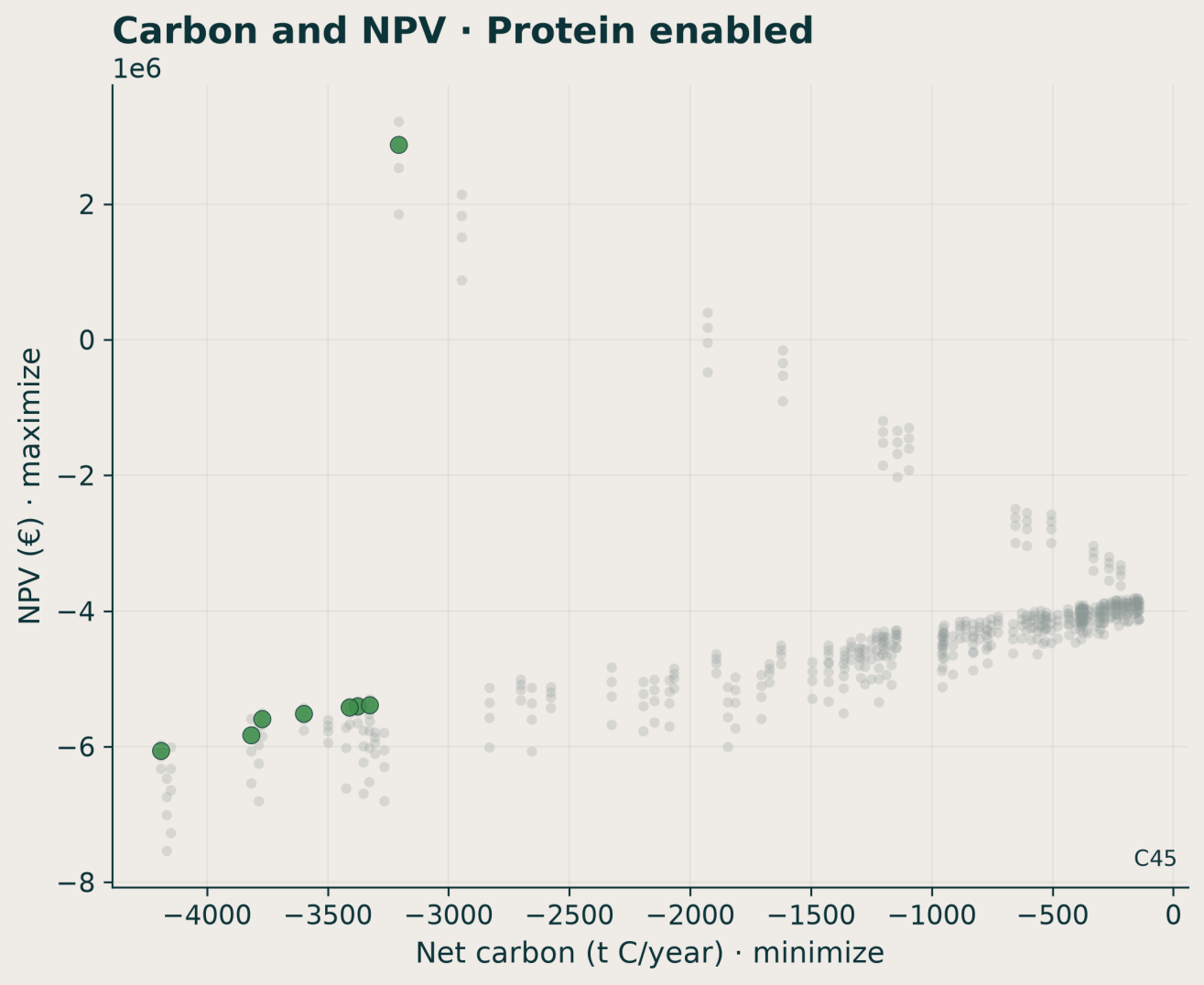


Figure 04. Carbon and NPV · Protein enabled

protein_enabled / case C45 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

pareto net carbon vs npv protein enabled case C60

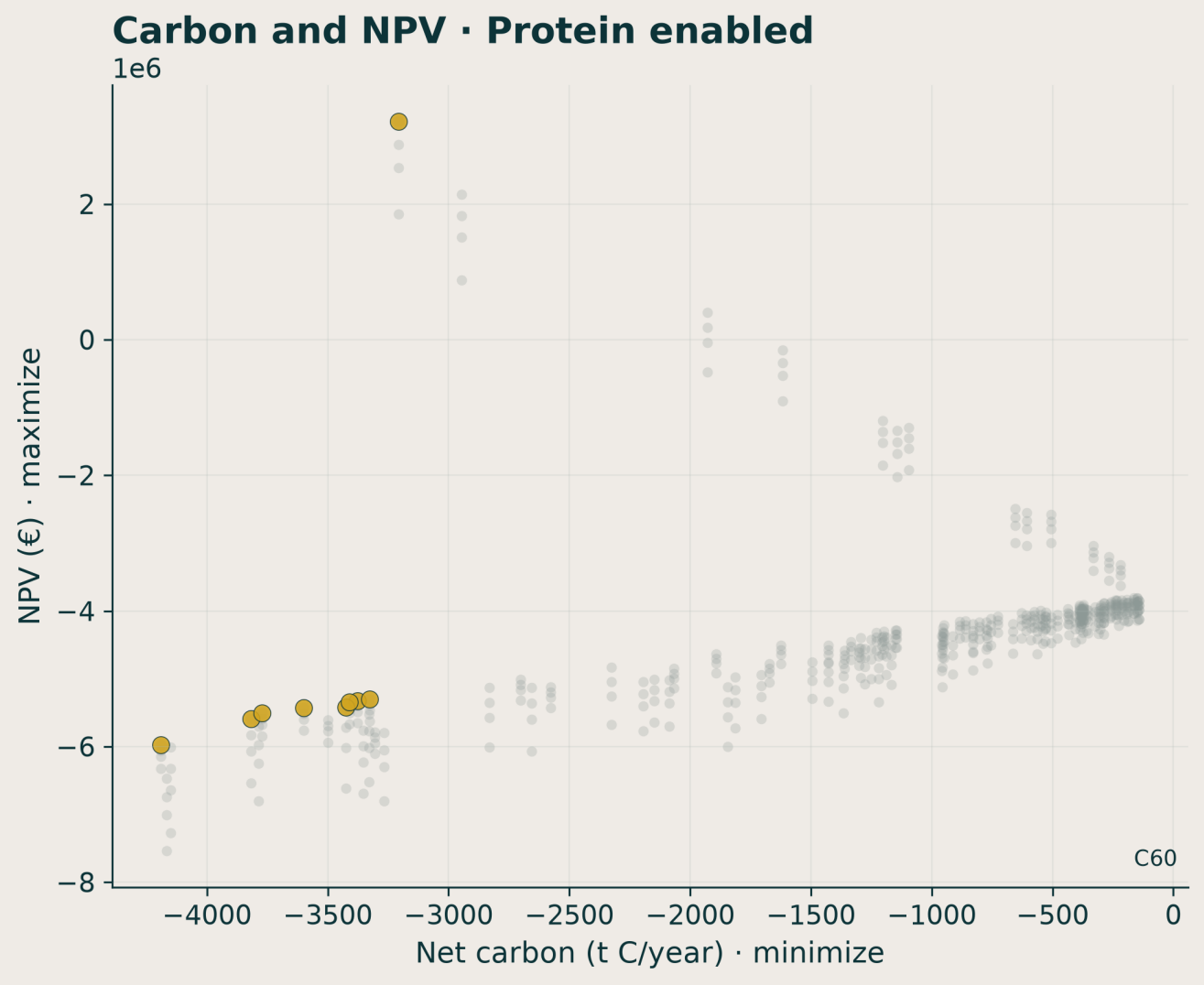


Figure 05. Carbon and NPV · Protein enabled

protein_enabled / case C60 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

pareto net carbon vs npv protein enabled case C0

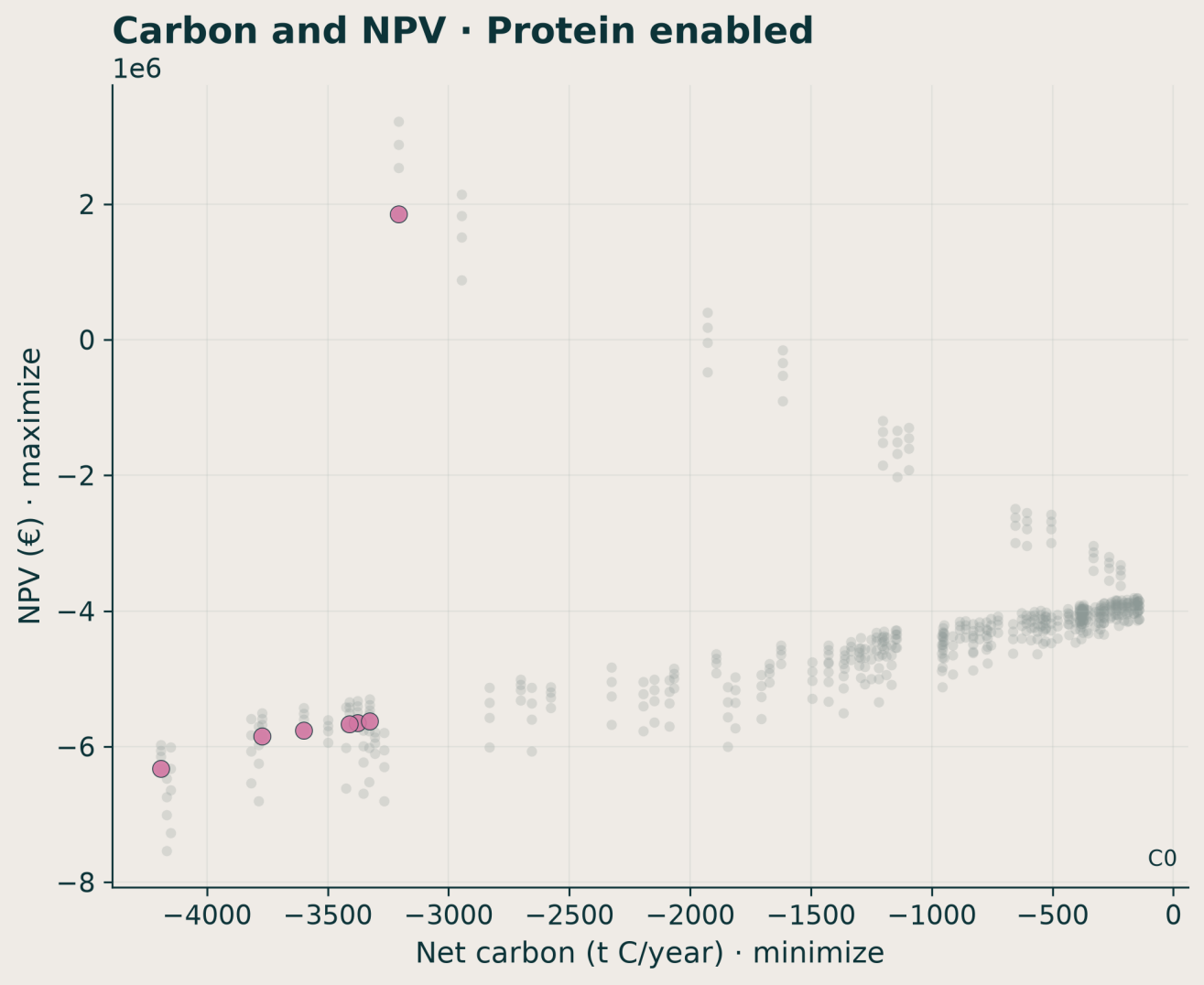


Figure 06. Carbon and NPV · Protein enabled

protein_enabled / case C0 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

pareto 3d carbon npv support protein enabled

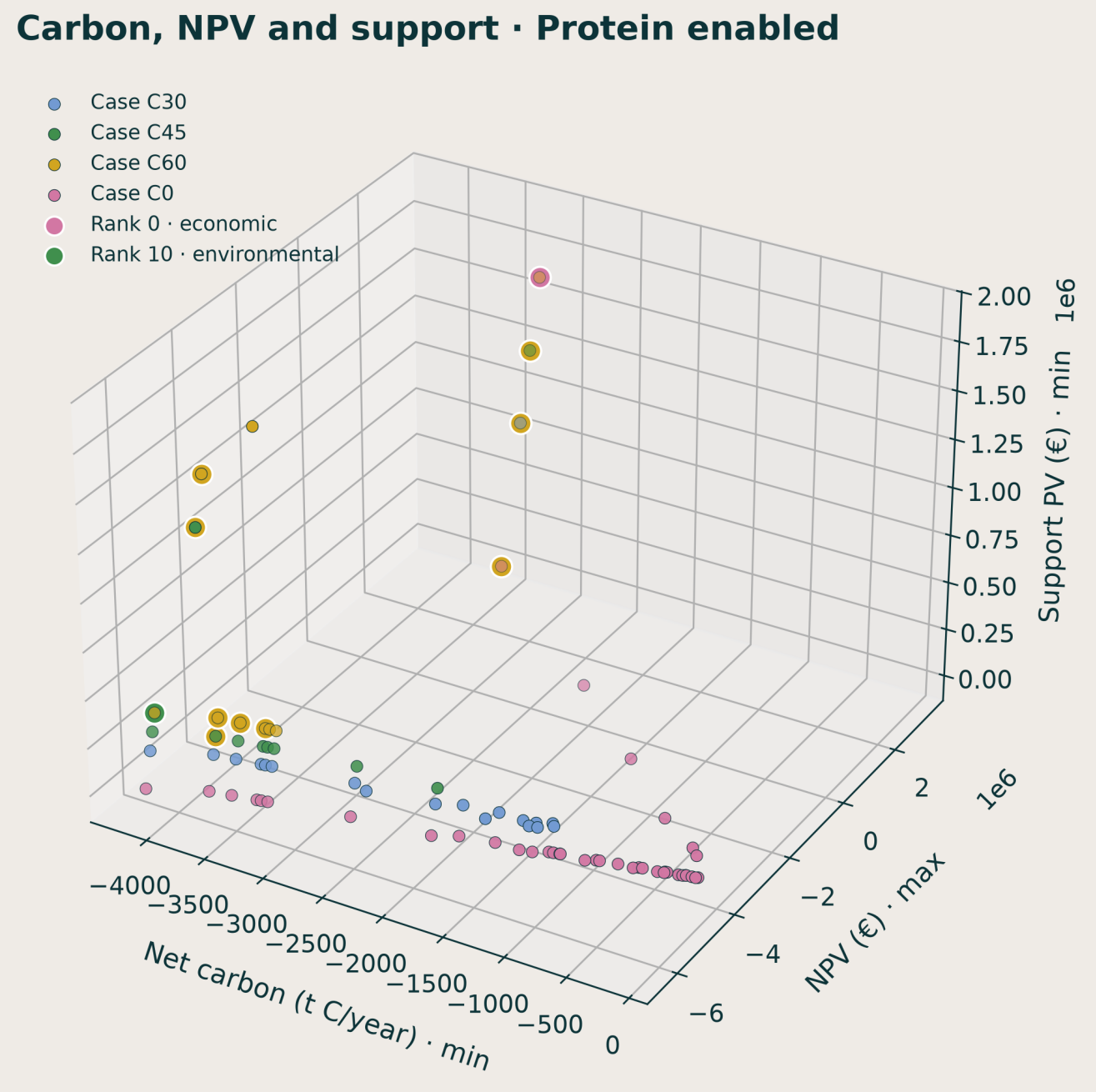


Figure 07. Three-objective Pareto front

Protein enabled | t C/year; €; € present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpi heatmap protein enabled

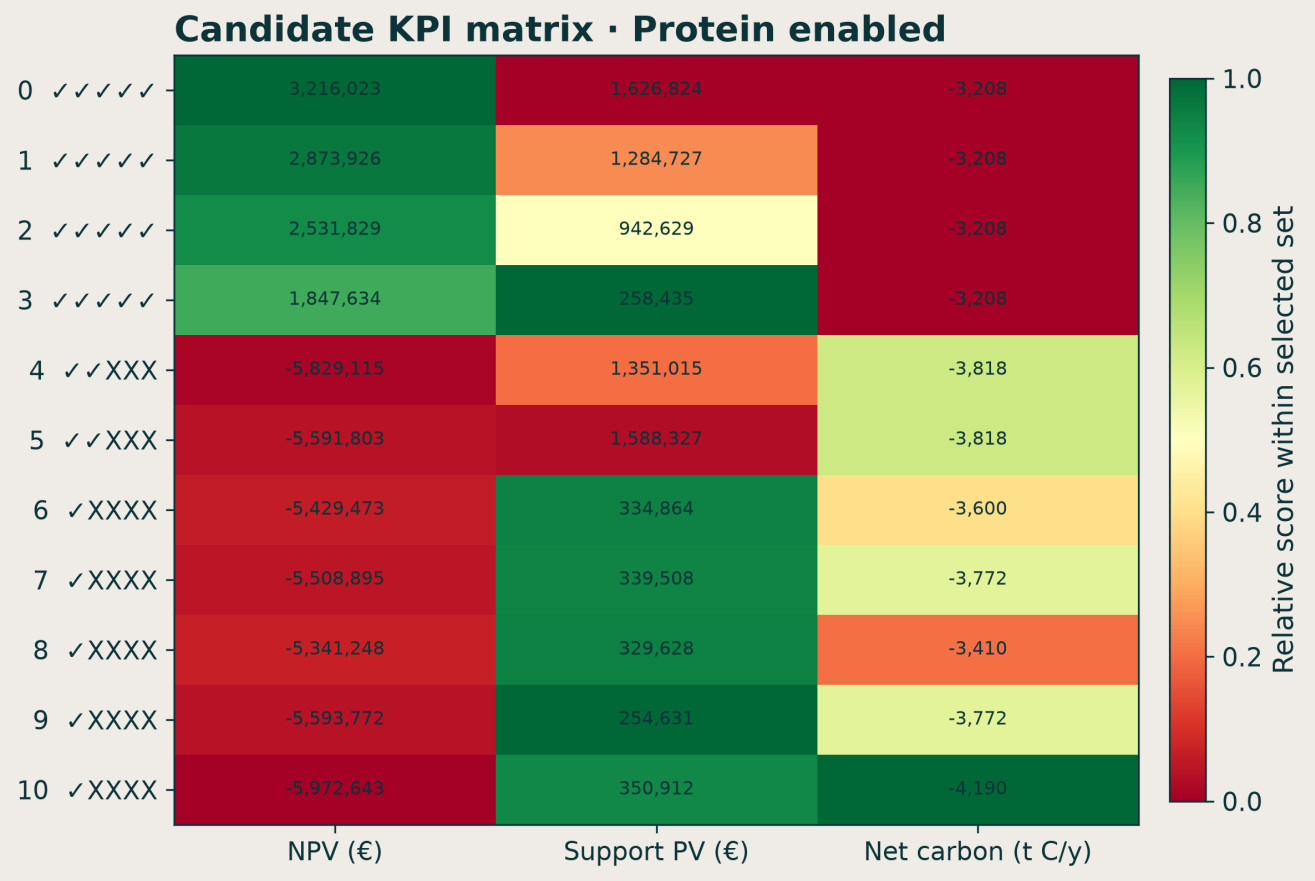


Figure 08. Selected candidate KPI comparison; color normalized by objective direction

Protein enabled | €, € present value, t C/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate economics protein enabled

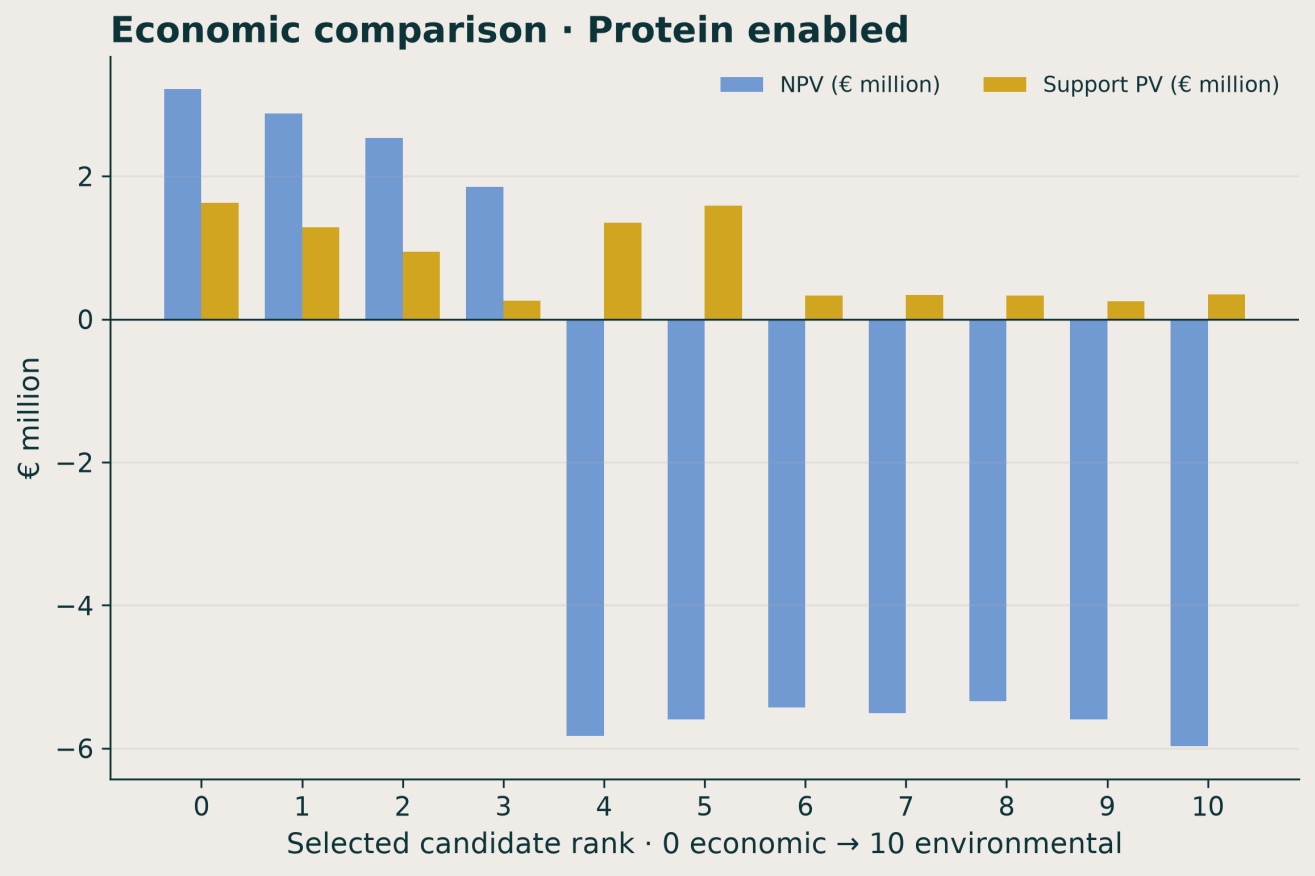


Figure 09. NPV and assumed public support PV for selected Pareto candidates

Protein enabled | € million | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate environment protein enabled

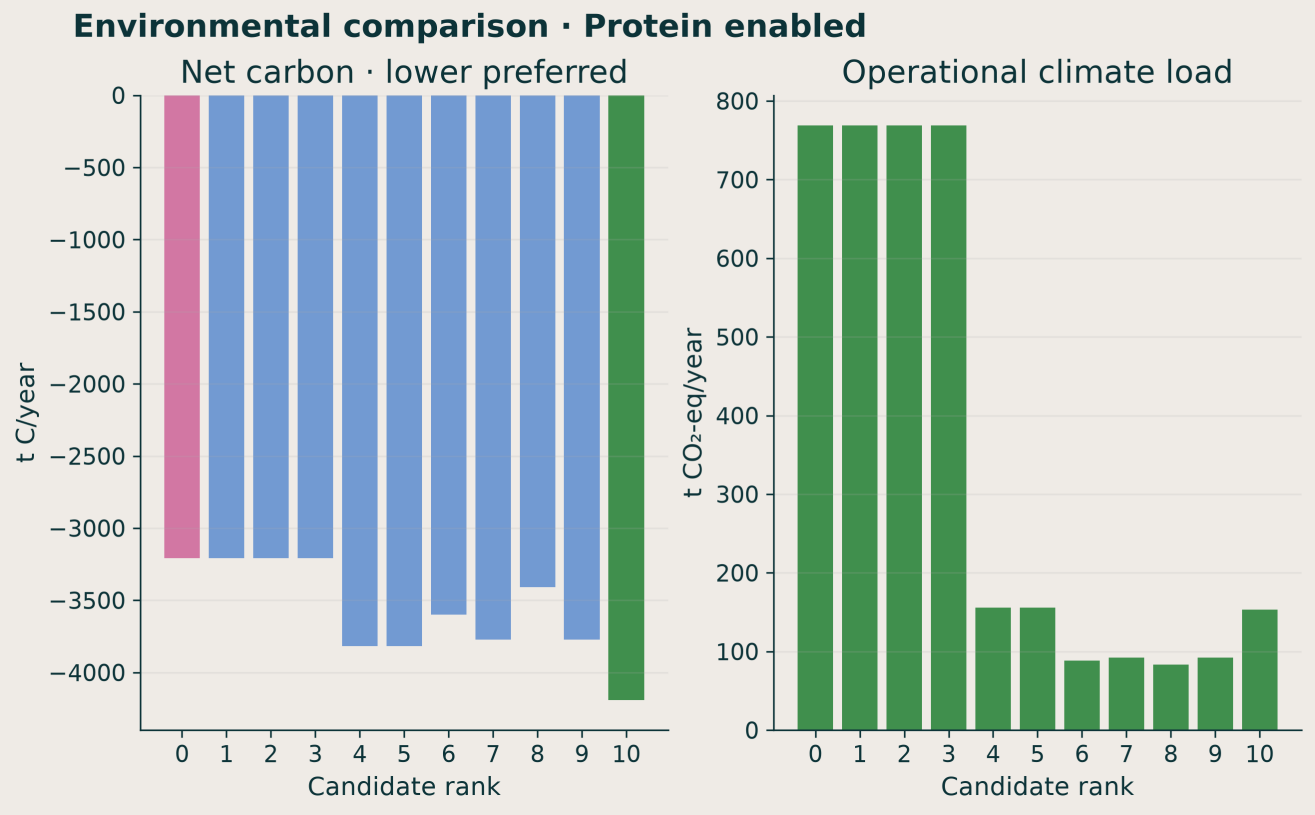


Figure 10. Net stored carbon objective and attributional operational climate load

Protein enabled | t C/year; t CO■-eq/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

carbon load vs stored carbon protein enabled

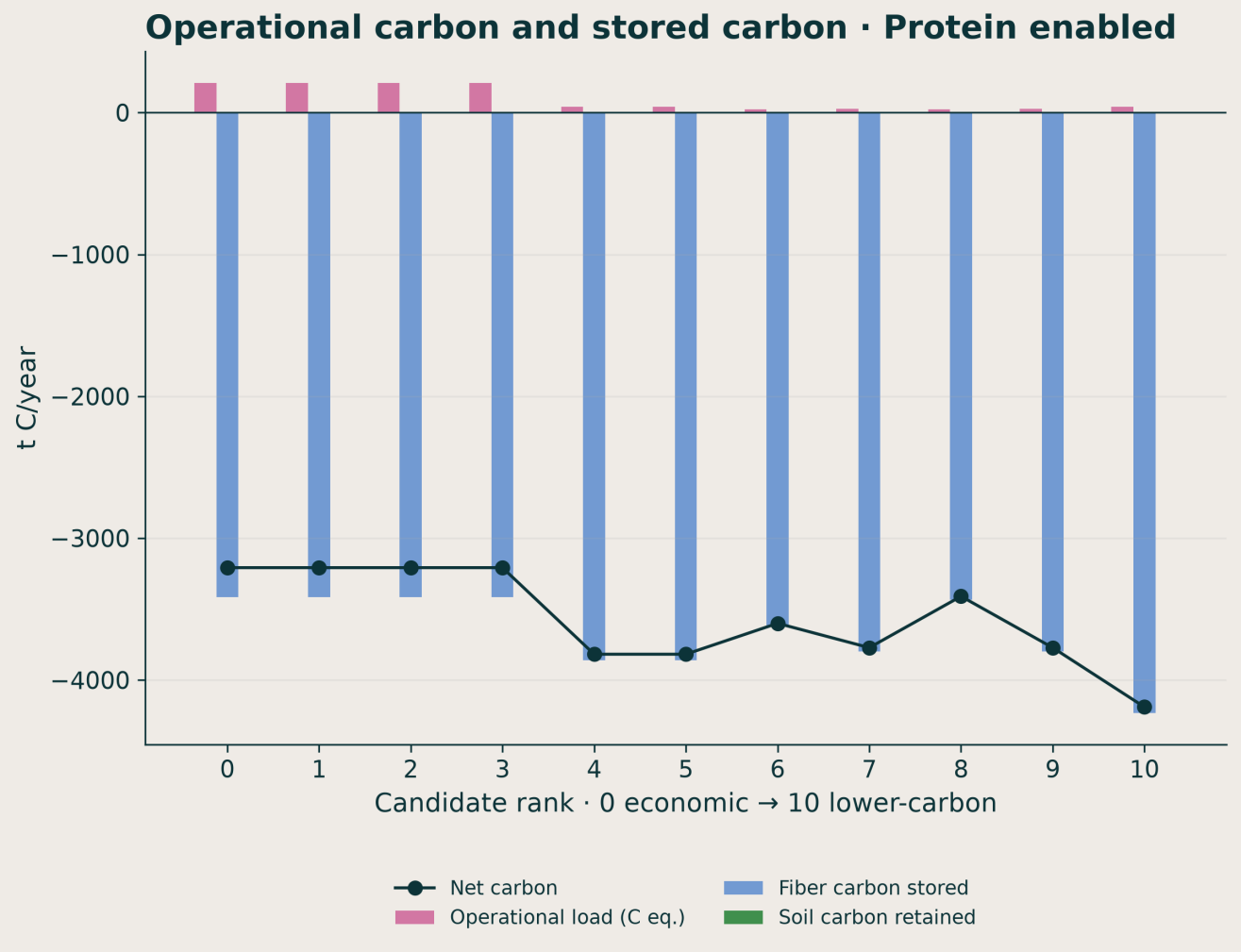


Figure 11. Attributional operational carbon-equivalent load minus fiber and soil retained product carbon

Protein enabled | t C/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

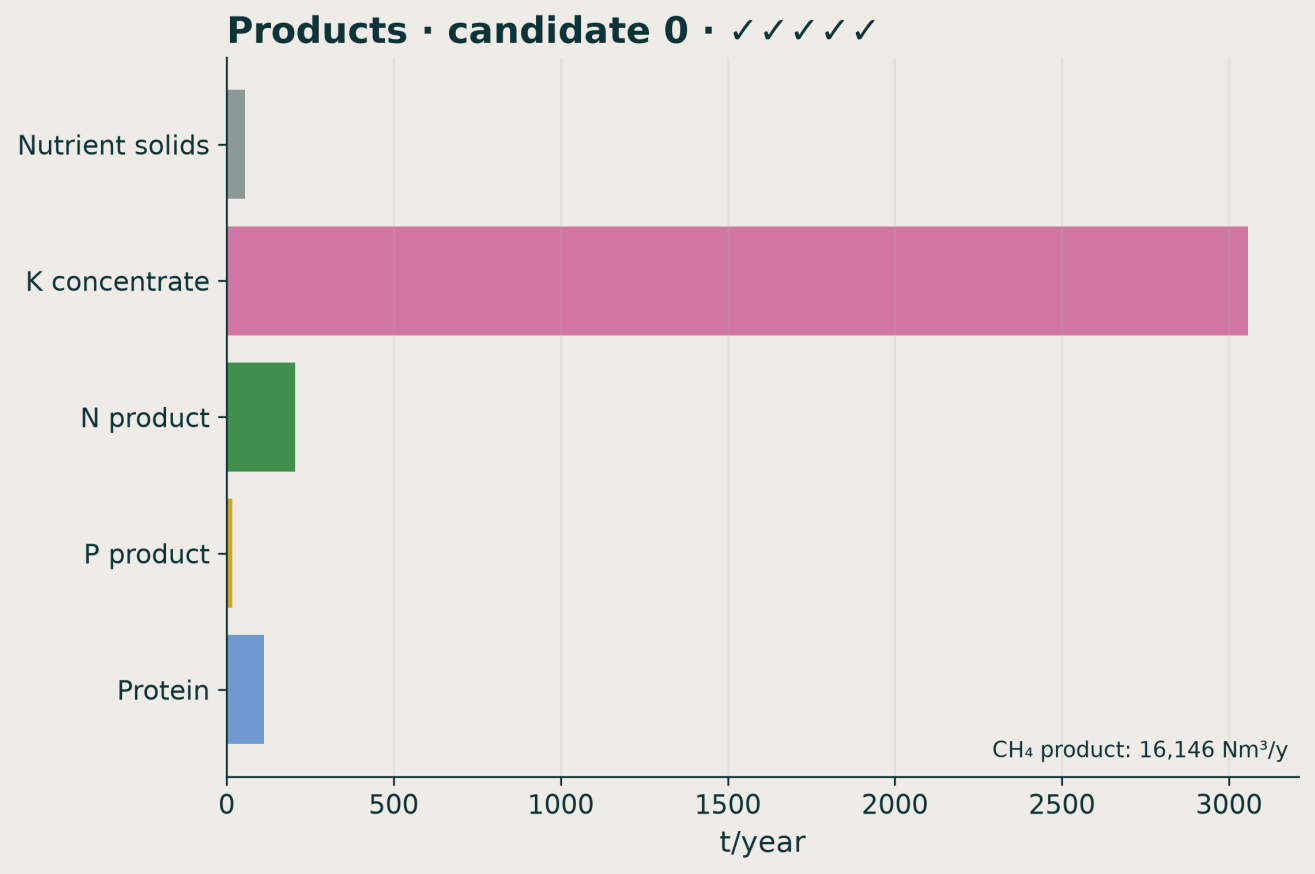


Figure 12. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

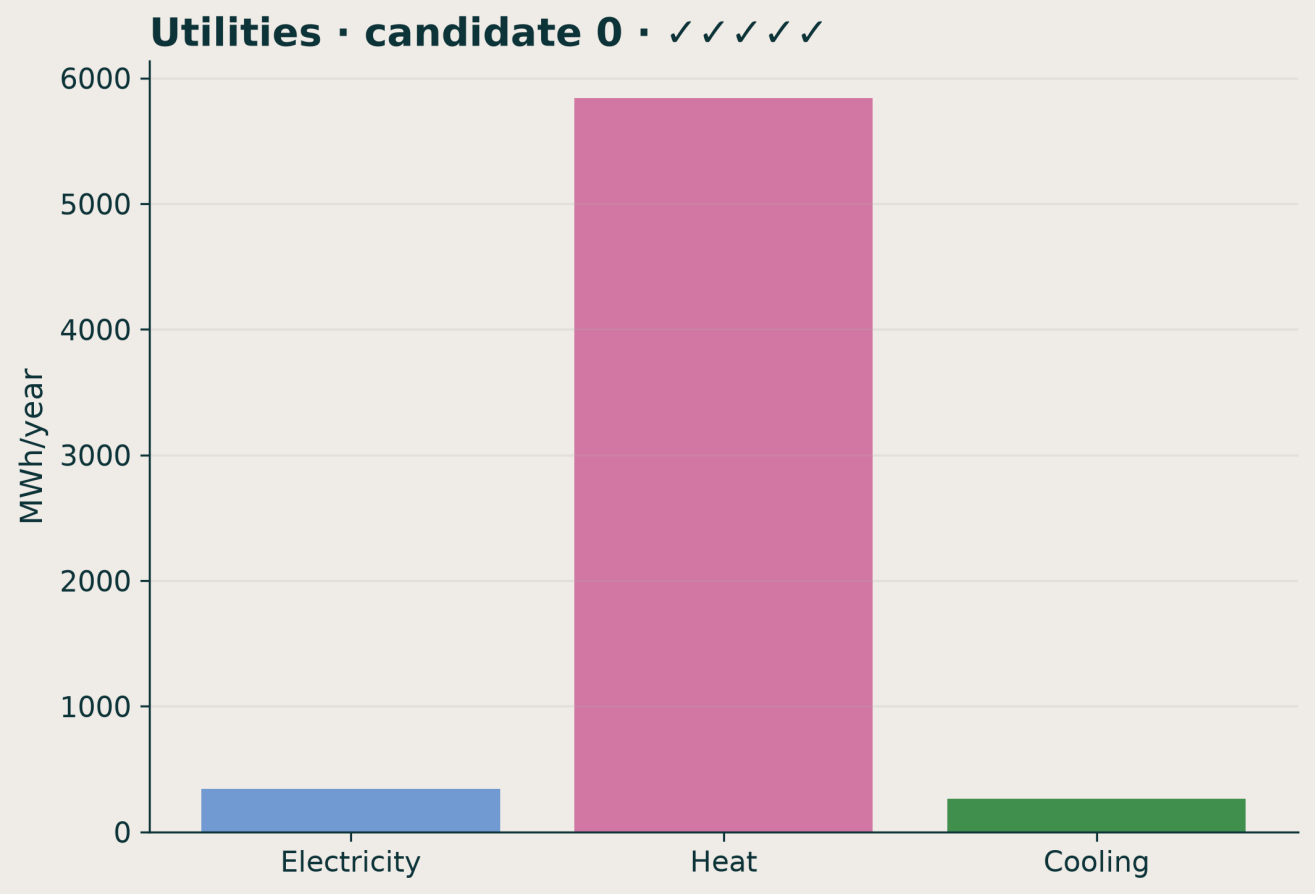


Figure 13. Annual electricity, heating and cooling duty

protein_enabled; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

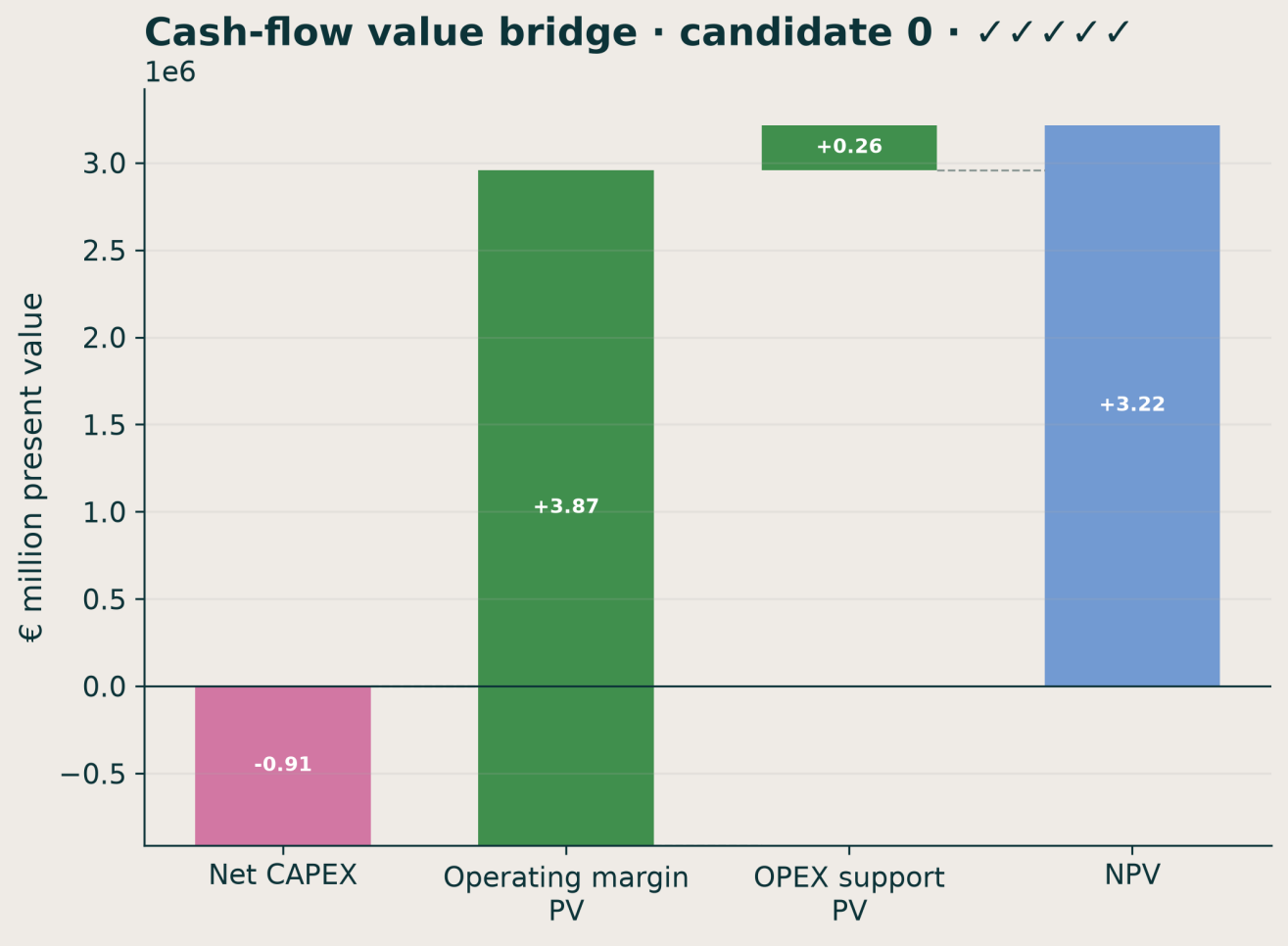


Figure 14. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 0; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

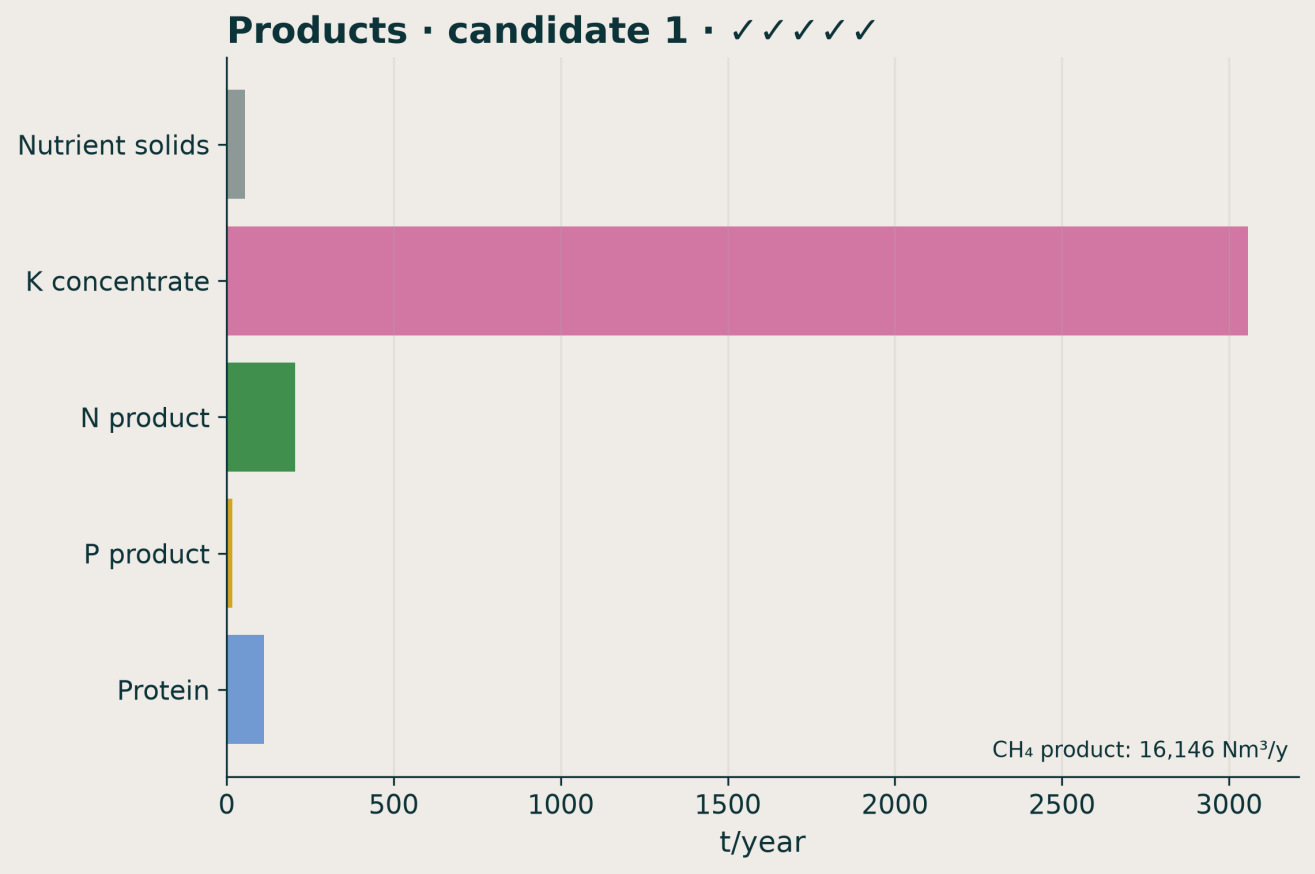


Figure 15. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

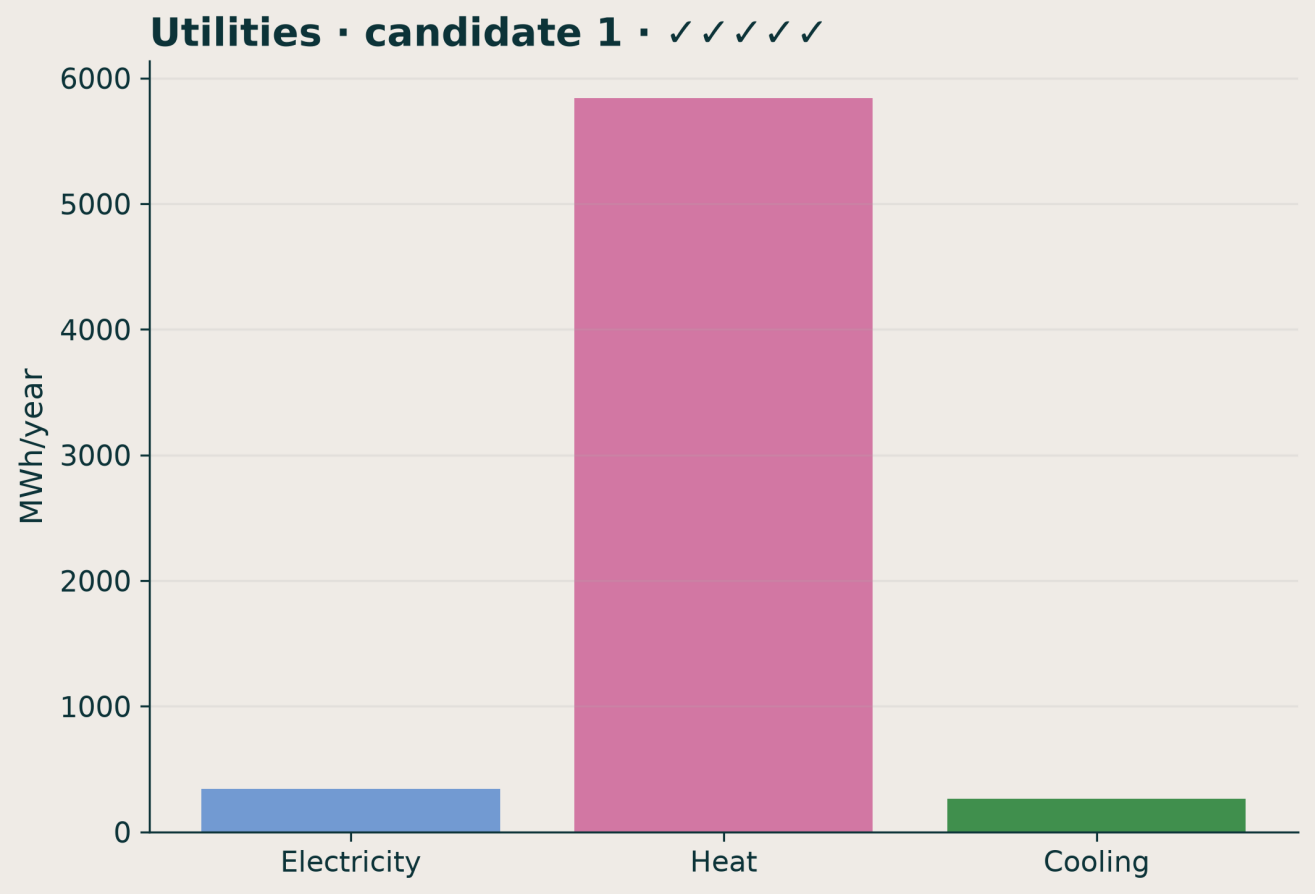


Figure 16. Annual electricity, heating and cooling duty

protein_enabled; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

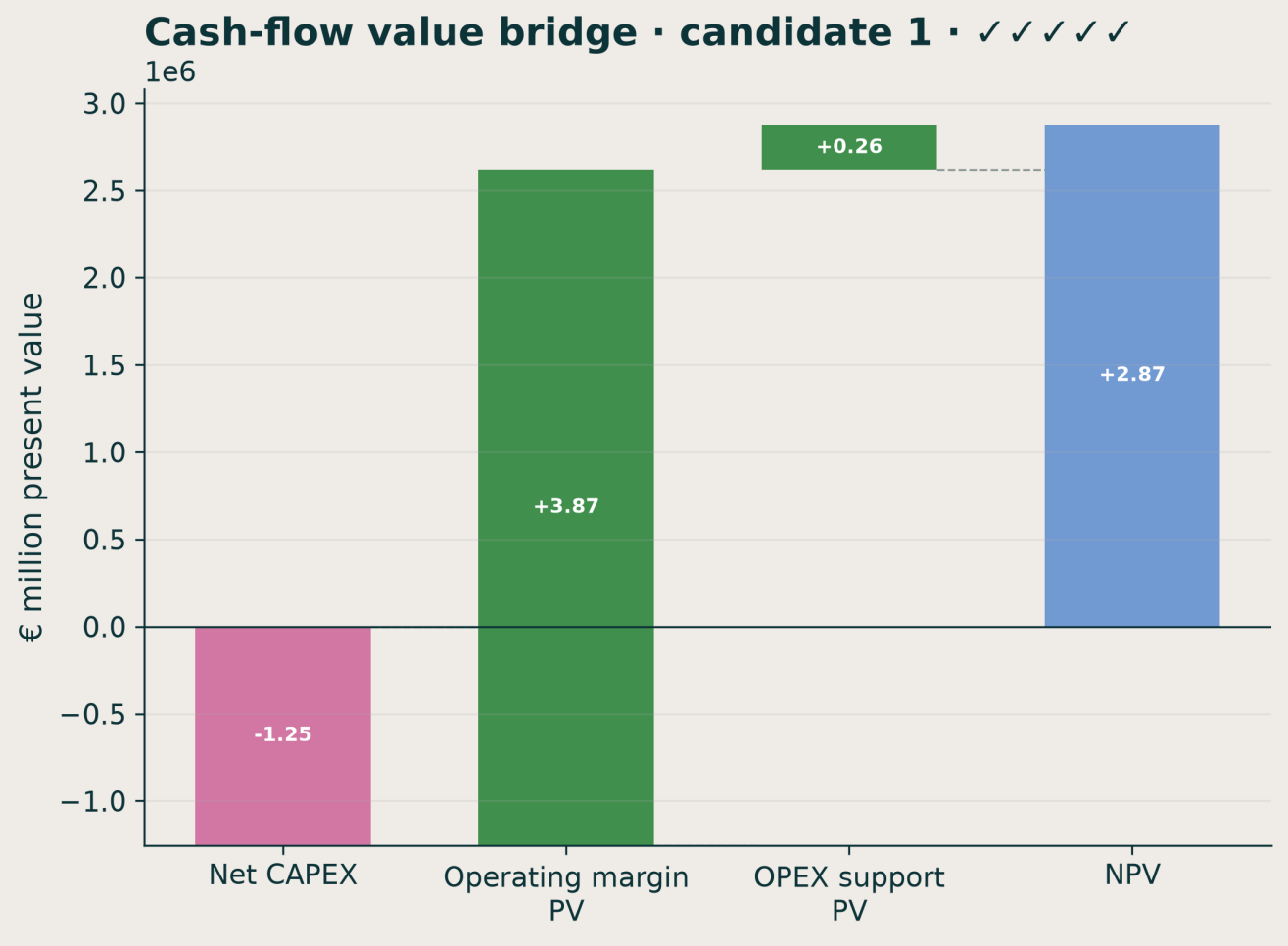


Figure 17. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 1; support C45 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

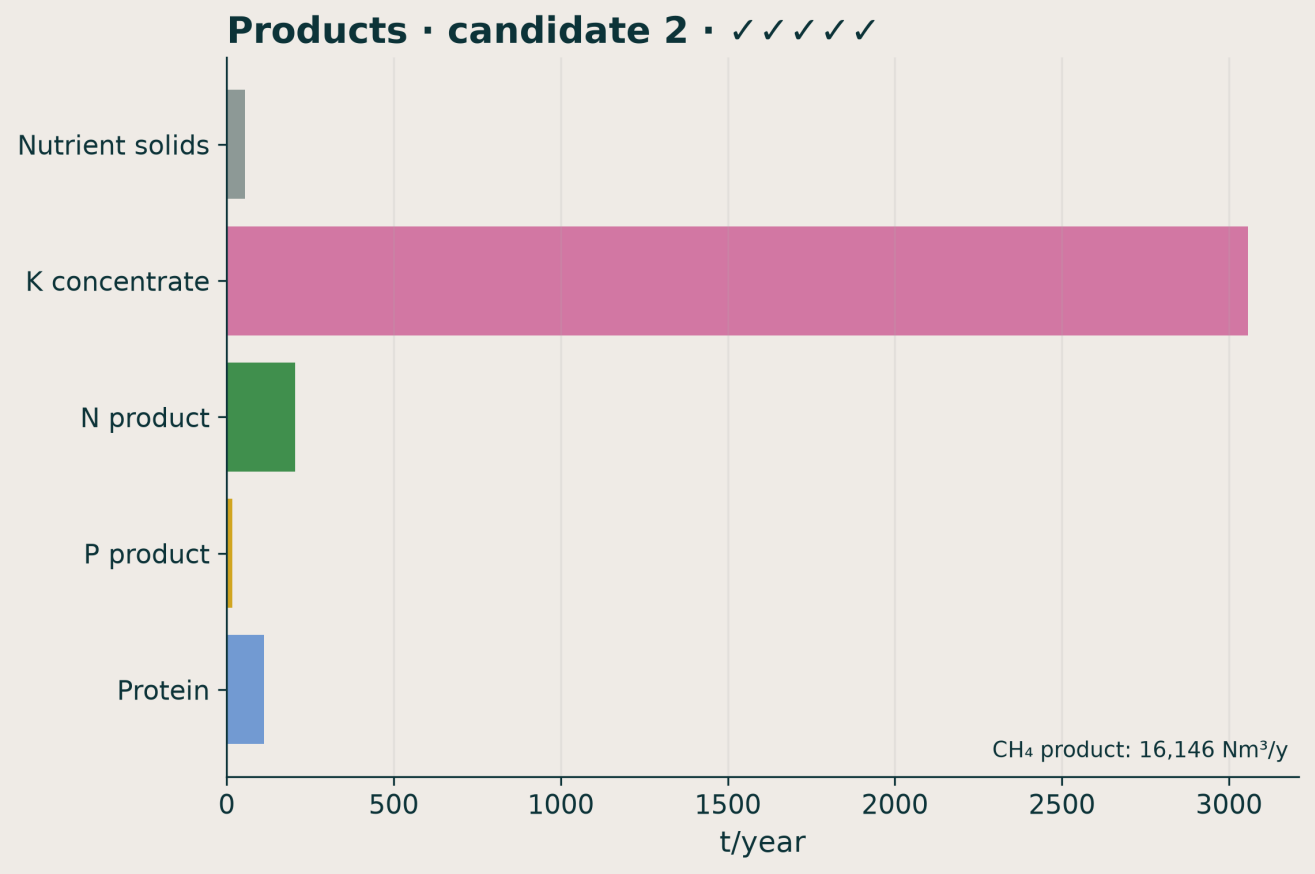


Figure 18. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

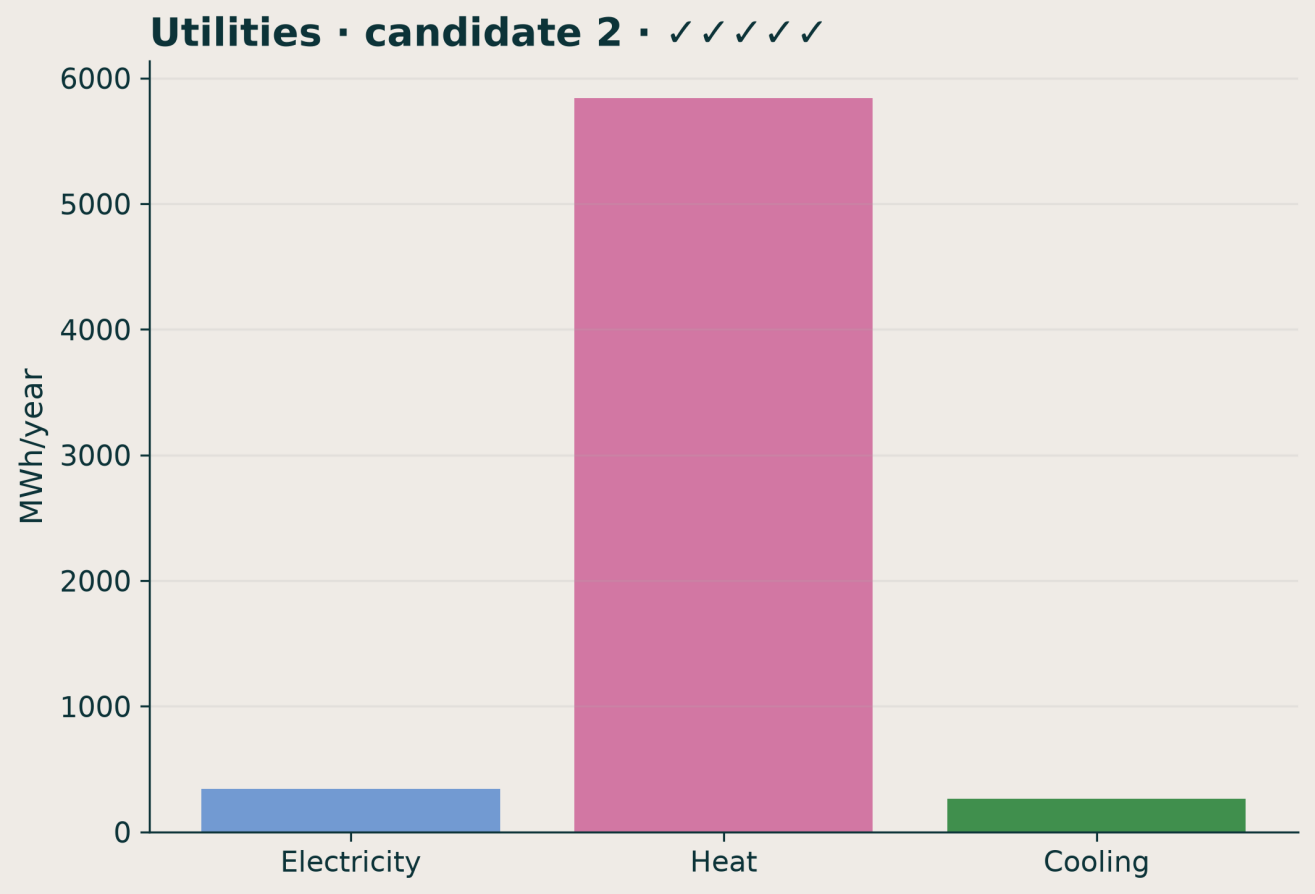


Figure 19. Annual electricity, heating and cooling duty

protein_enabled; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

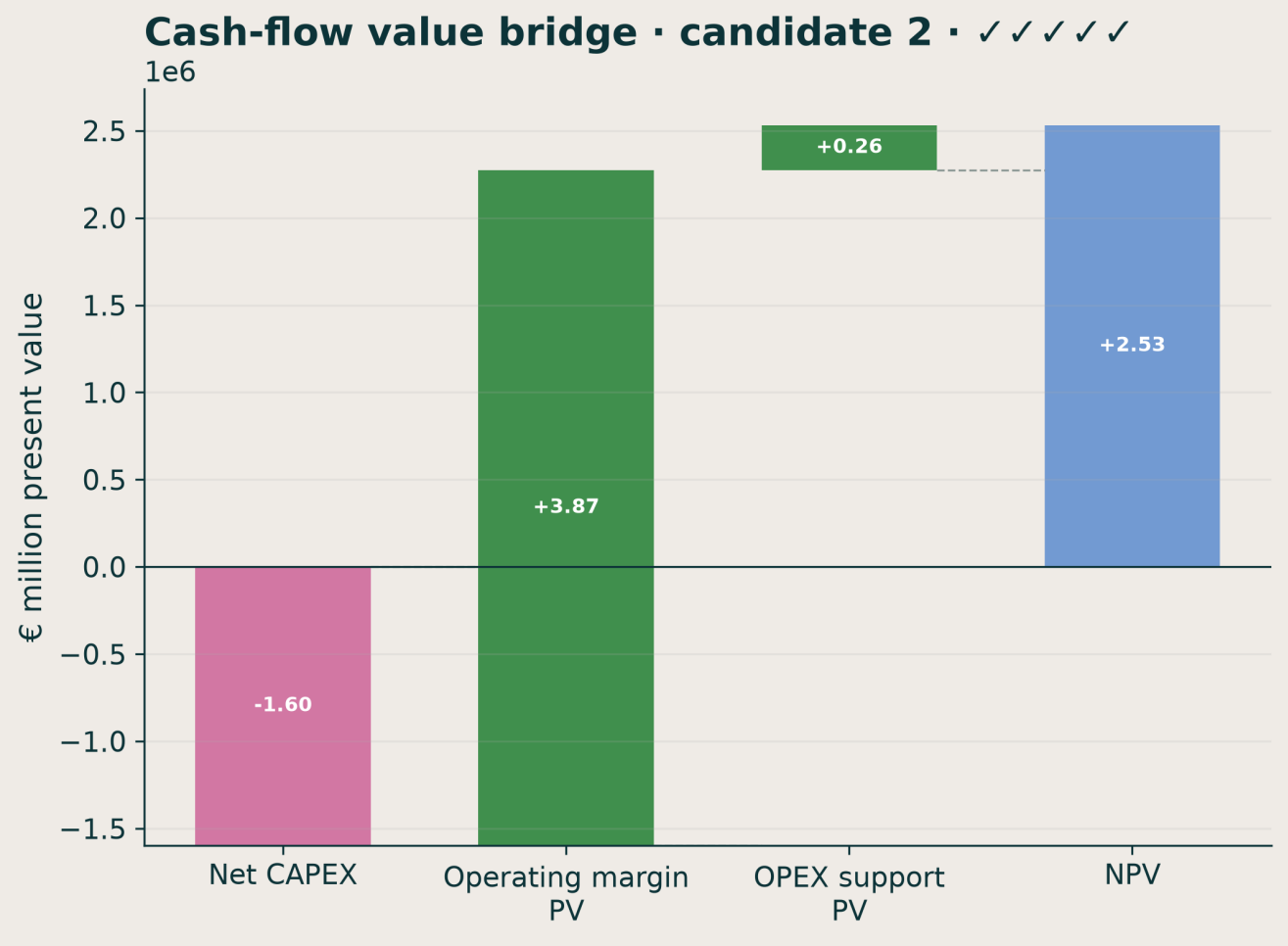


Figure 20. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 2; support C30 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

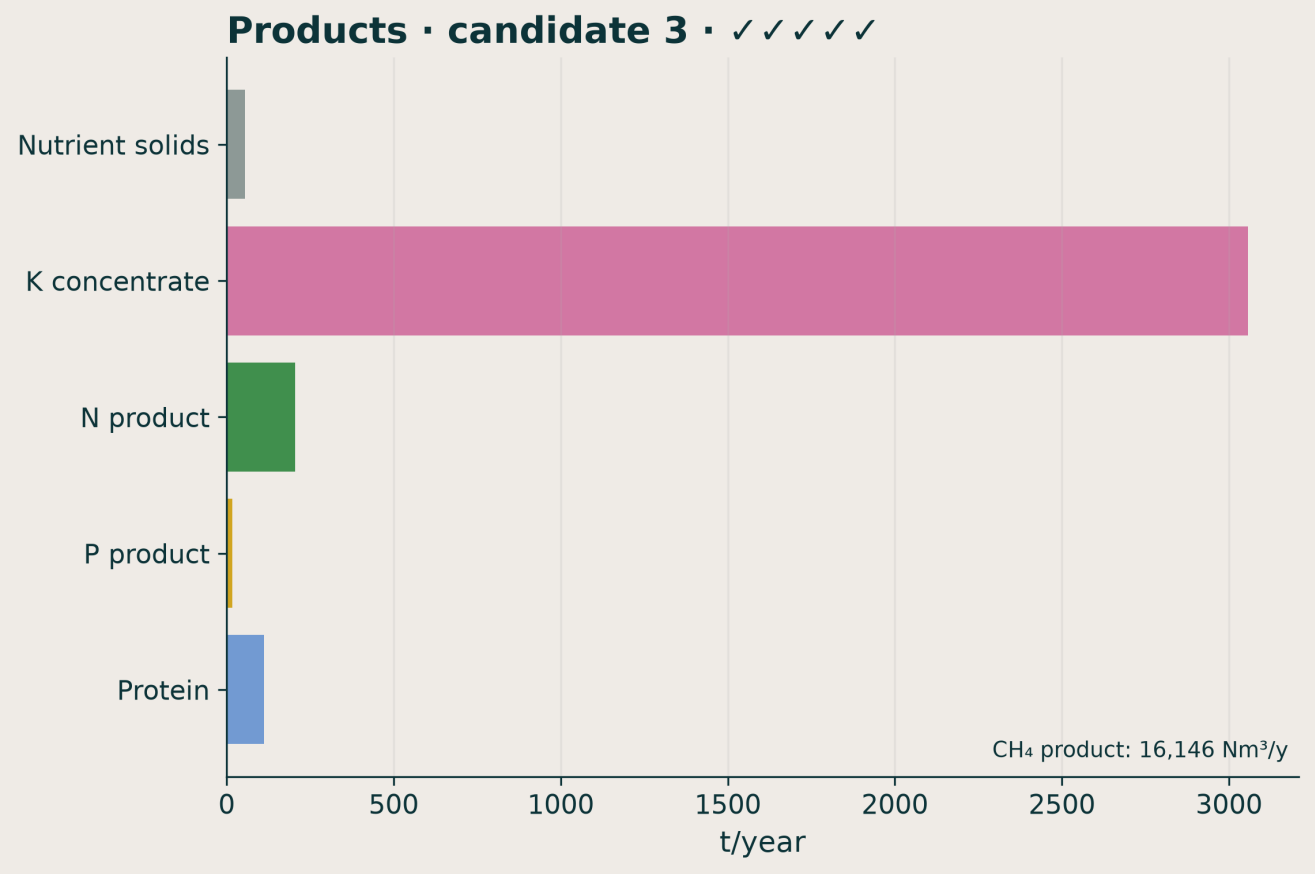


Figure 21. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

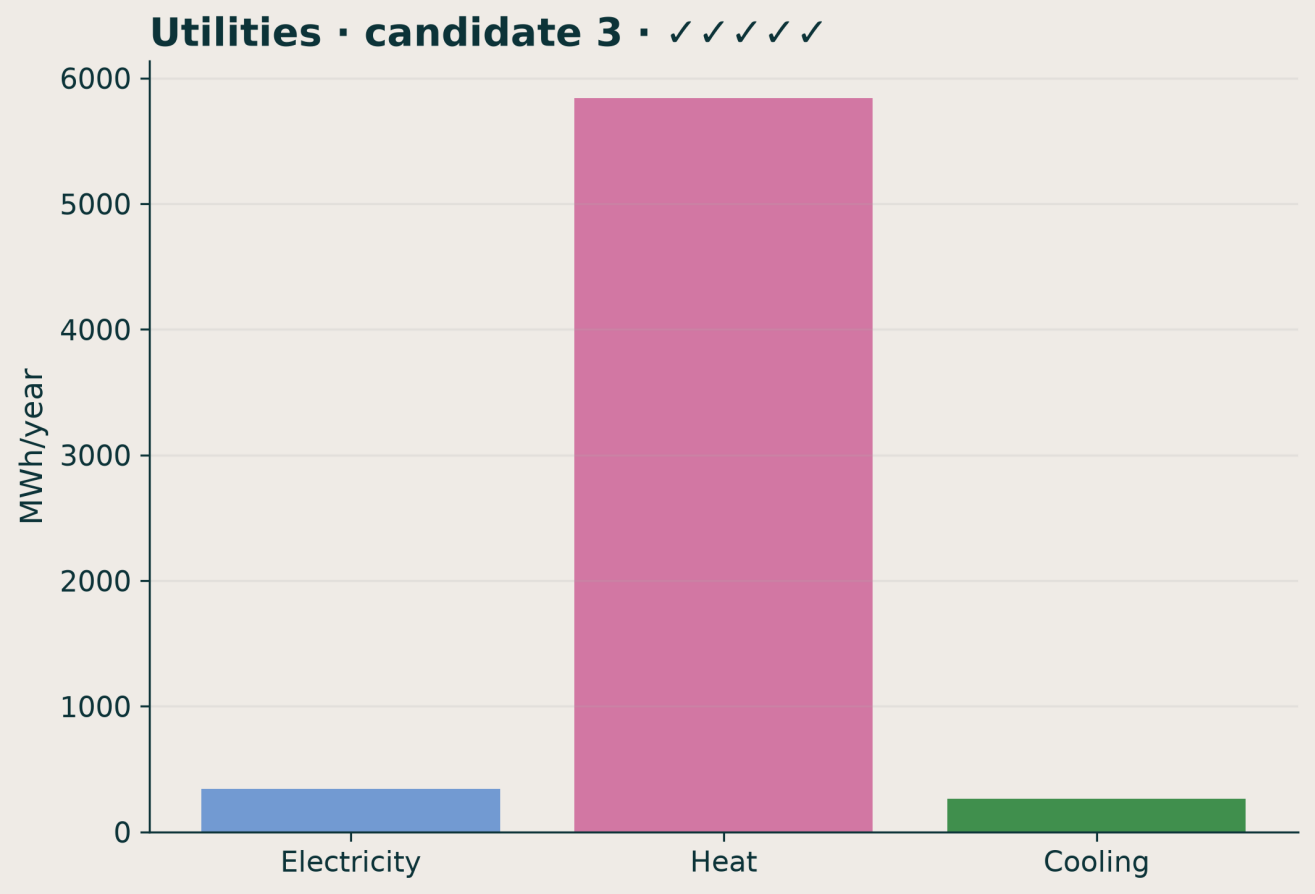


Figure 22. Annual electricity, heating and cooling duty

protein_enabled; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD1', 'biogas_upgrading': 'BG4', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

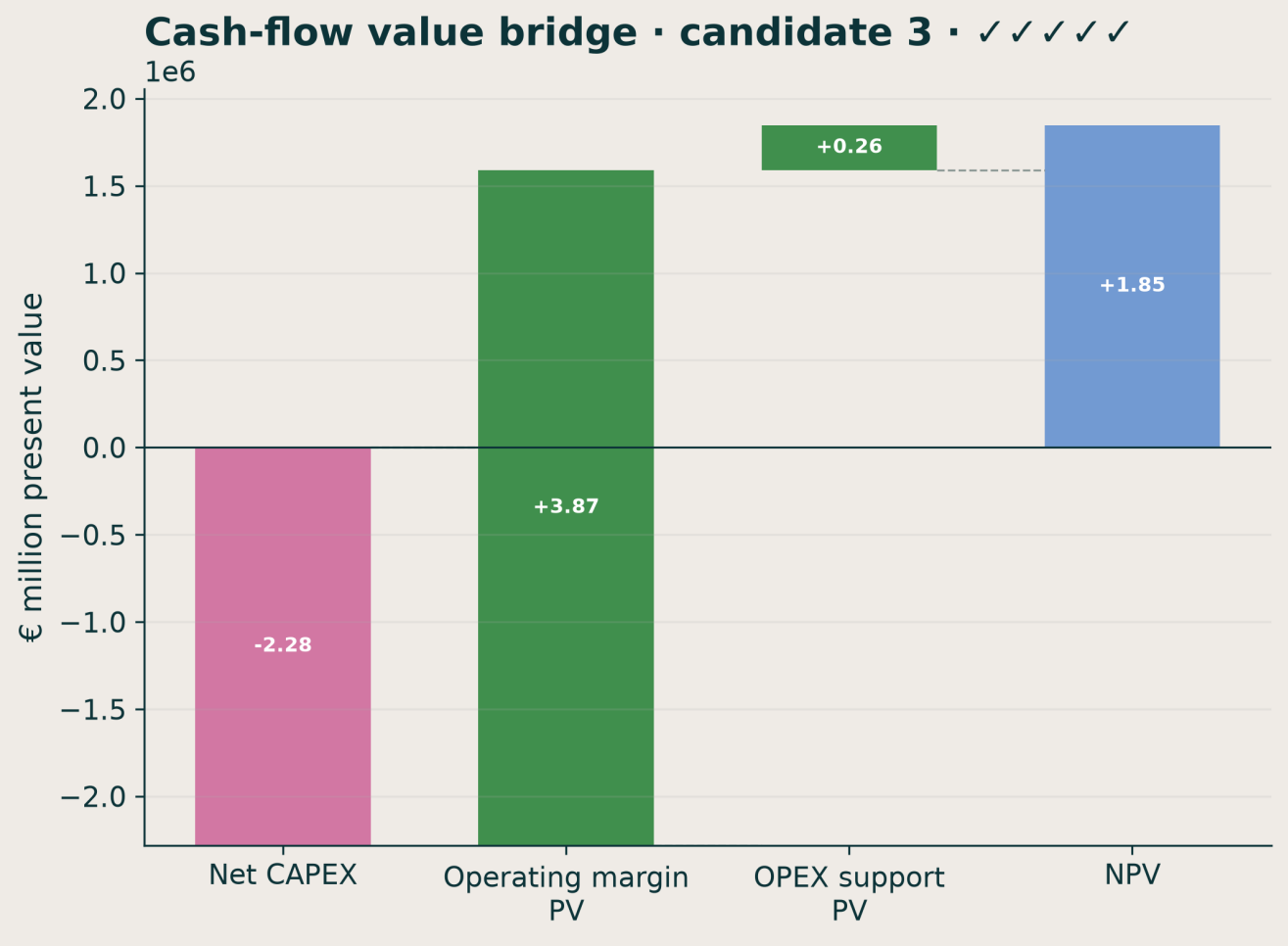


Figure 23. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 3; support C0 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

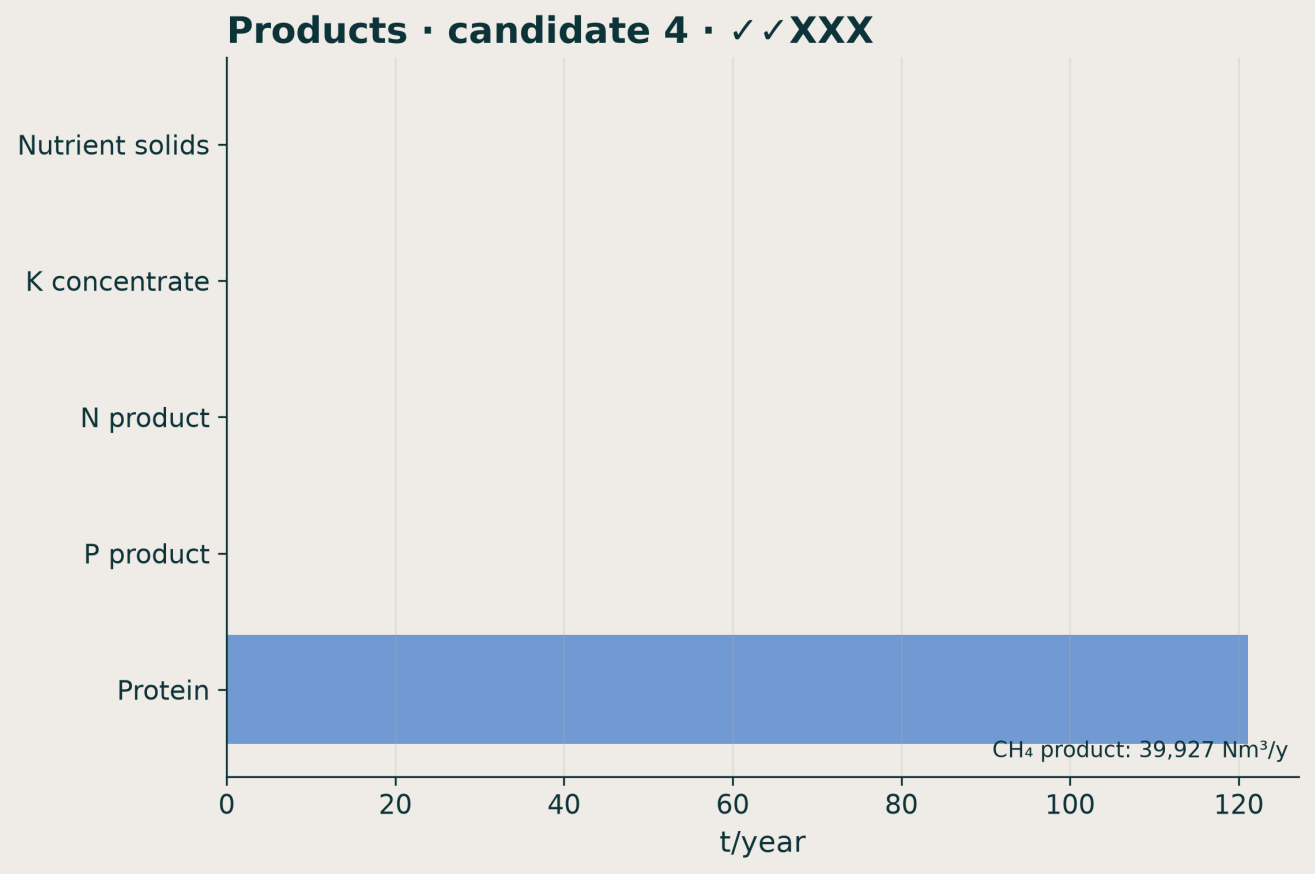


Figure 24. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

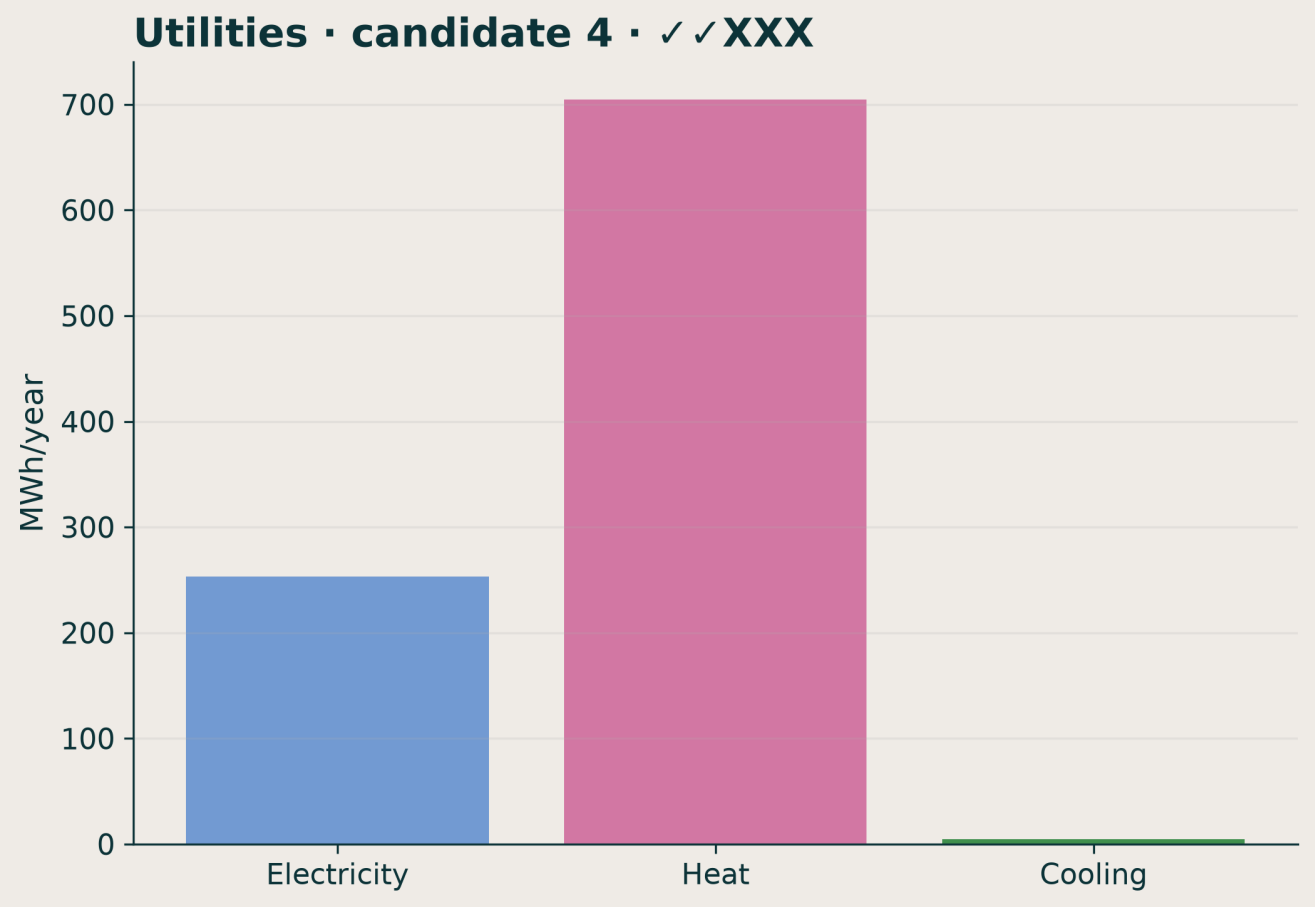


Figure 25. Annual electricity, heating and cooling duty

protein_enabled; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

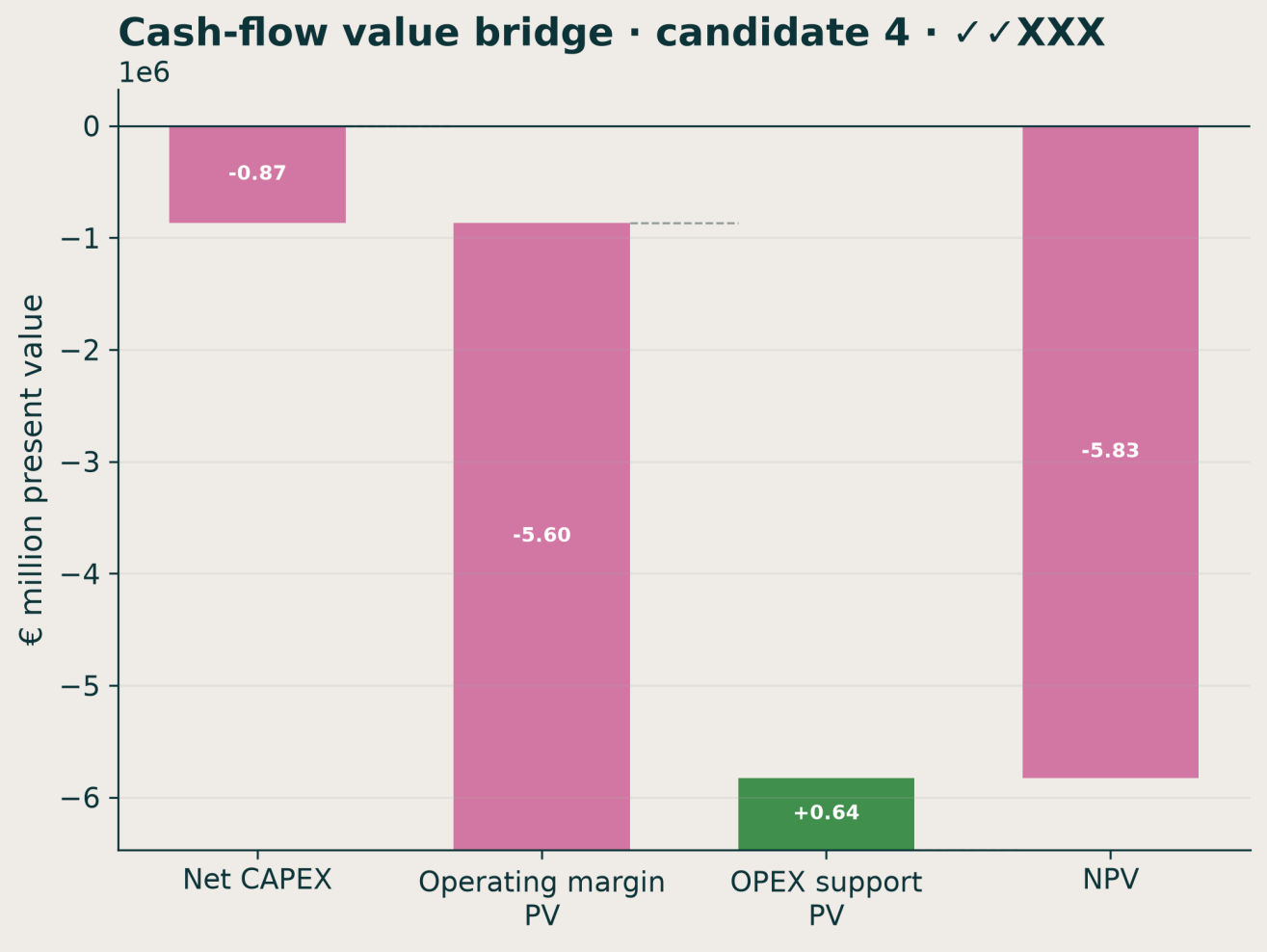


Figure 26. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 4; support C45 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

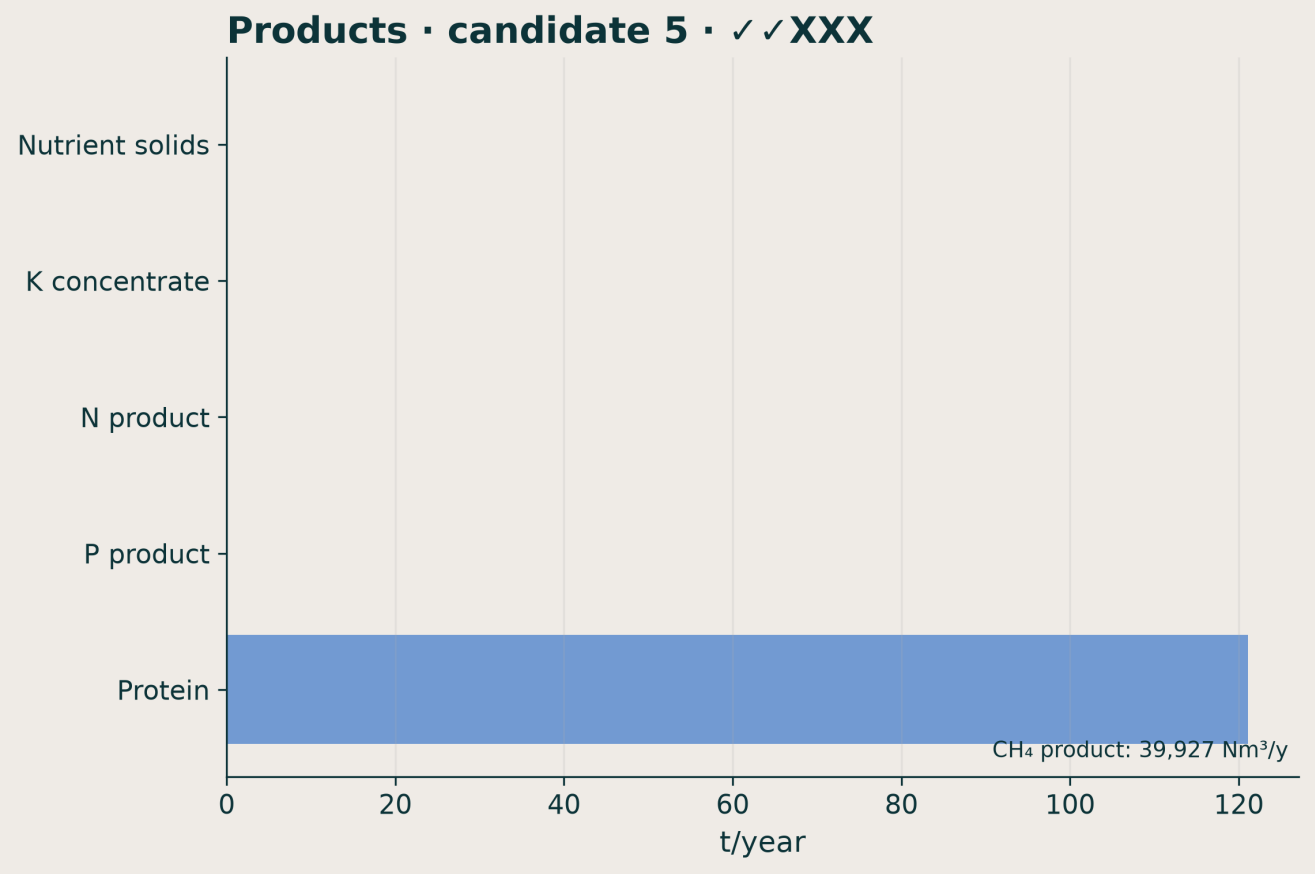


Figure 27. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

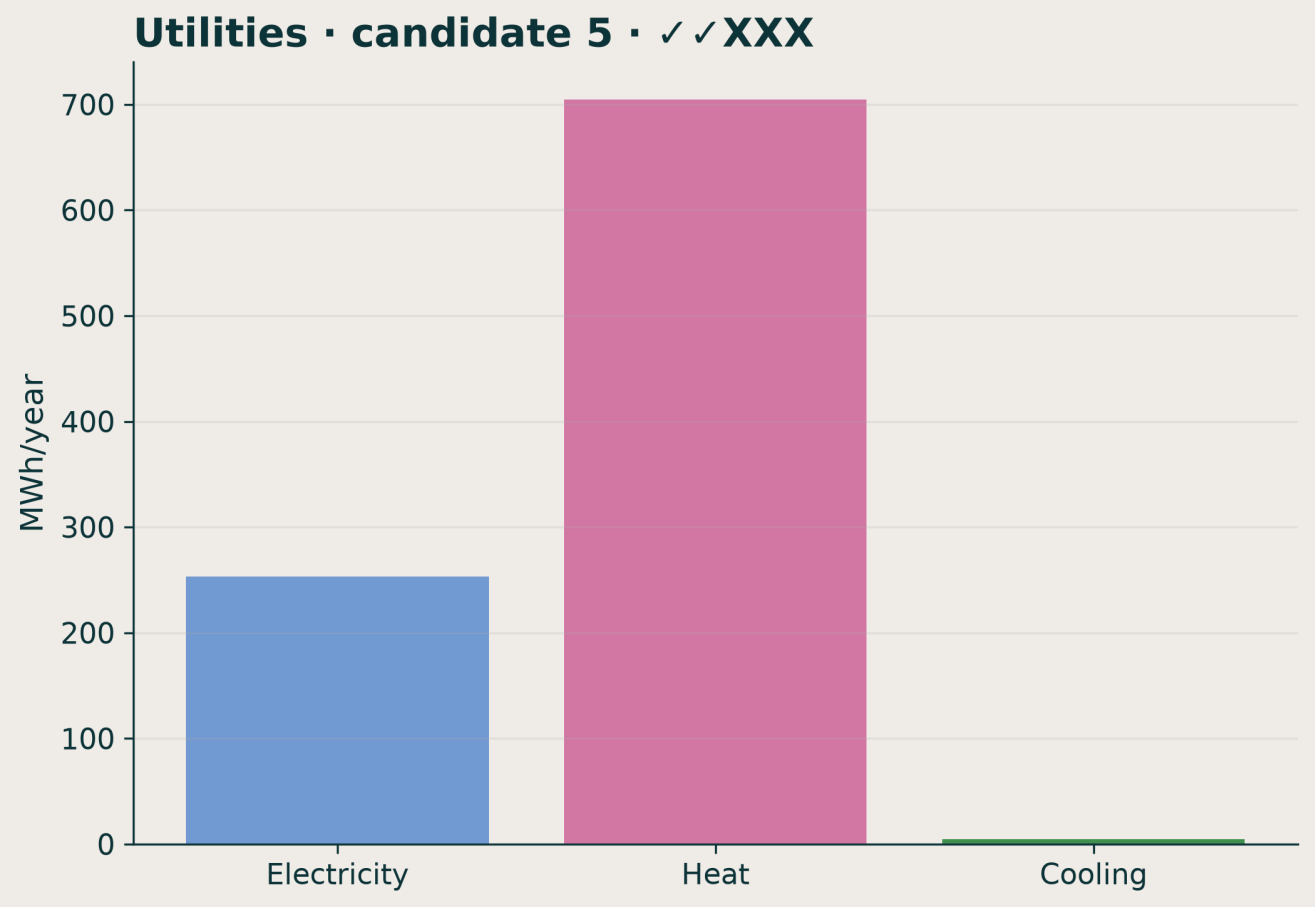


Figure 28. Annual electricity, heating and cooling duty

protein_enabled; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

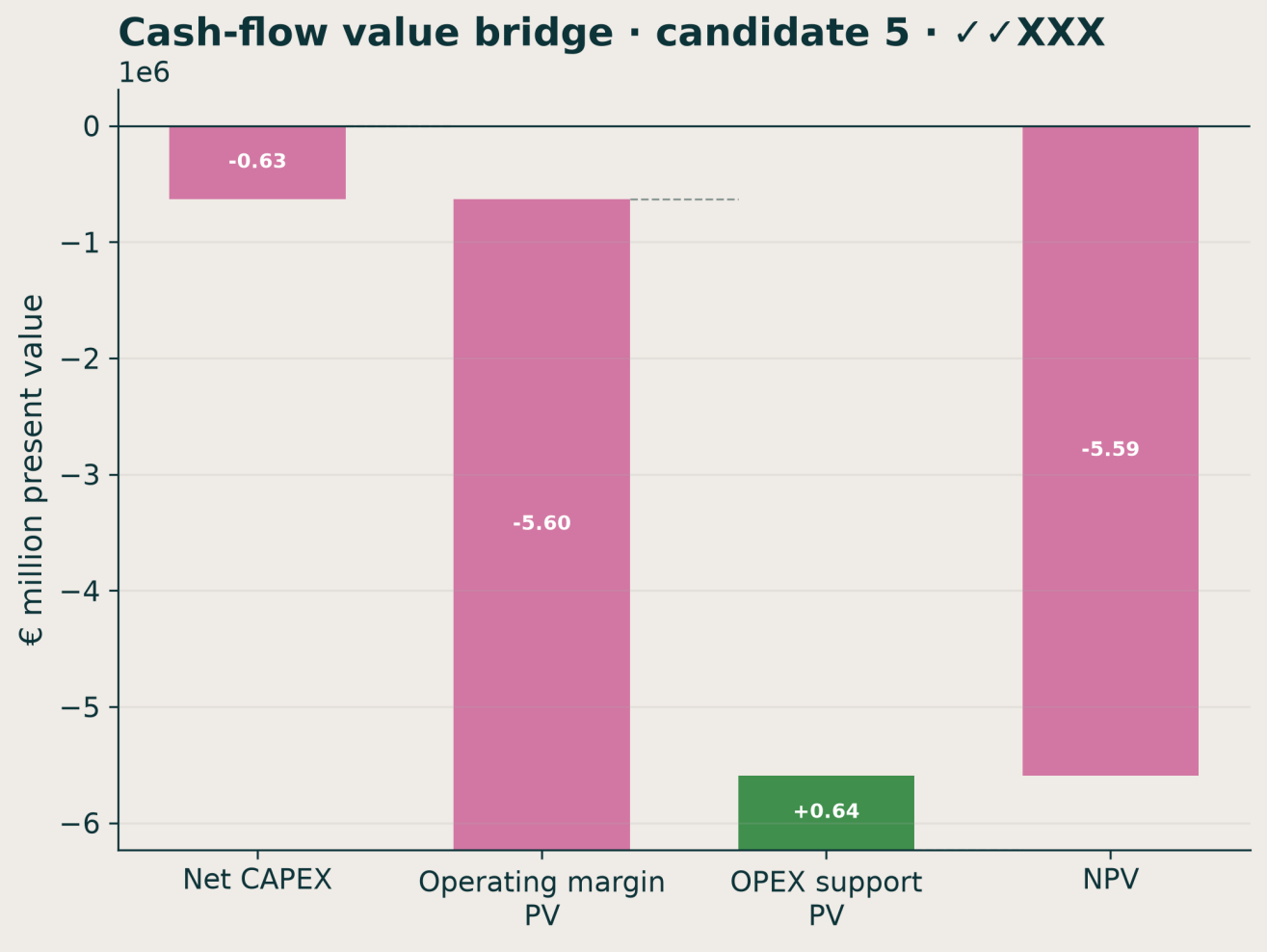


Figure 29. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 5; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

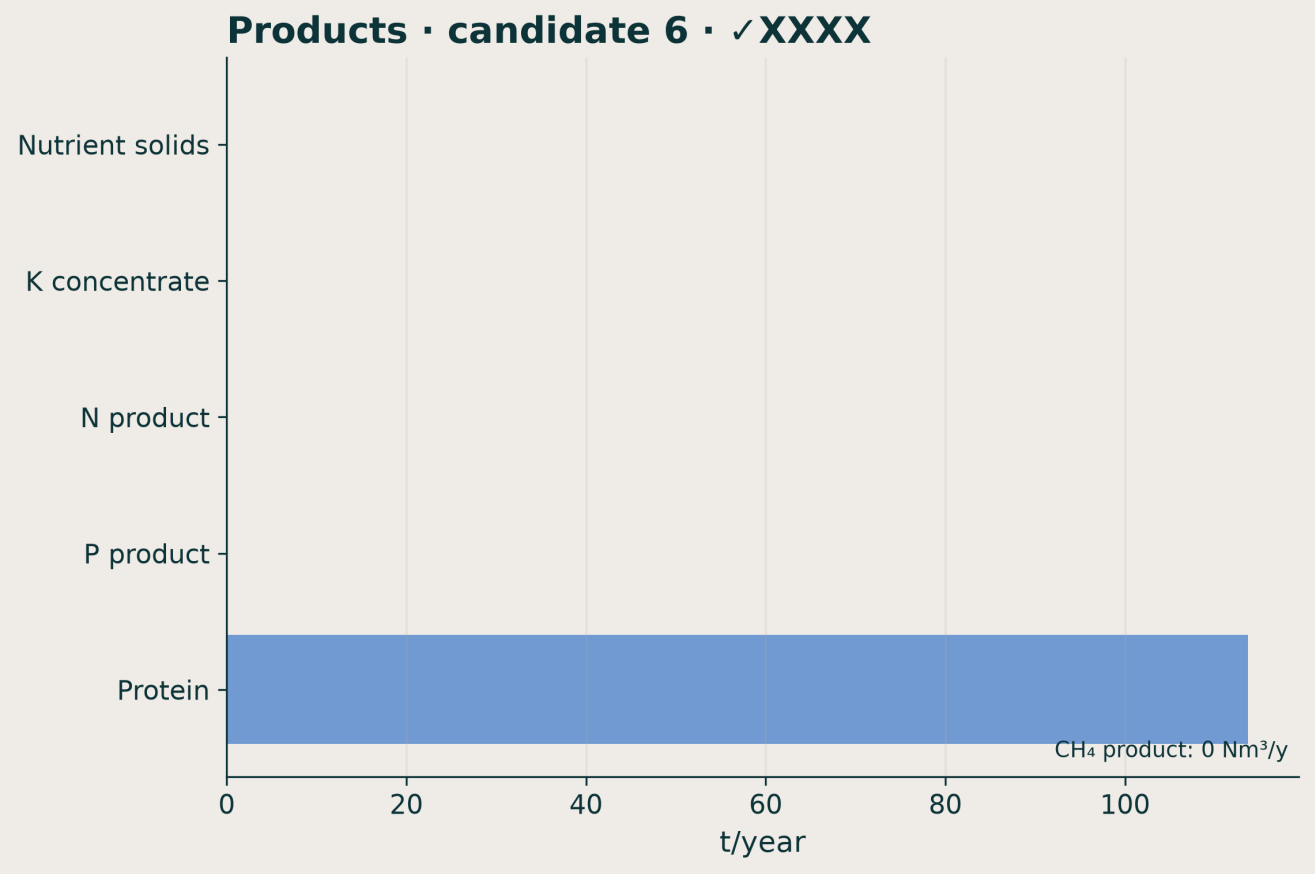


Figure 30. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

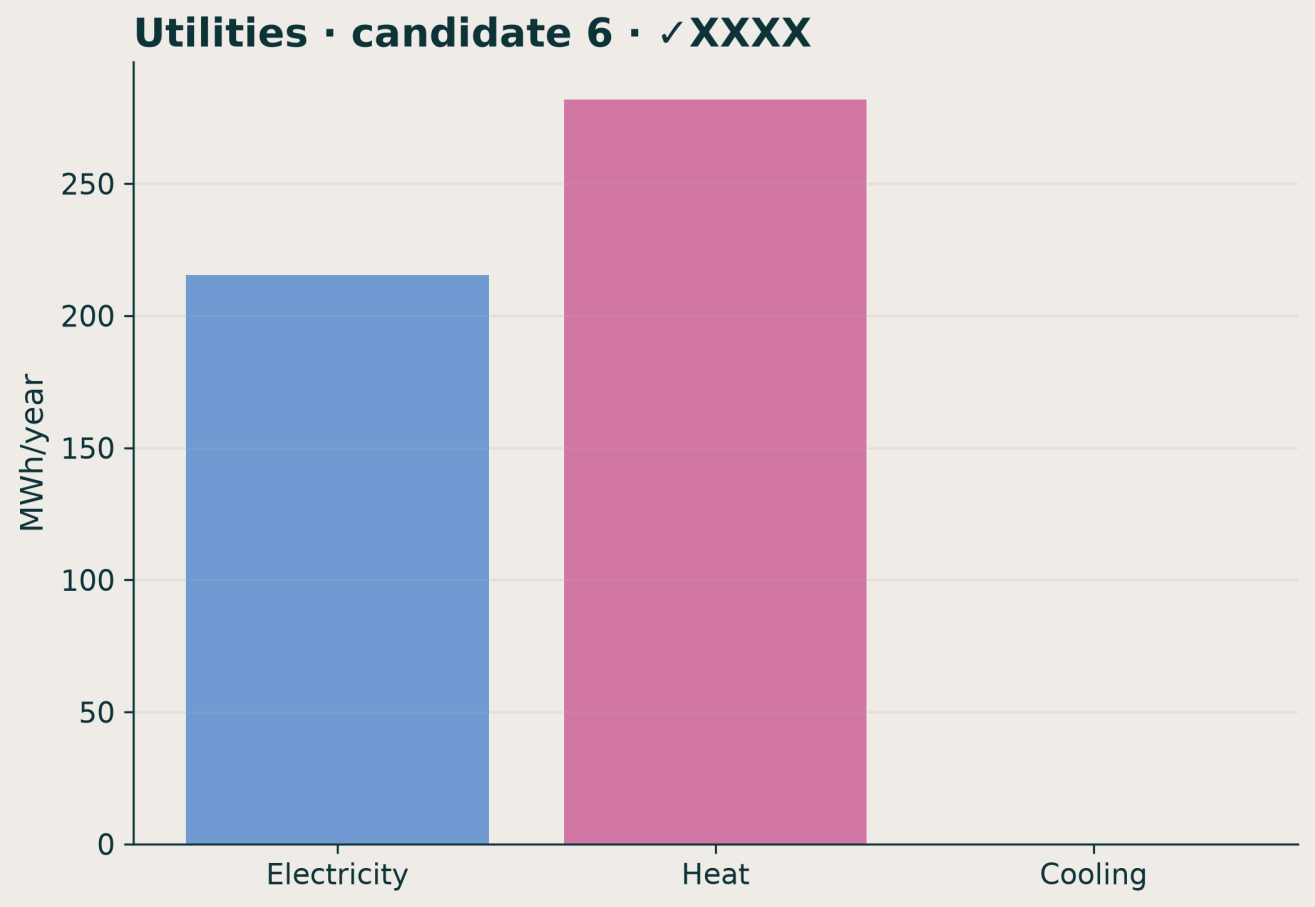


Figure 31. Annual electricity, heating and cooling duty

protein_enabled; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

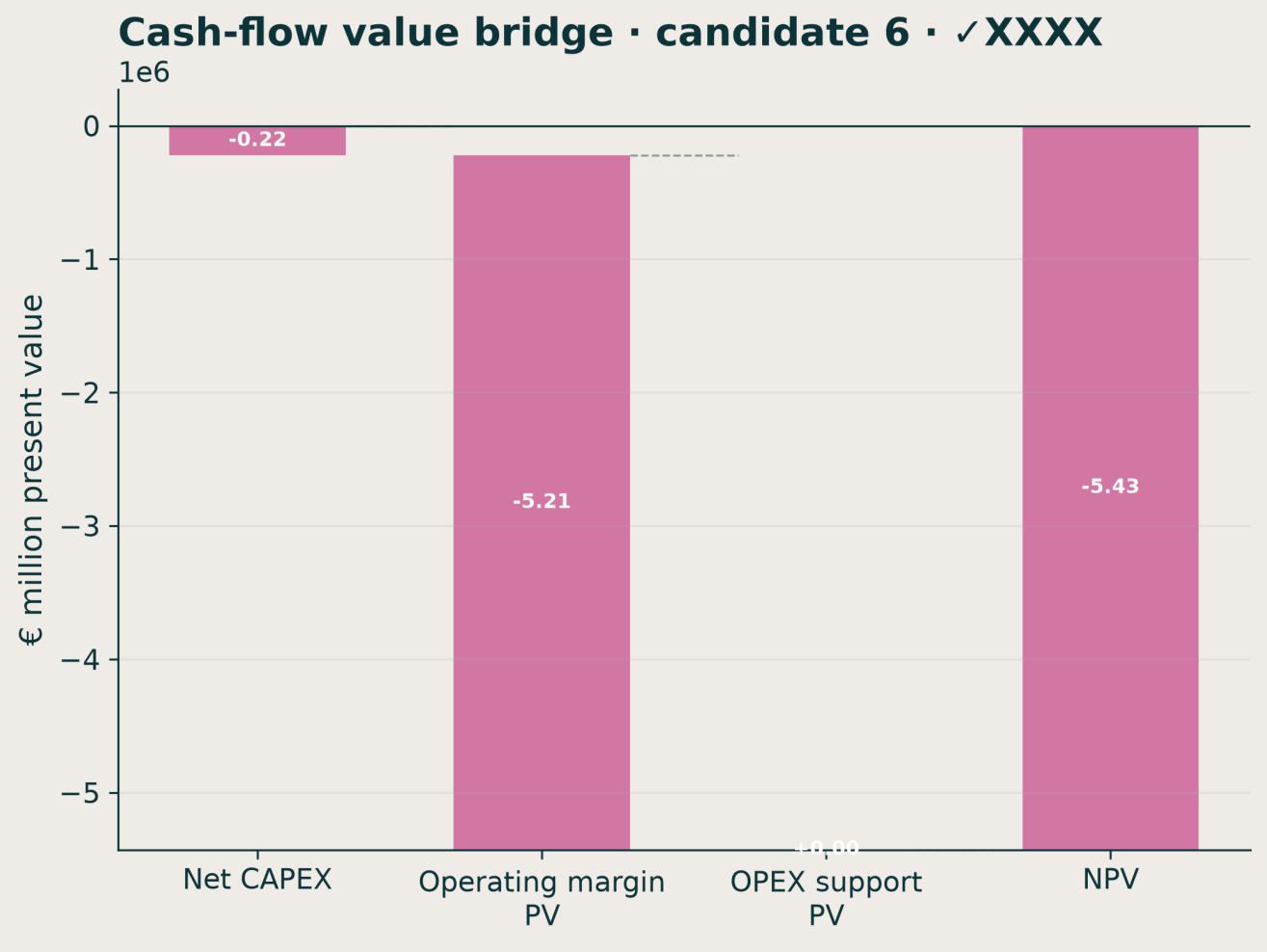


Figure 32. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 6; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

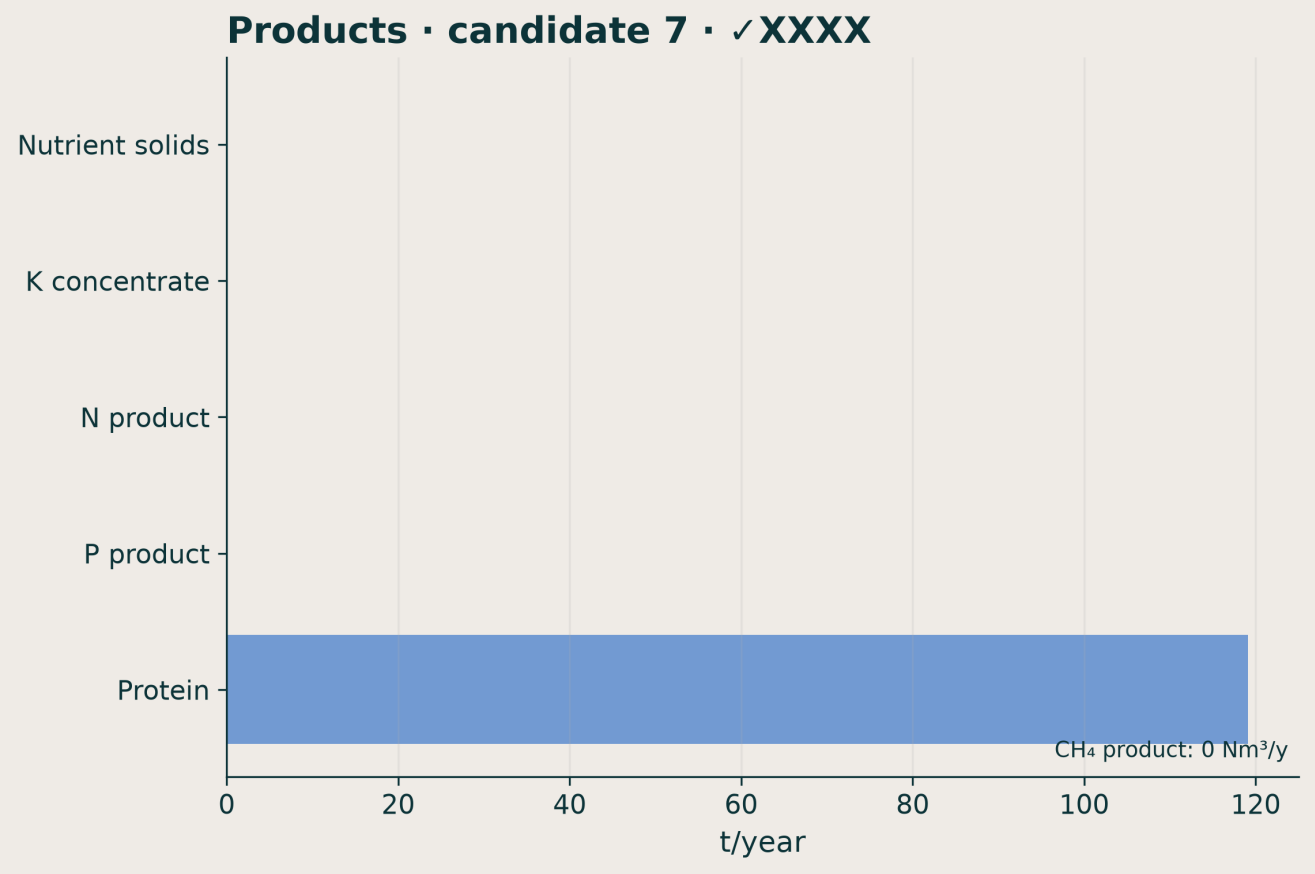


Figure 33. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_enabled; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

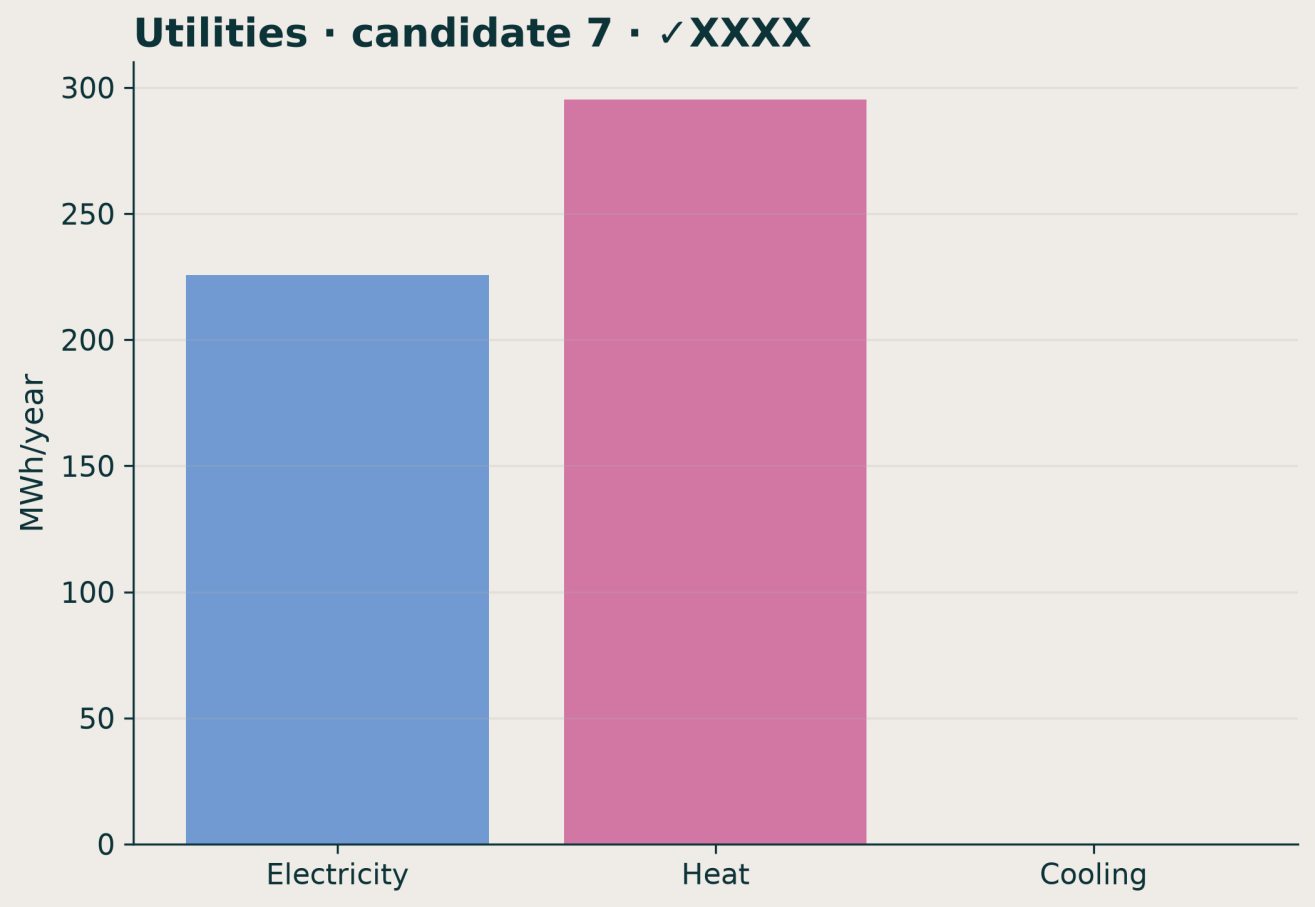


Figure 34. Annual electricity, heating and cooling duty

protein_enabled; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

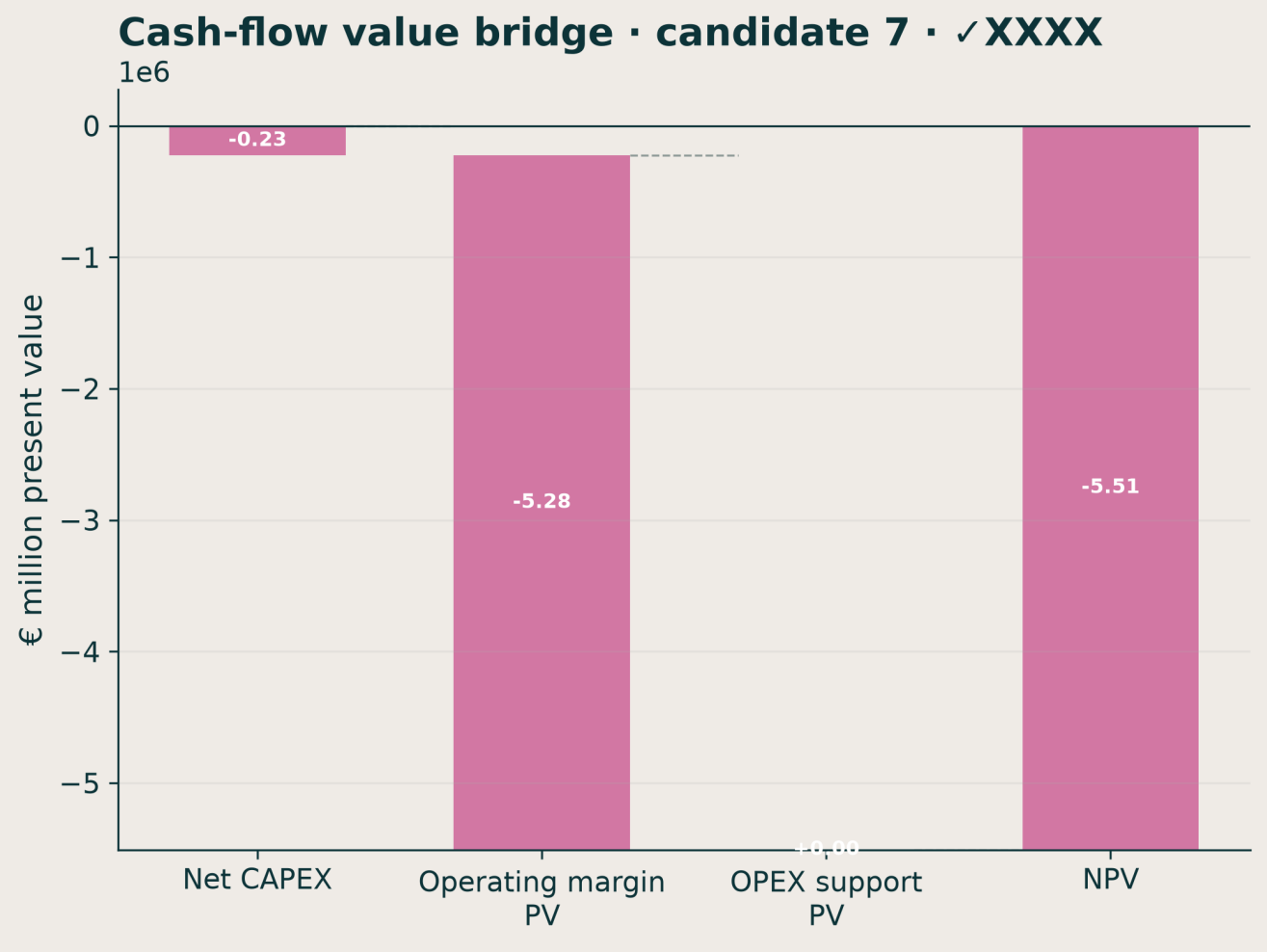


Figure 35. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 7; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

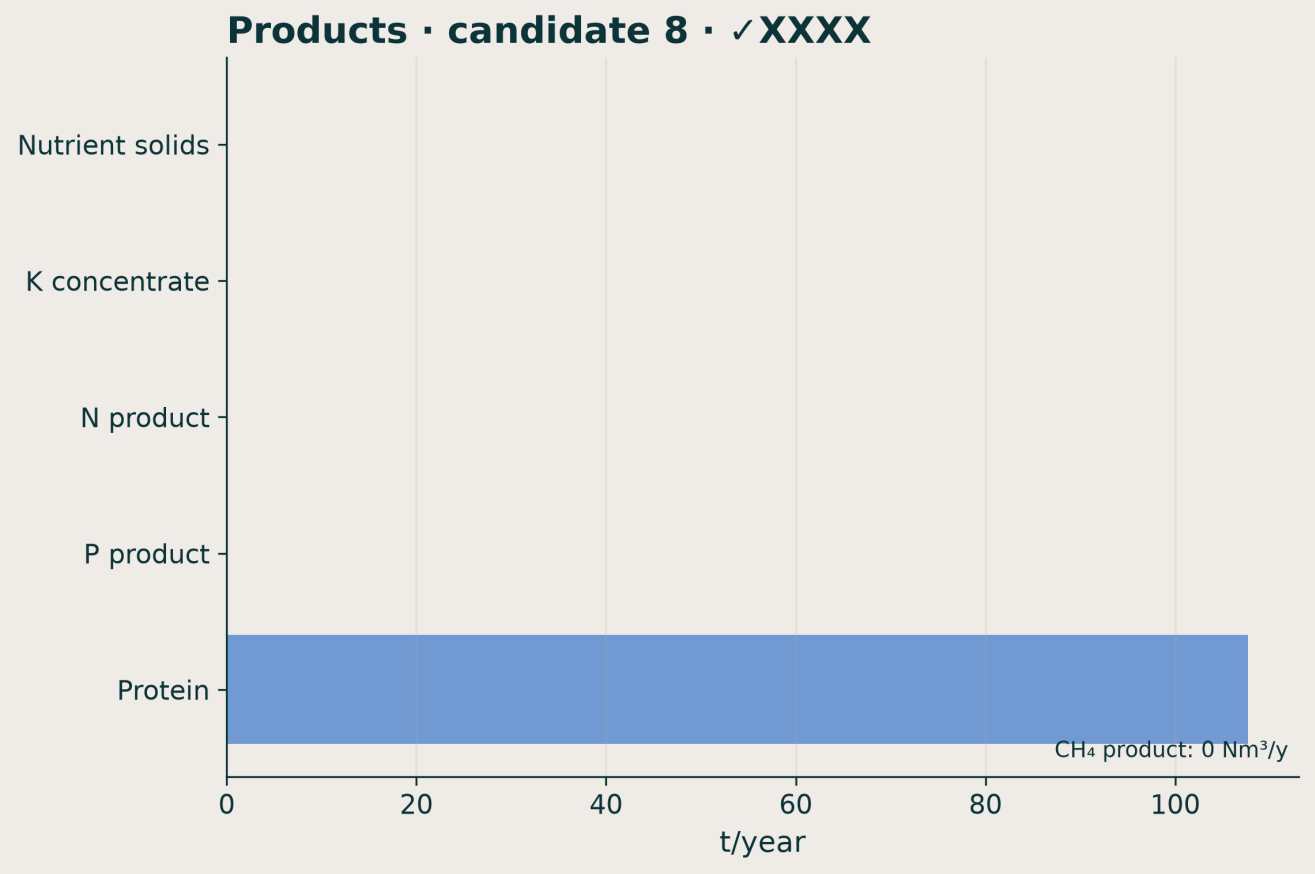


Figure 36. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

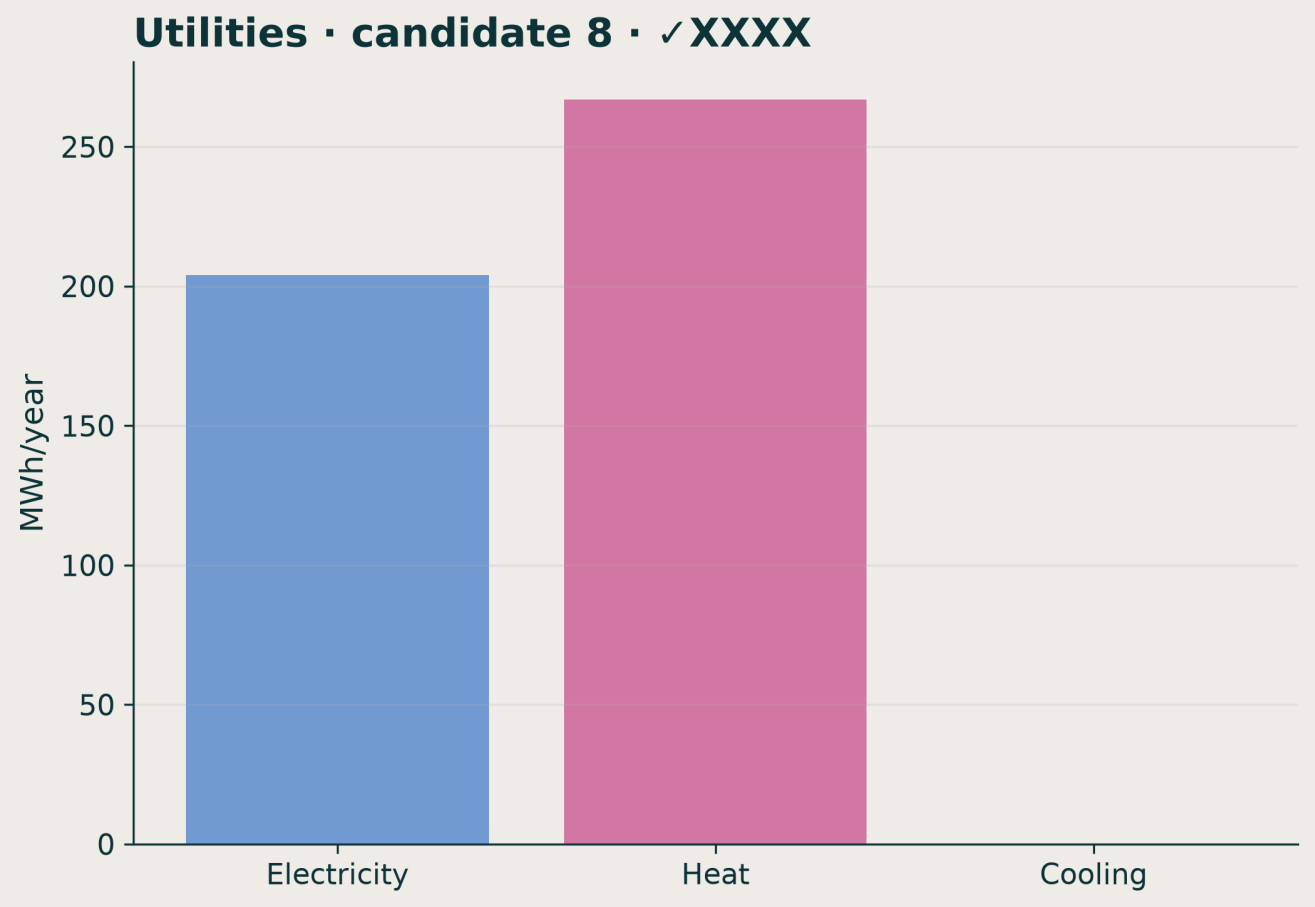


Figure 37. Annual electricity, heating and cooling duty

protein_enabled; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

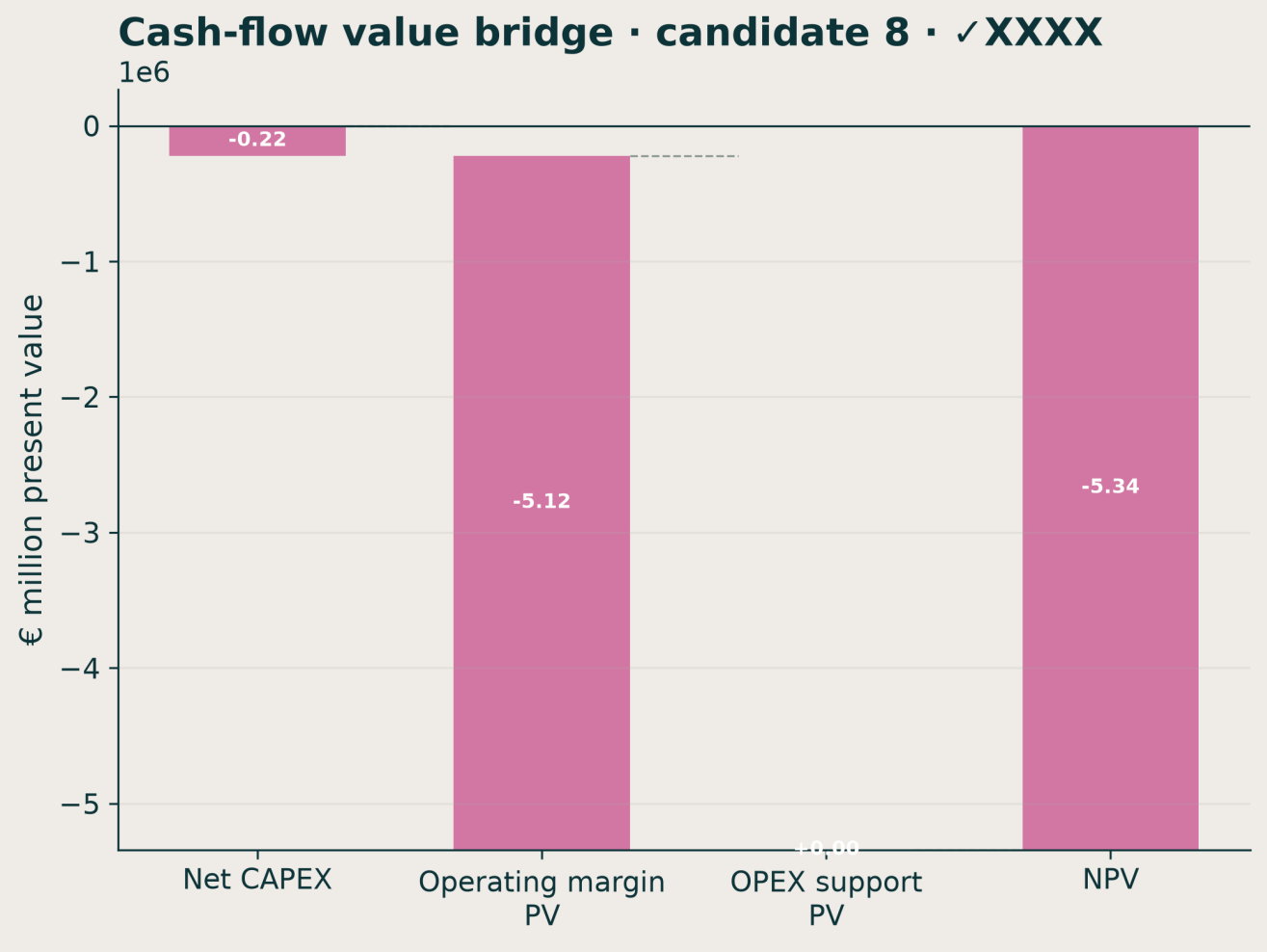


Figure 38. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 8; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

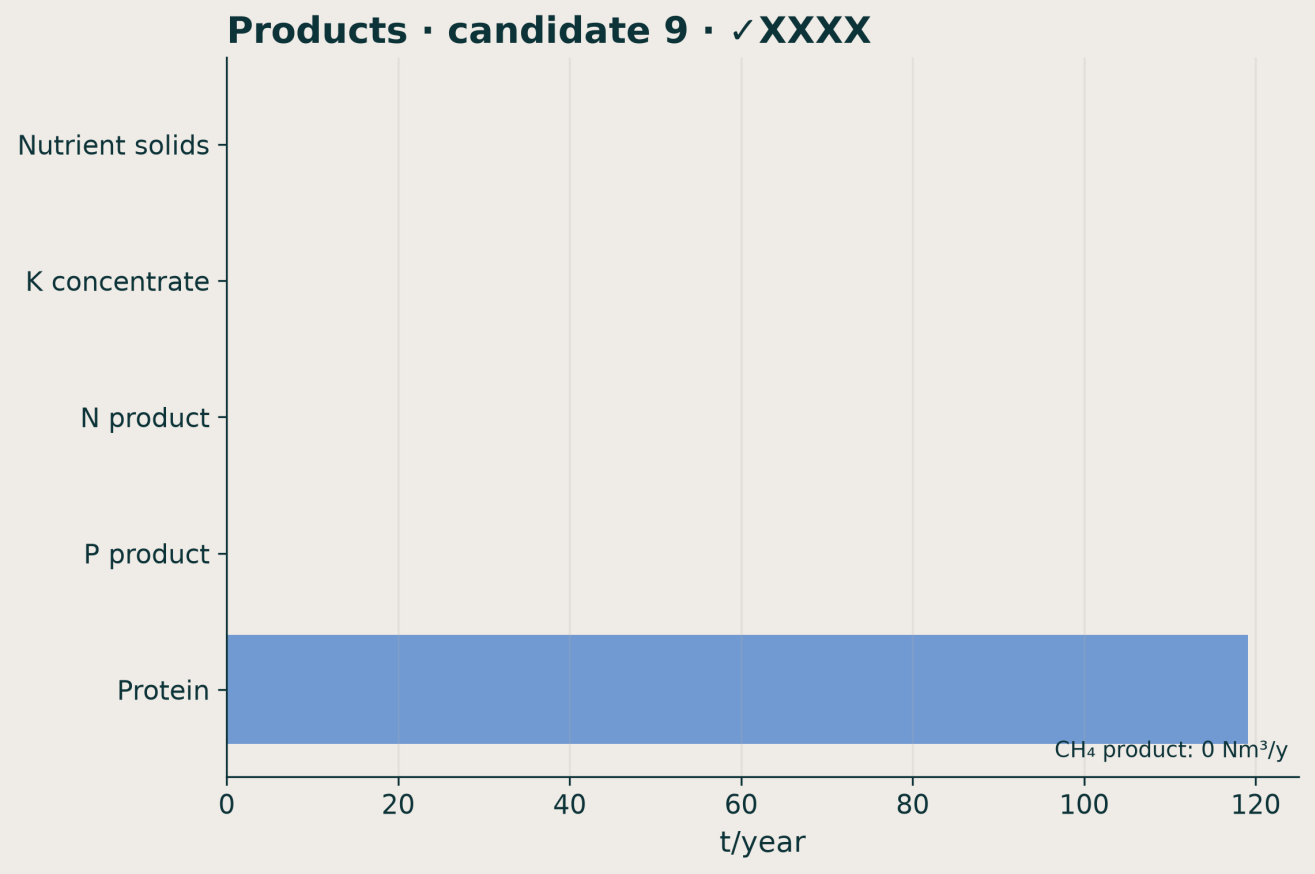


Figure 39. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

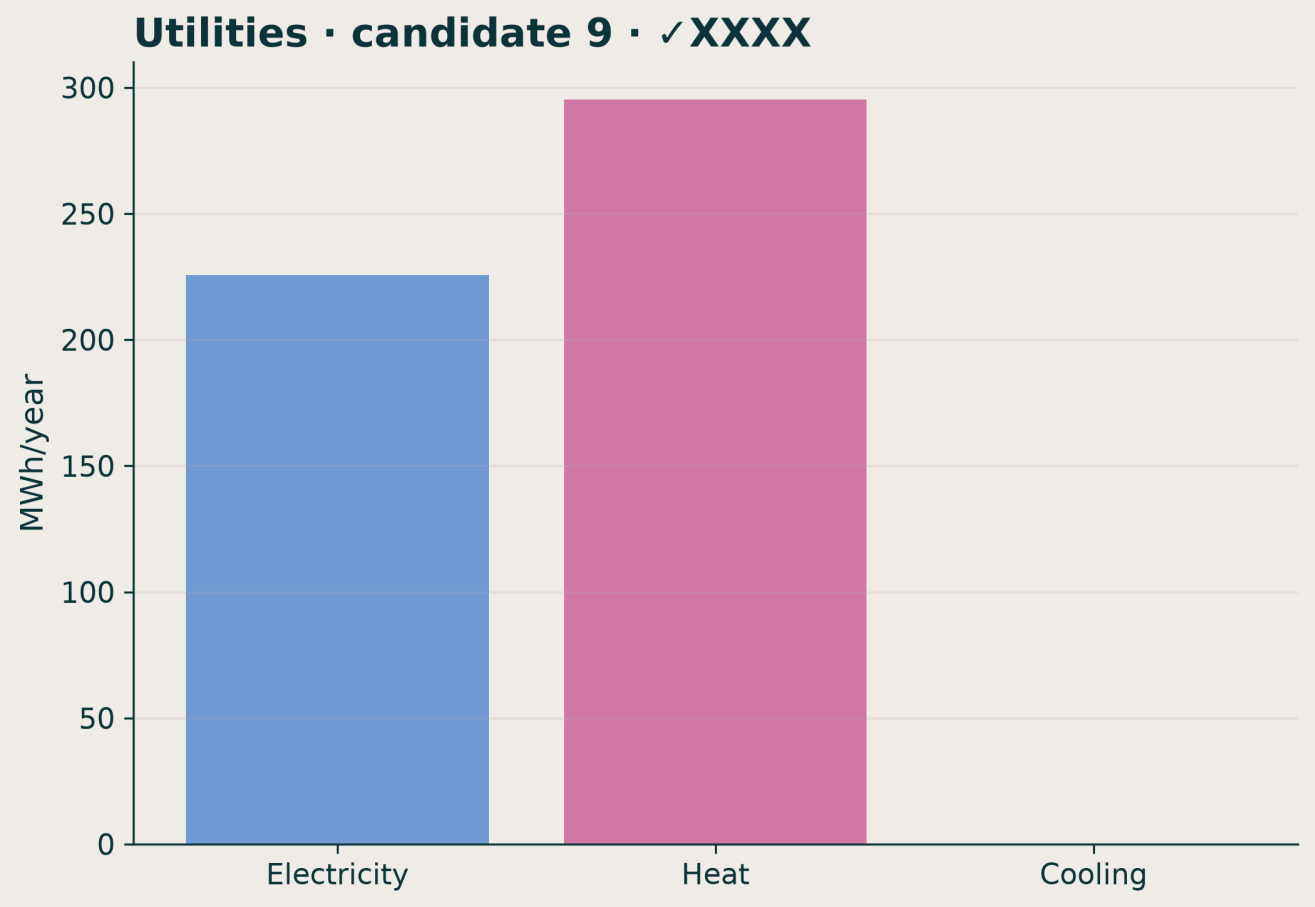


Figure 40. Annual electricity, heating and cooling duty

protein_enabled; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR1', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

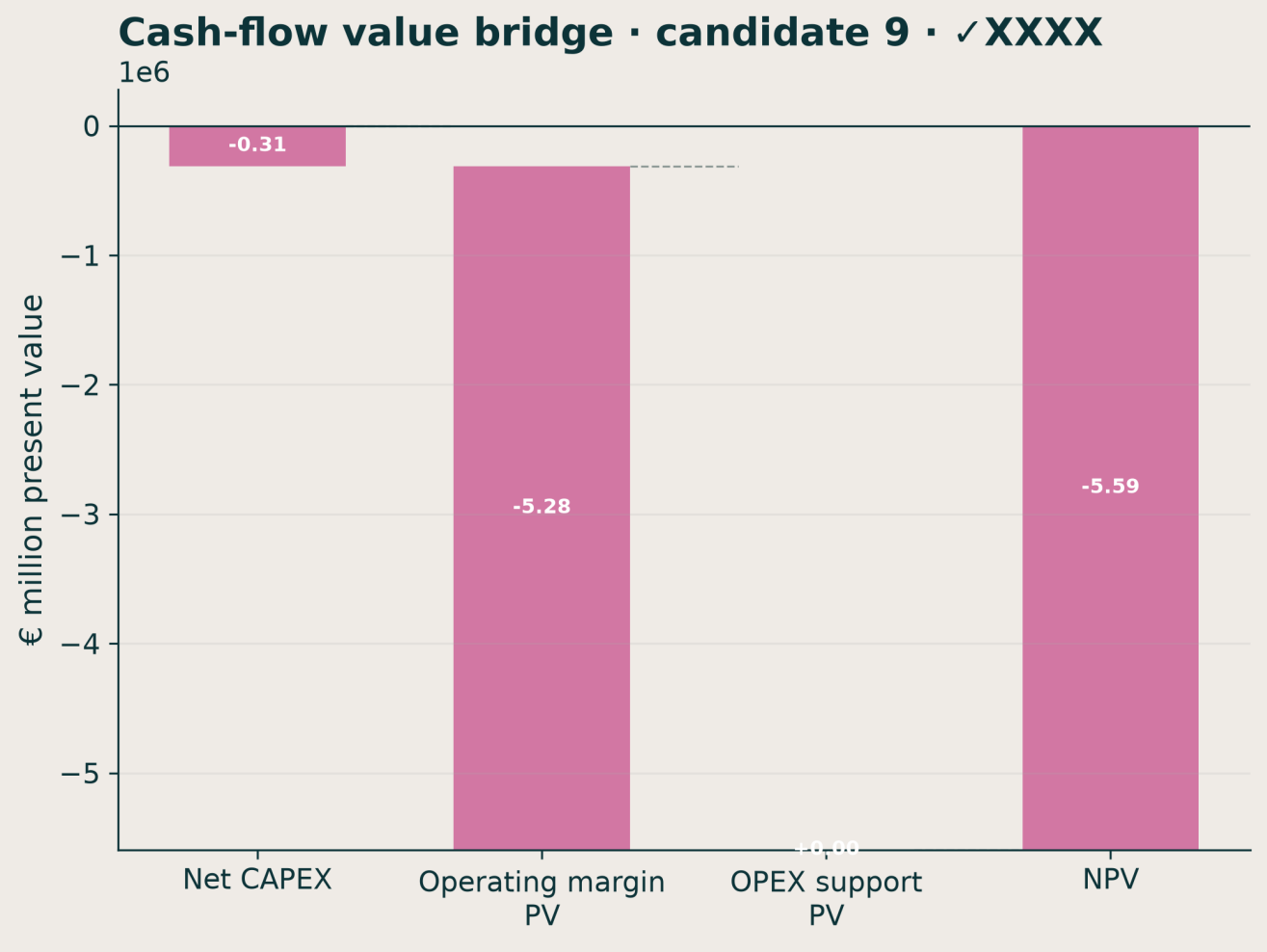


Figure 41. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 9; support C45 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate products

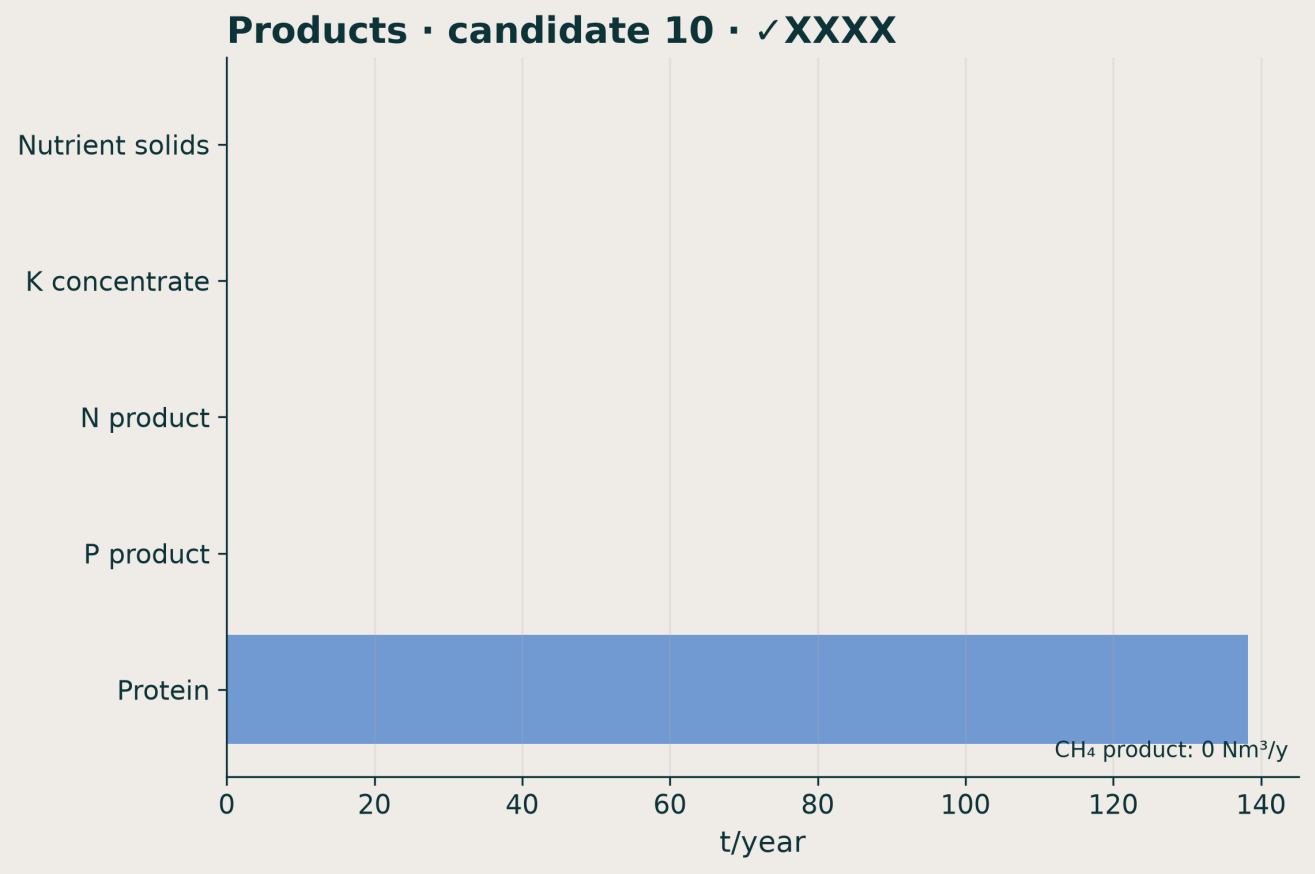


Figure 42. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_enabled; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate utilities

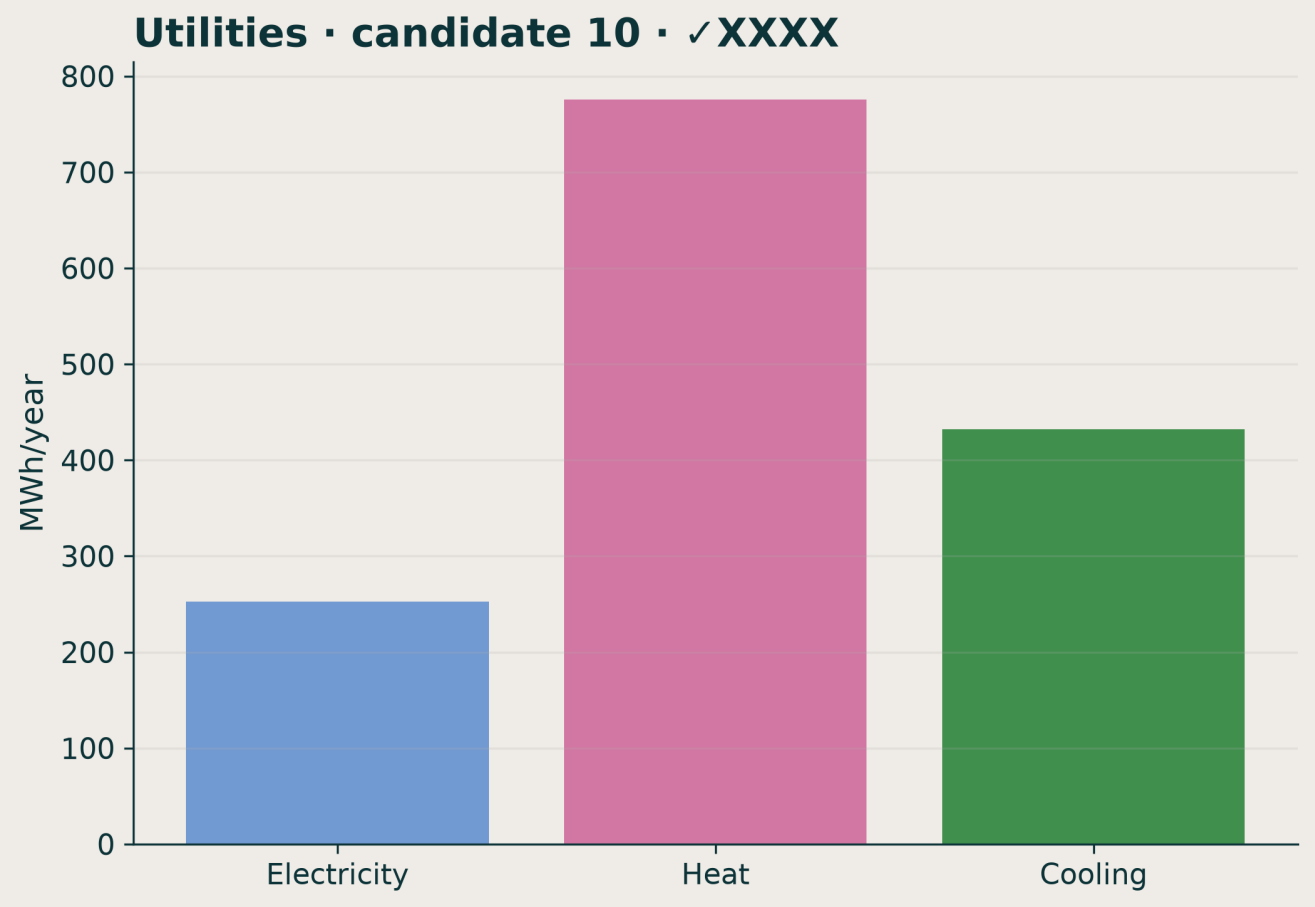


Figure 43. Annual electricity, heating and cooling duty

protein_enabled; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR3', 'anaerobic_digestion': 'AD0', 'biogas_upgrading': 'BG0', 'nutrient_recovery': 'NR0', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

candidate kpis

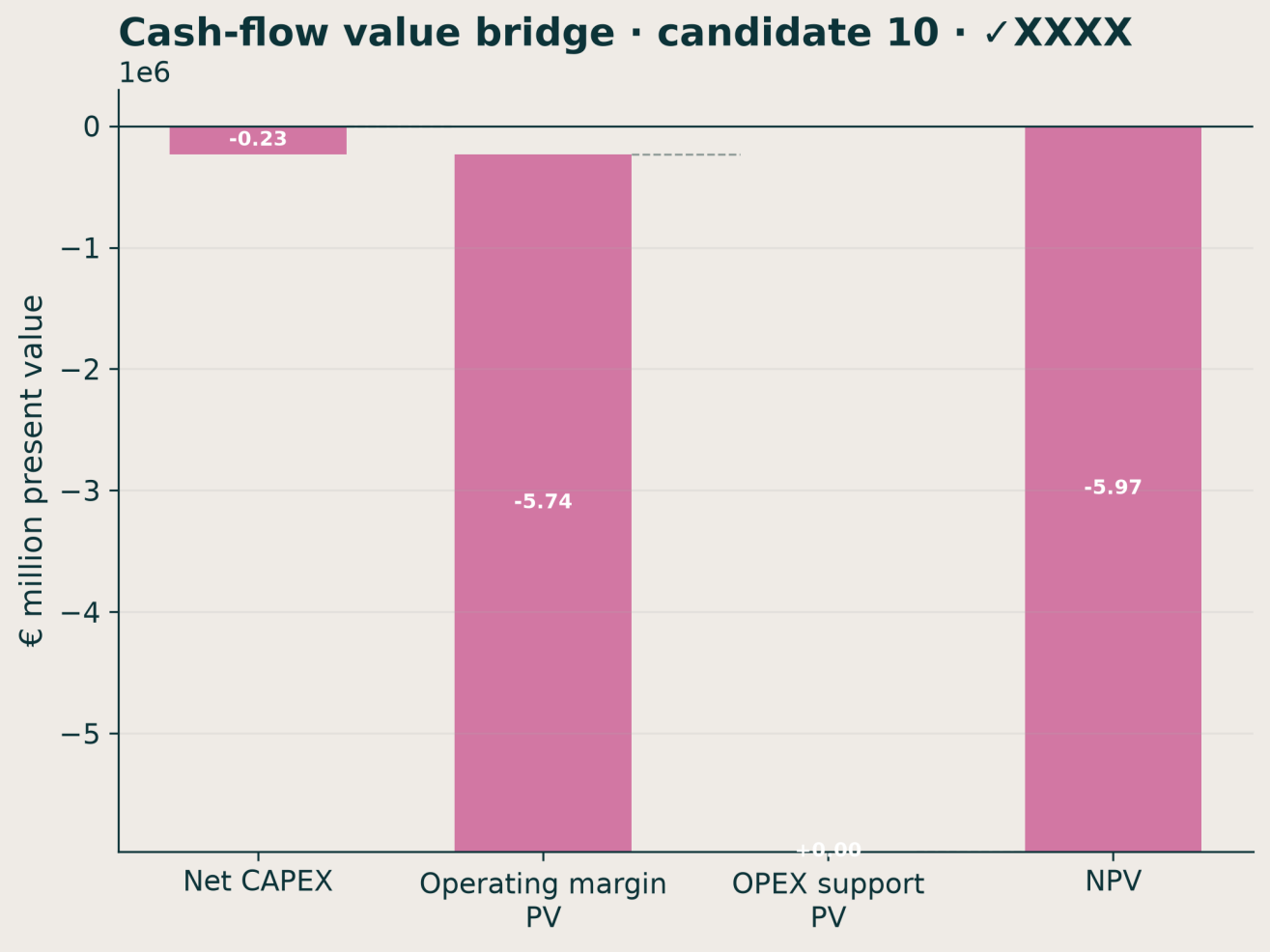


Figure 44. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_enabled; rank 10; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_enabled_batch.json

pareto net carbon vs npv protein bypass

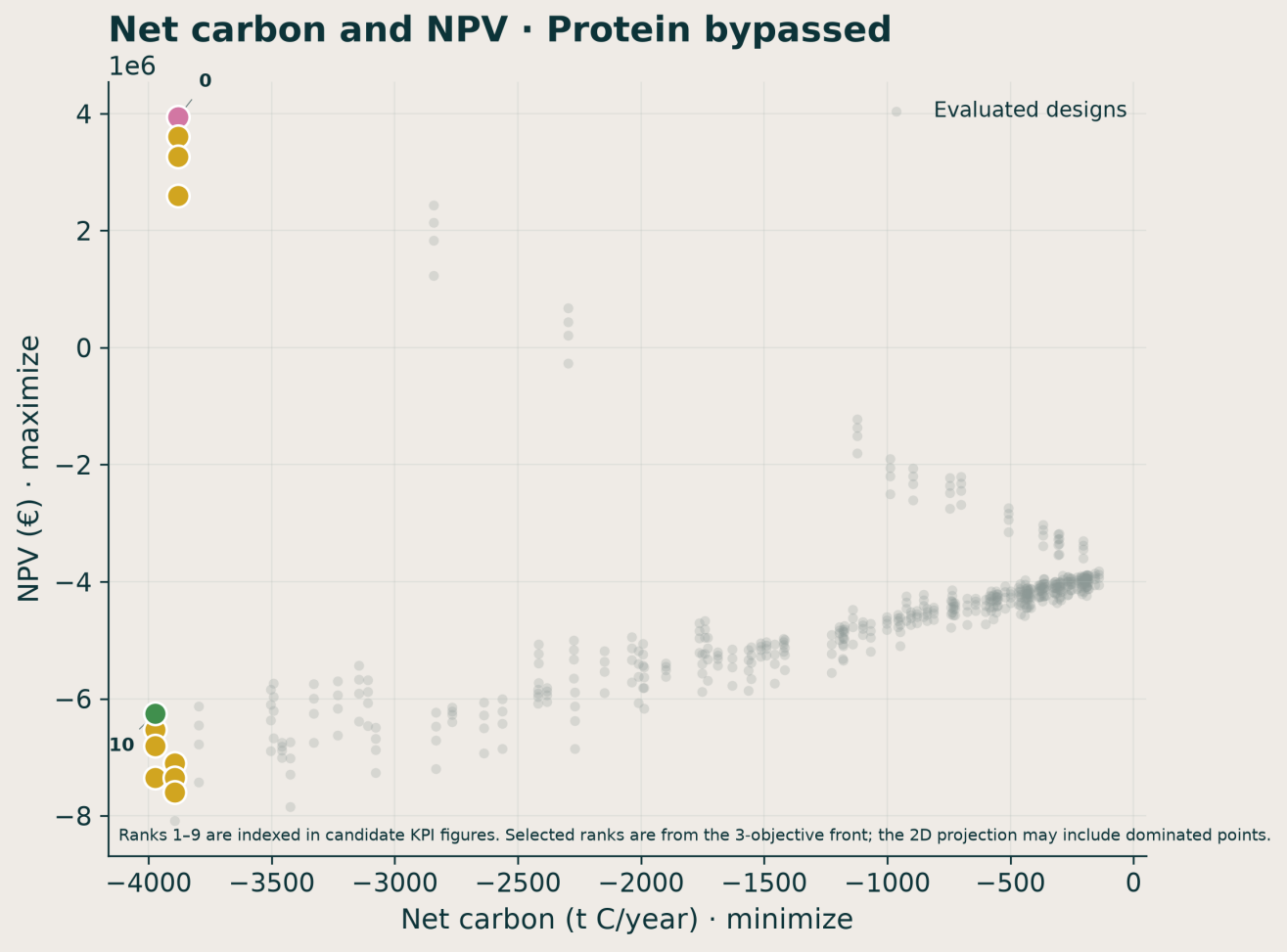


Figure 45. Net carbon and NPV · Protein bypassed

protein_bypass / all support cases | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

pareto subsidy granted vs npv protein bypass

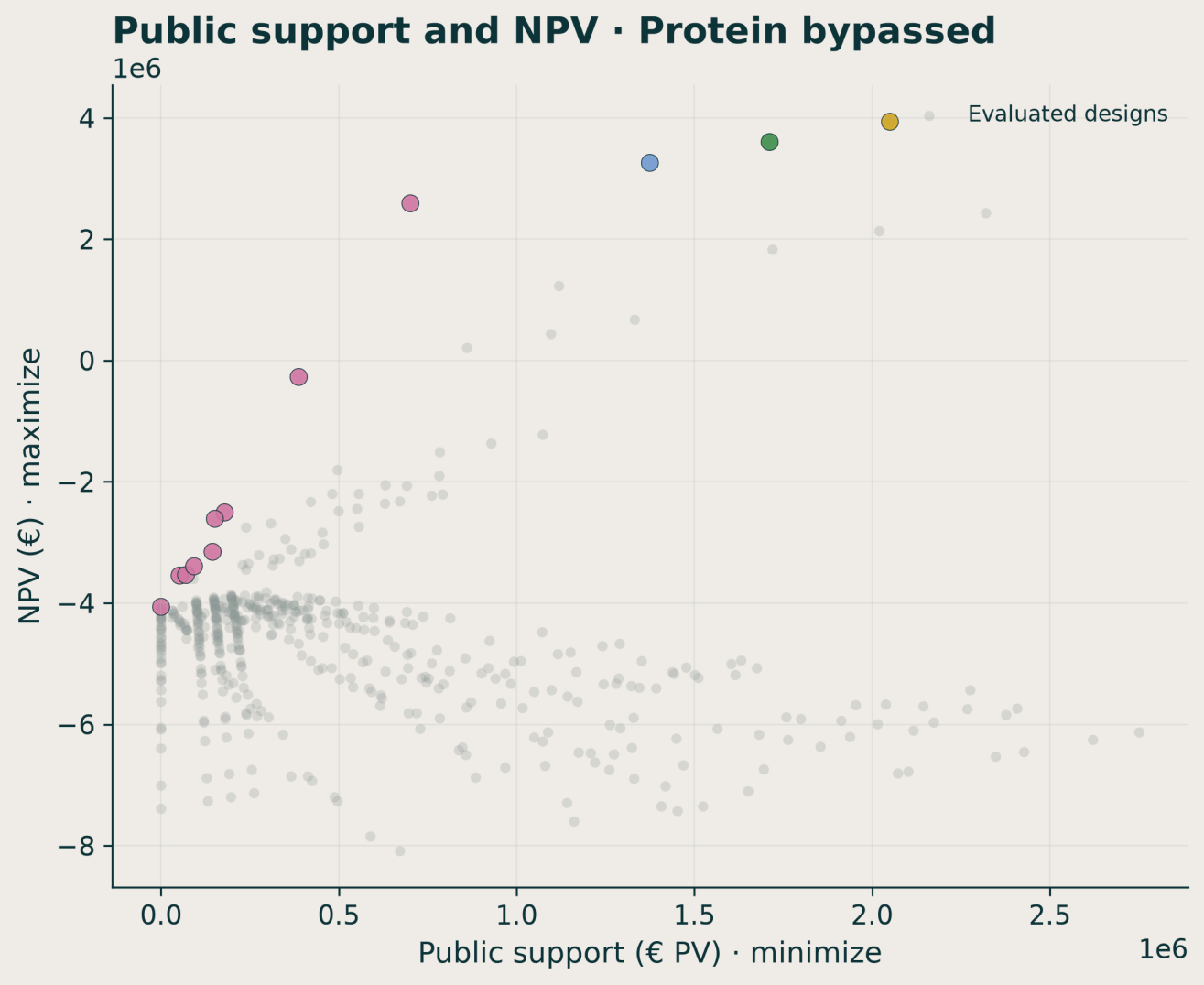


Figure 46. Public support and NPV · Protein bypassed

protein_bypass / all support cases | Public support: € PV; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

pareto net carbon vs npv protein bypass case C30

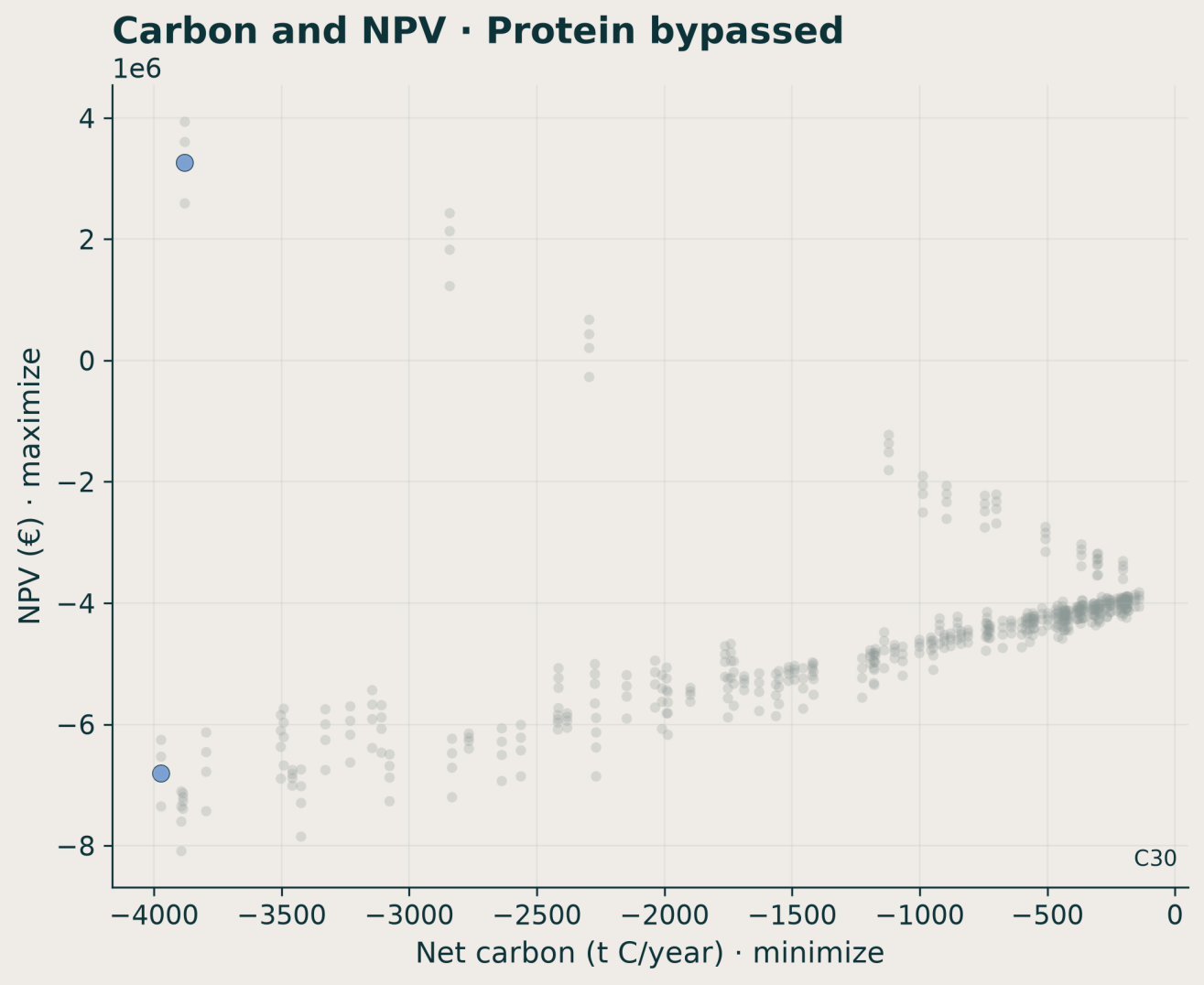


Figure 47. Carbon and NPV · Protein bypassed

protein_bypass / case C30 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

pareto net carbon vs npv protein bypass case C45

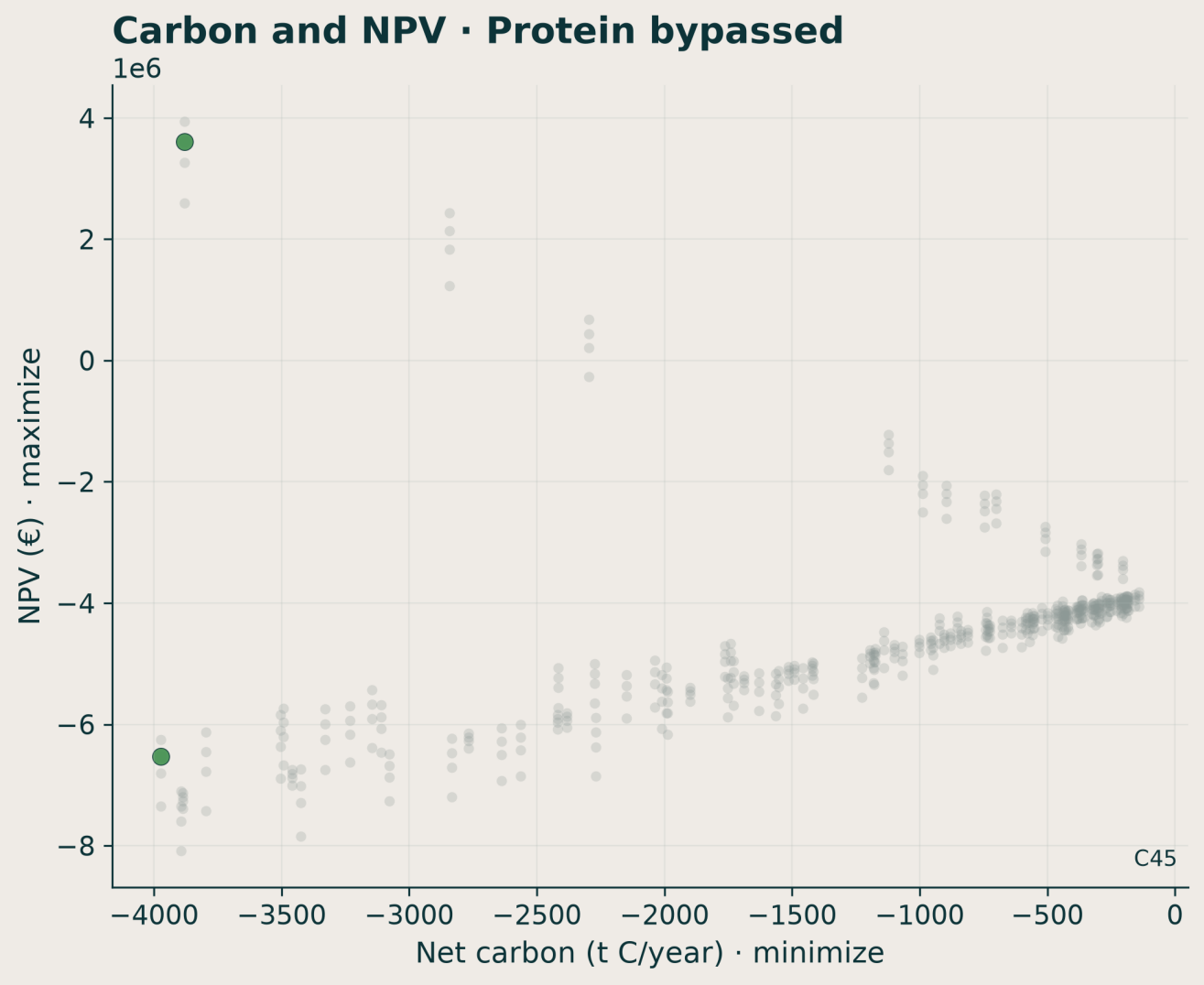


Figure 48. Carbon and NPV · Protein bypassed

protein_bypass / case C45 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

pareto net carbon vs npv protein bypass case C60

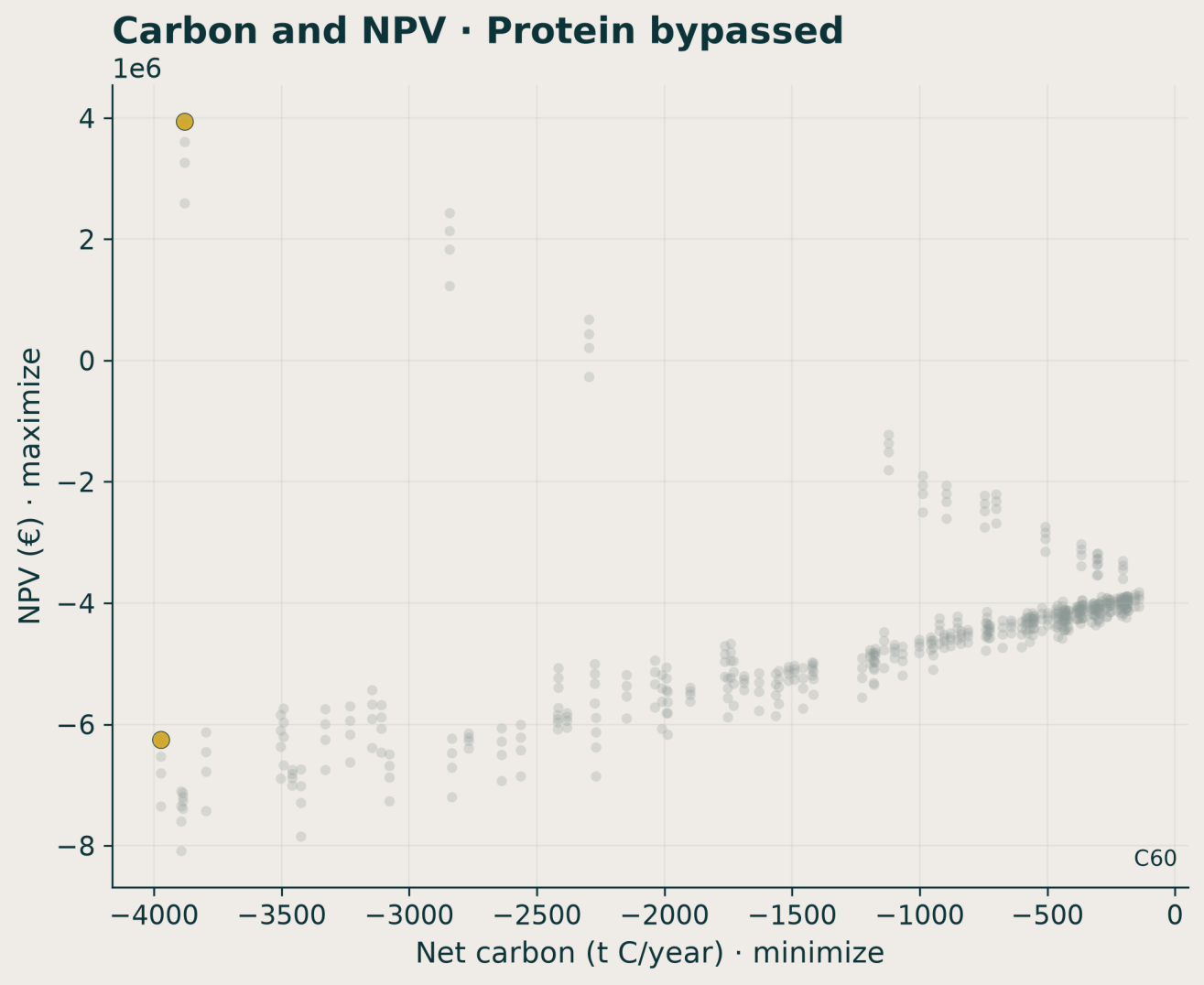


Figure 49. Carbon and NPV · Protein bypassed

protein_bypass / case C60 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

pareto net carbon vs npv protein bypass case C0

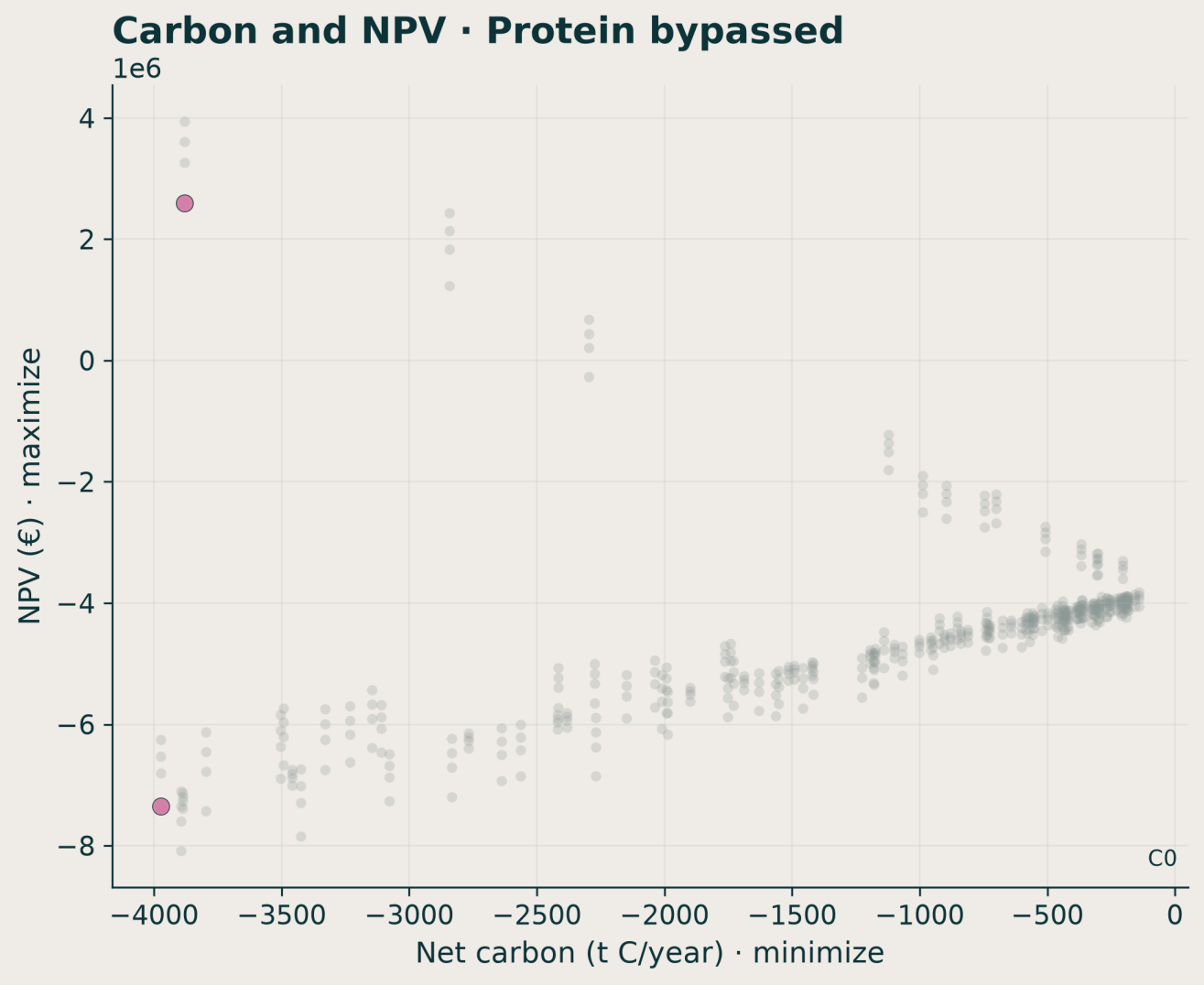


Figure 50. Carbon and NPV · Protein bypassed

protein_bypass / case C0 | Net carbon: t C/year; NPV: € | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

pareto 3d carbon npv support protein bypass

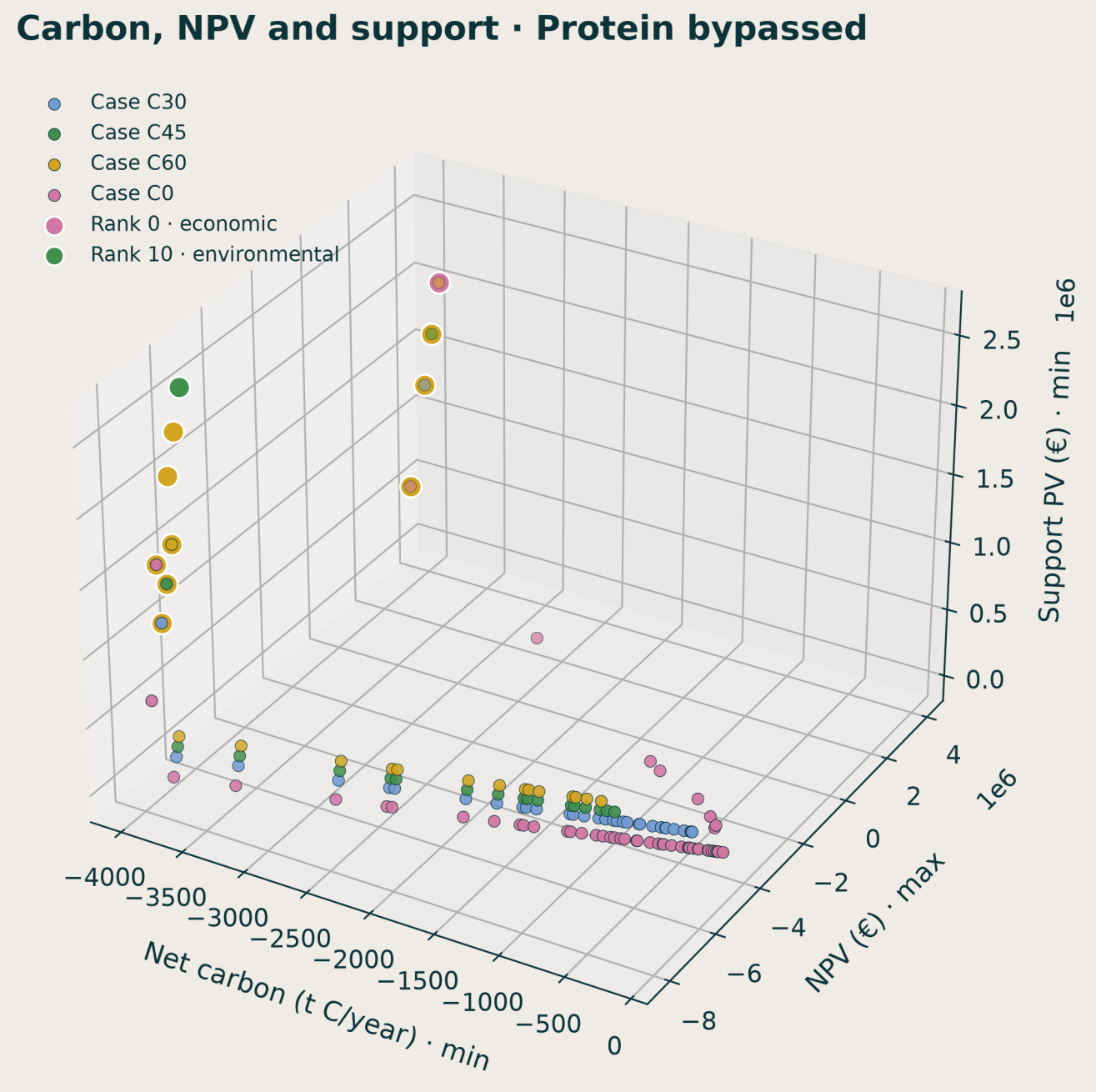


Figure 51. Three-objective Pareto front

Protein bypassed | t C/year; €; € present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpi heatmap protein bypass

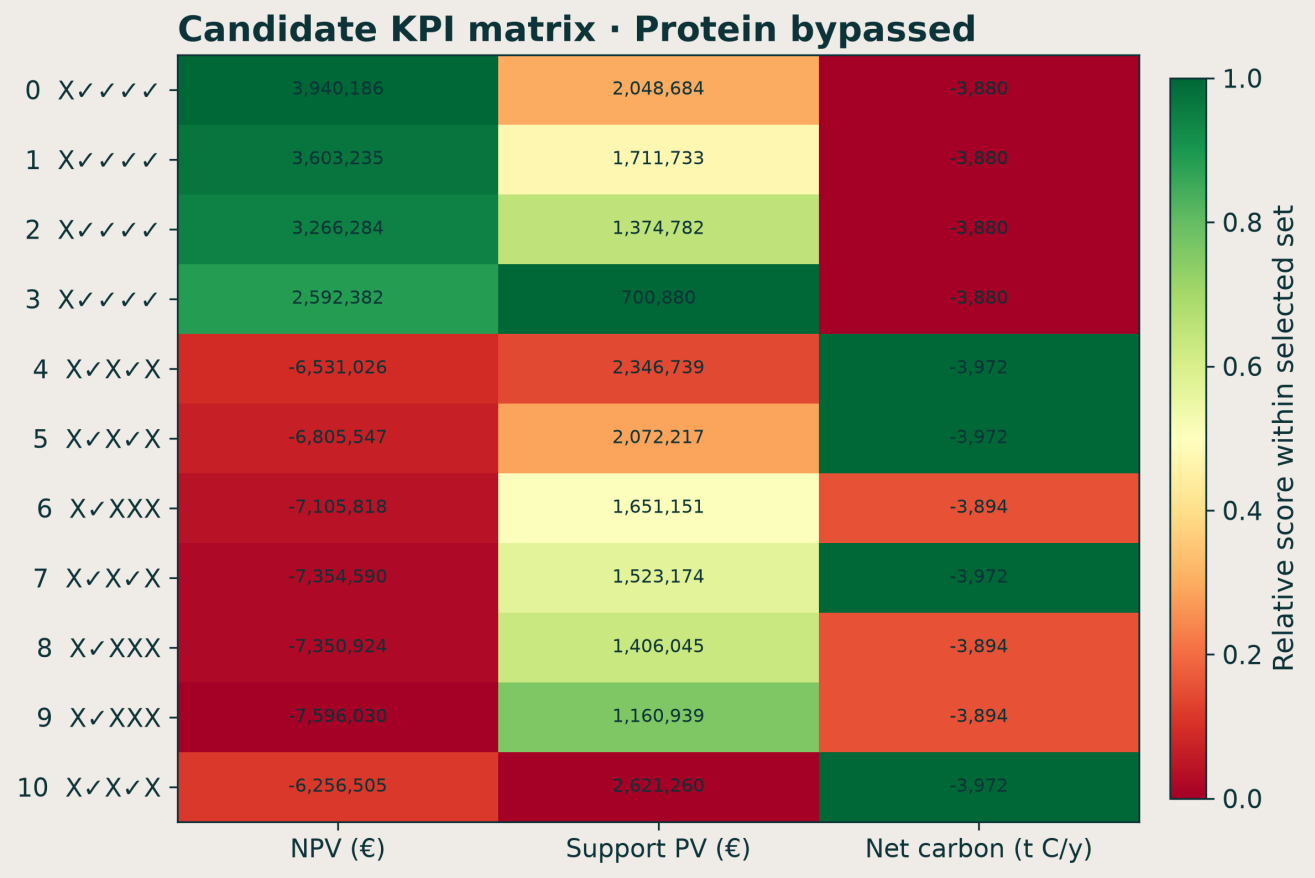


Figure 52. Selected candidate KPI comparison; color normalized by objective direction

Protein bypassed | €, € present value, t C/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate economics protein bypass

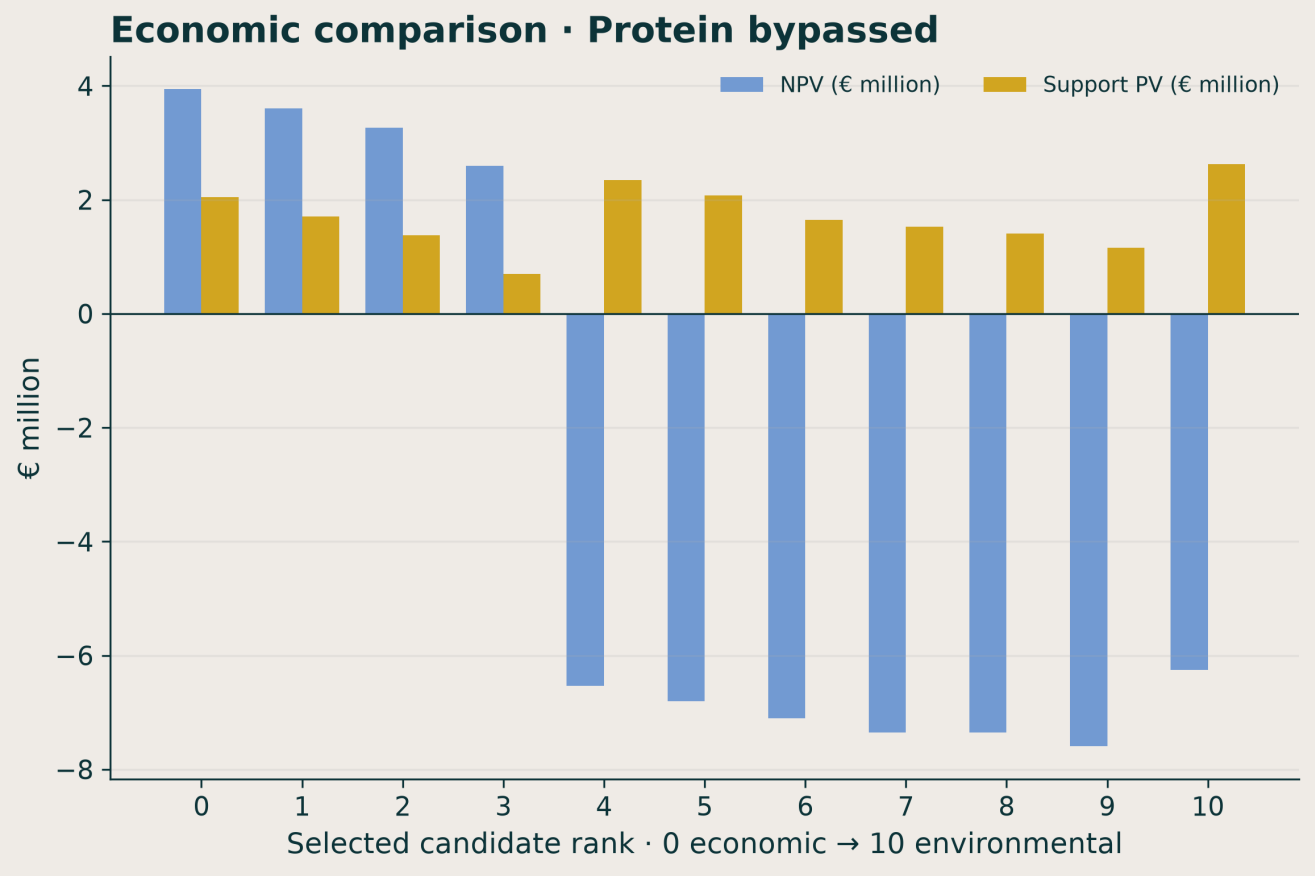


Figure 53. NPV and assumed public support PV for selected Pareto candidates

Protein bypassed | € million | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate environment protein bypass

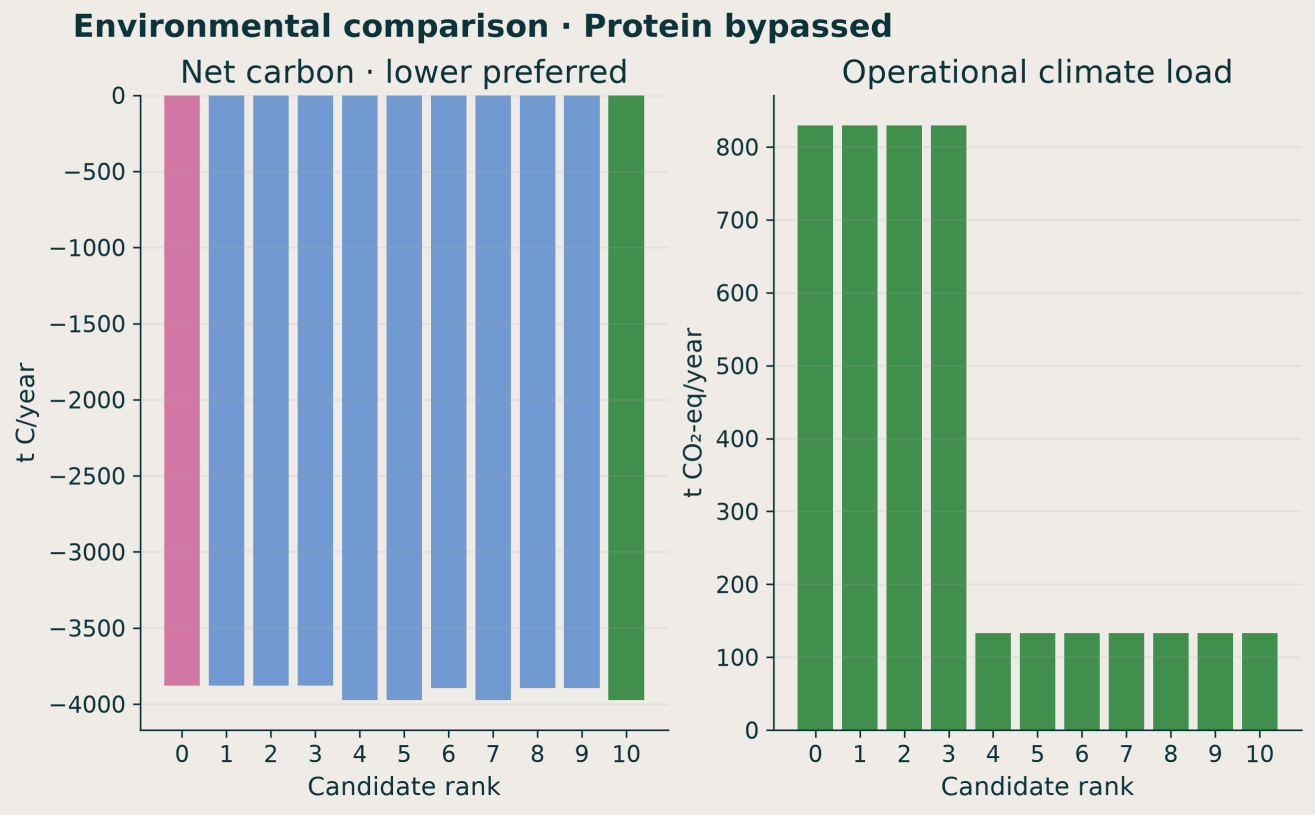


Figure 54. Net stored carbon objective and attributional operational climate load

Protein bypassed | t C/year; t CO■-eq/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

carbon load vs stored carbon protein bypass

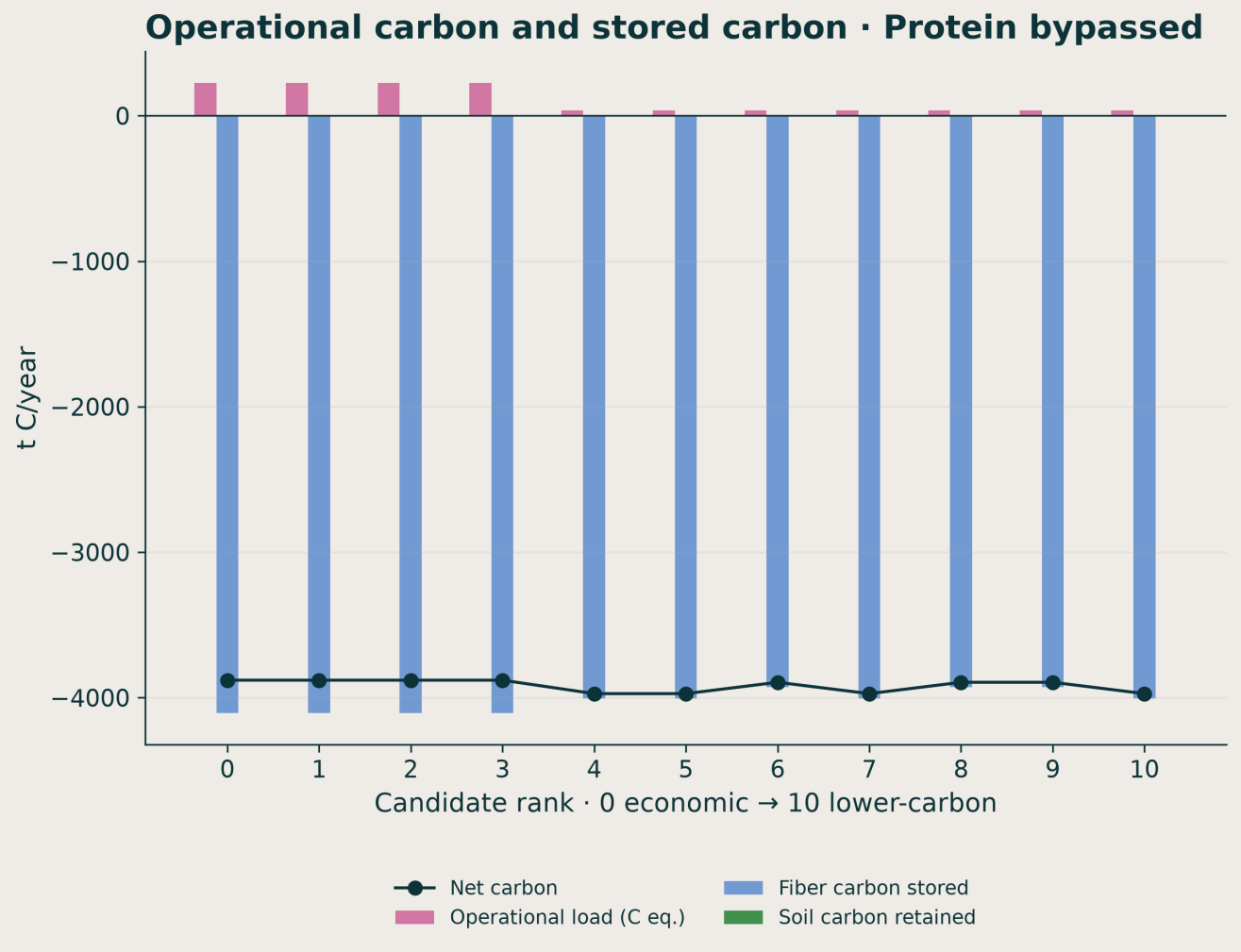


Figure 55. Attributional operational carbon-equivalent load minus fiber and soil retained product carbon

Protein bypassed | t C/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

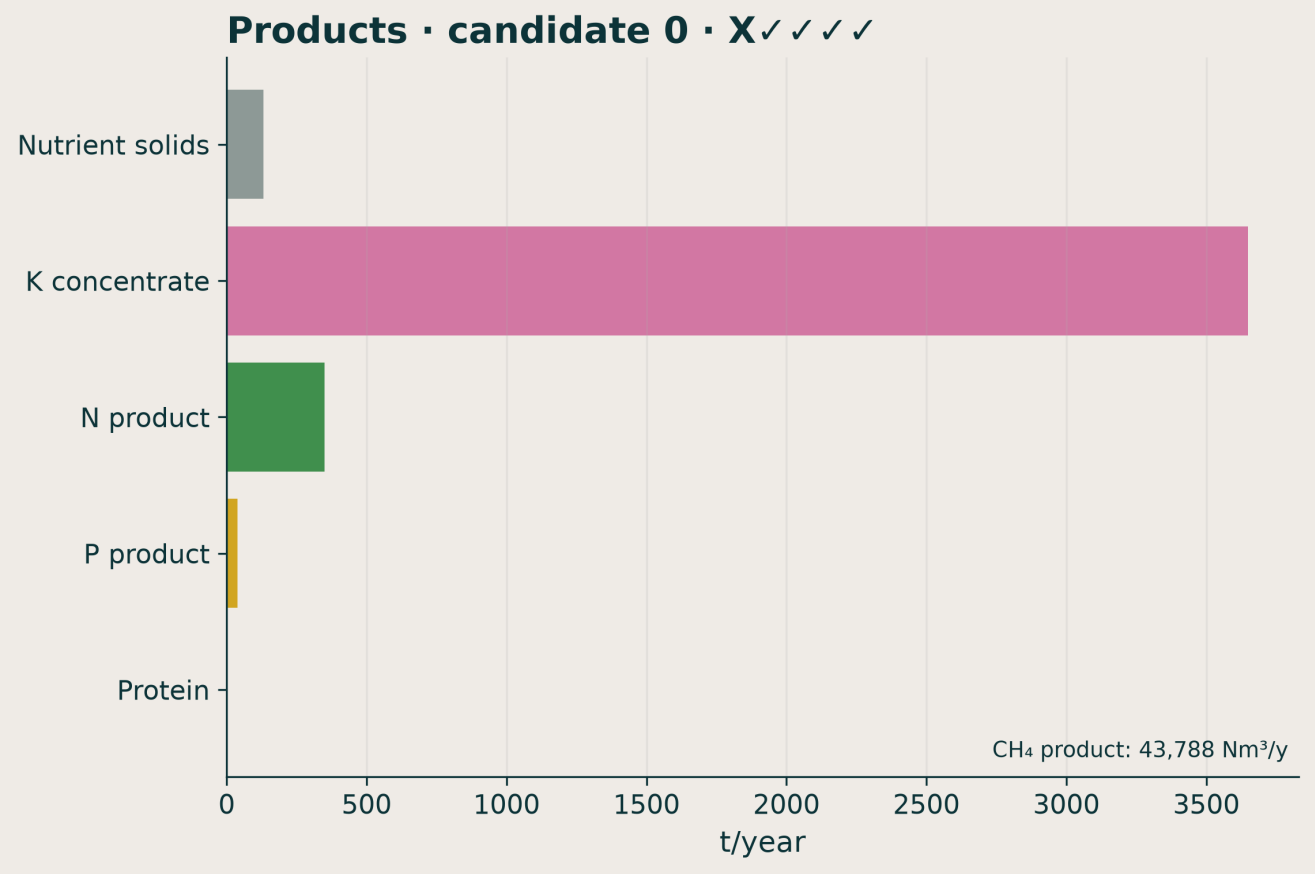


Figure 56. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

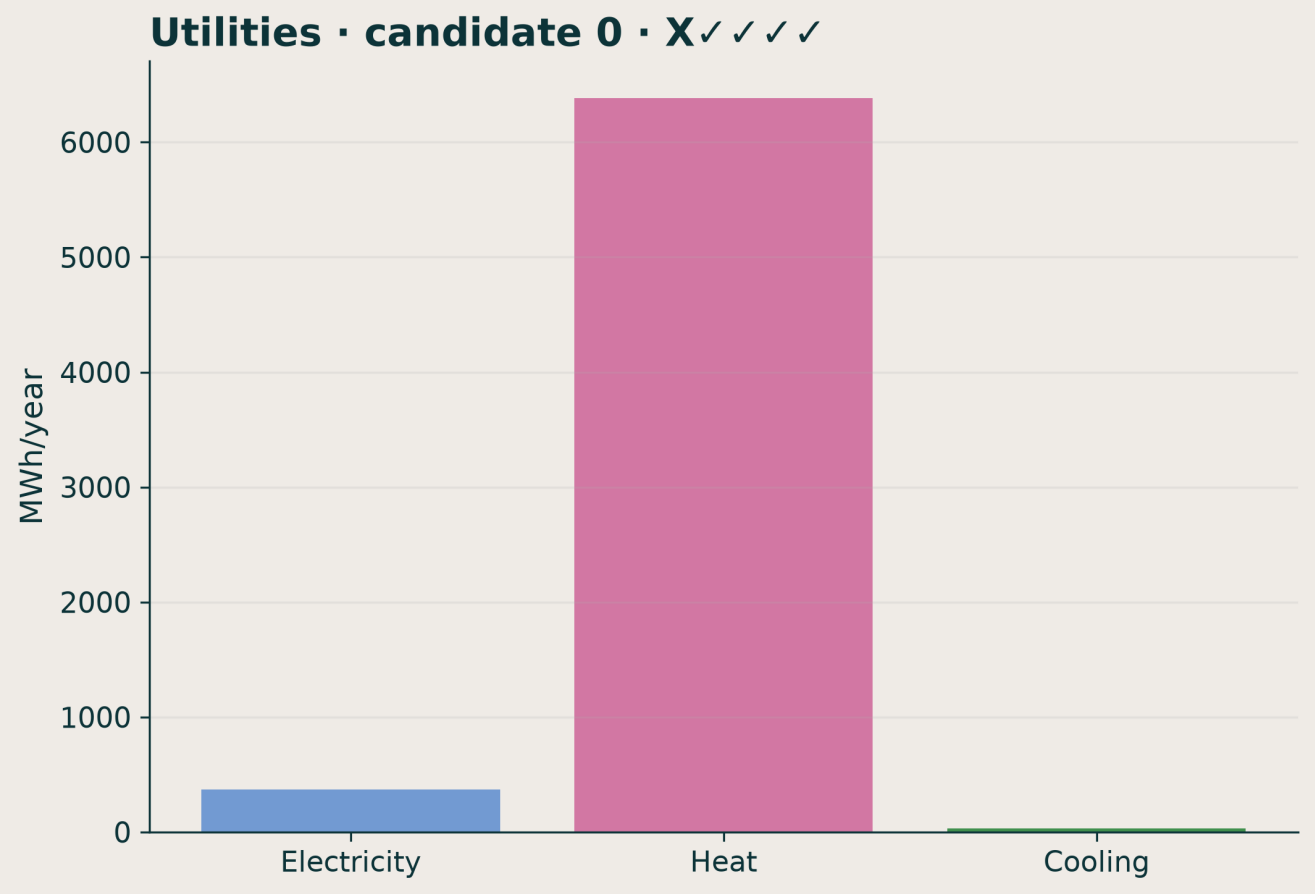


Figure 57. Annual electricity, heating and cooling duty

protein_bypass; rank 0; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

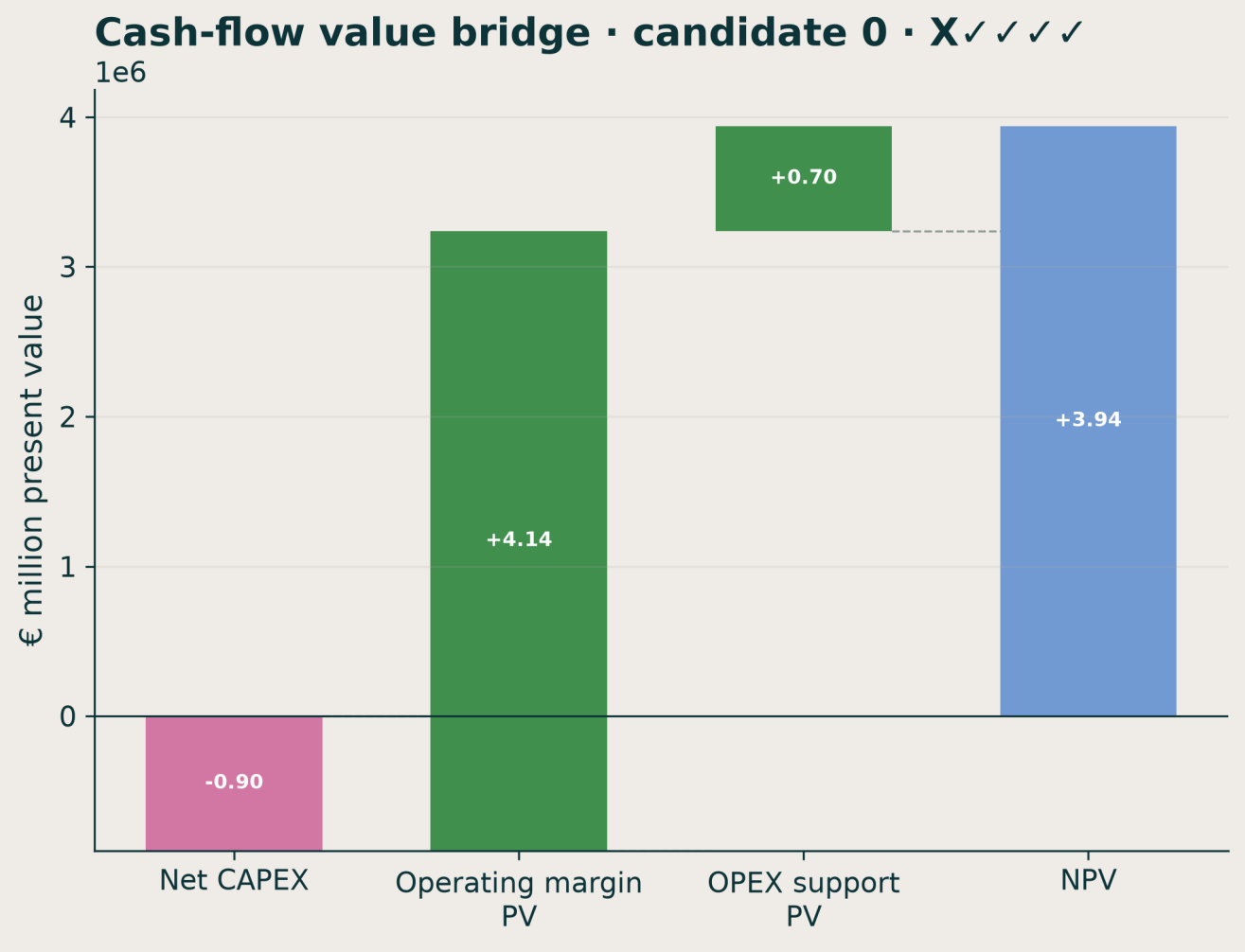


Figure 58. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 0; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

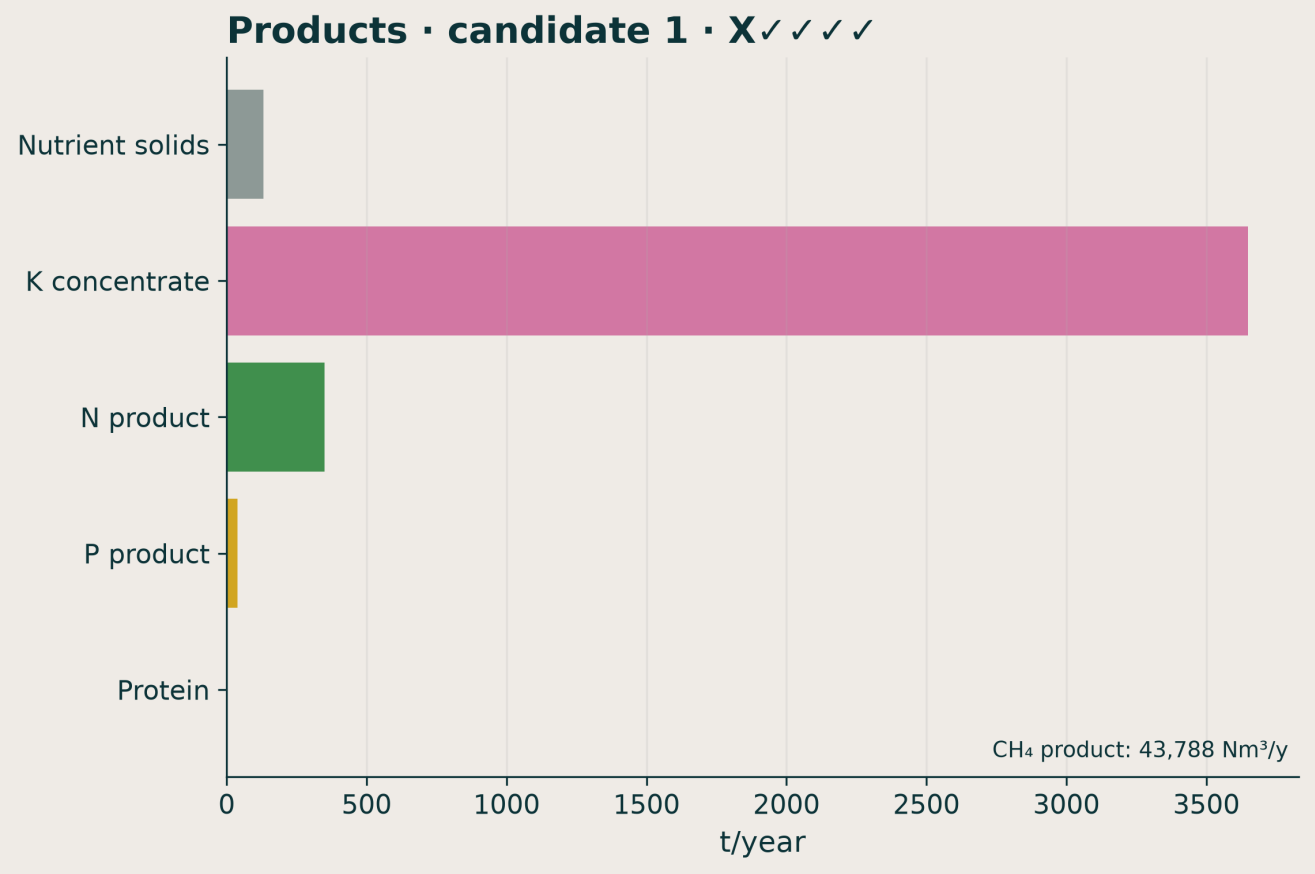


Figure 59. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

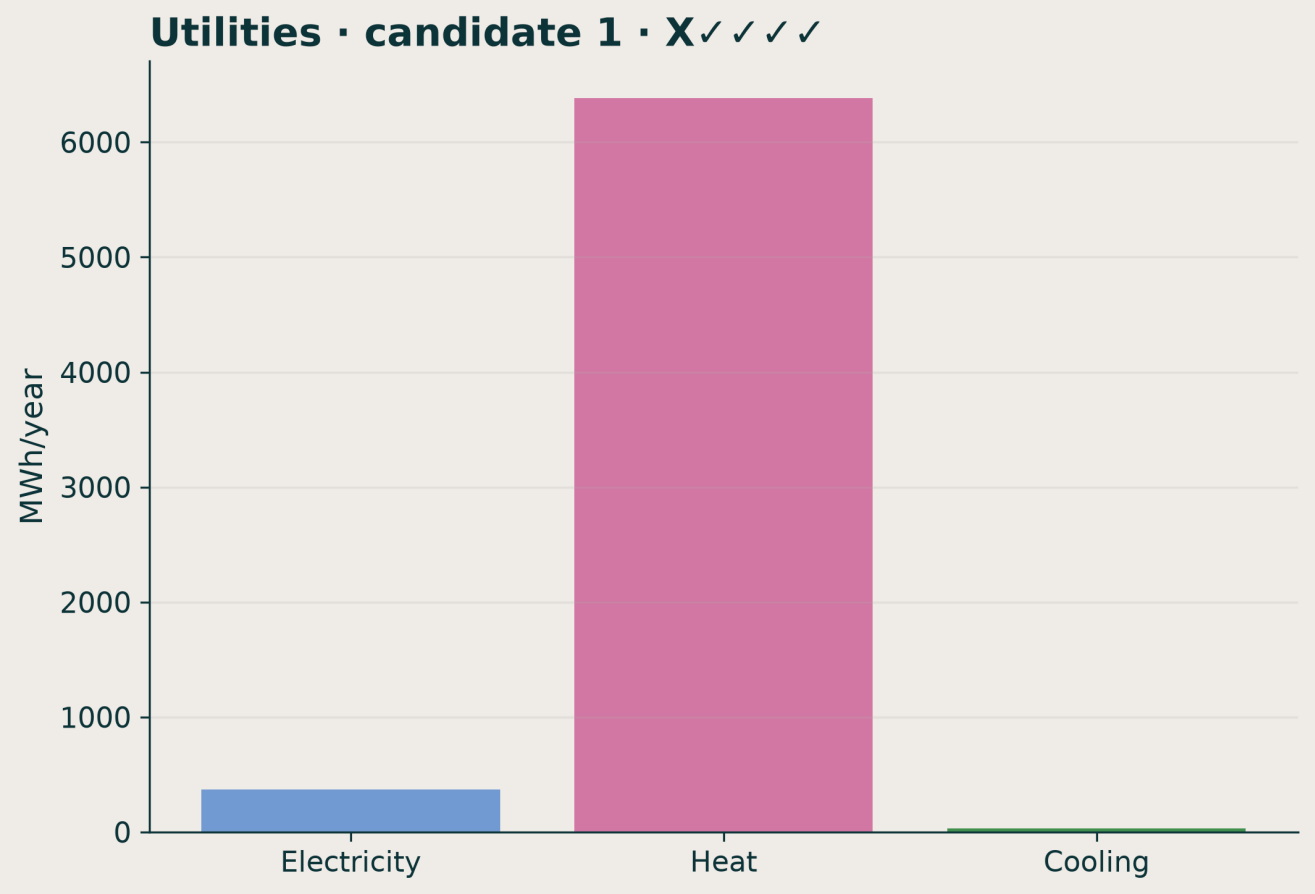


Figure 60. Annual electricity, heating and cooling duty

protein_bypass; rank 1; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

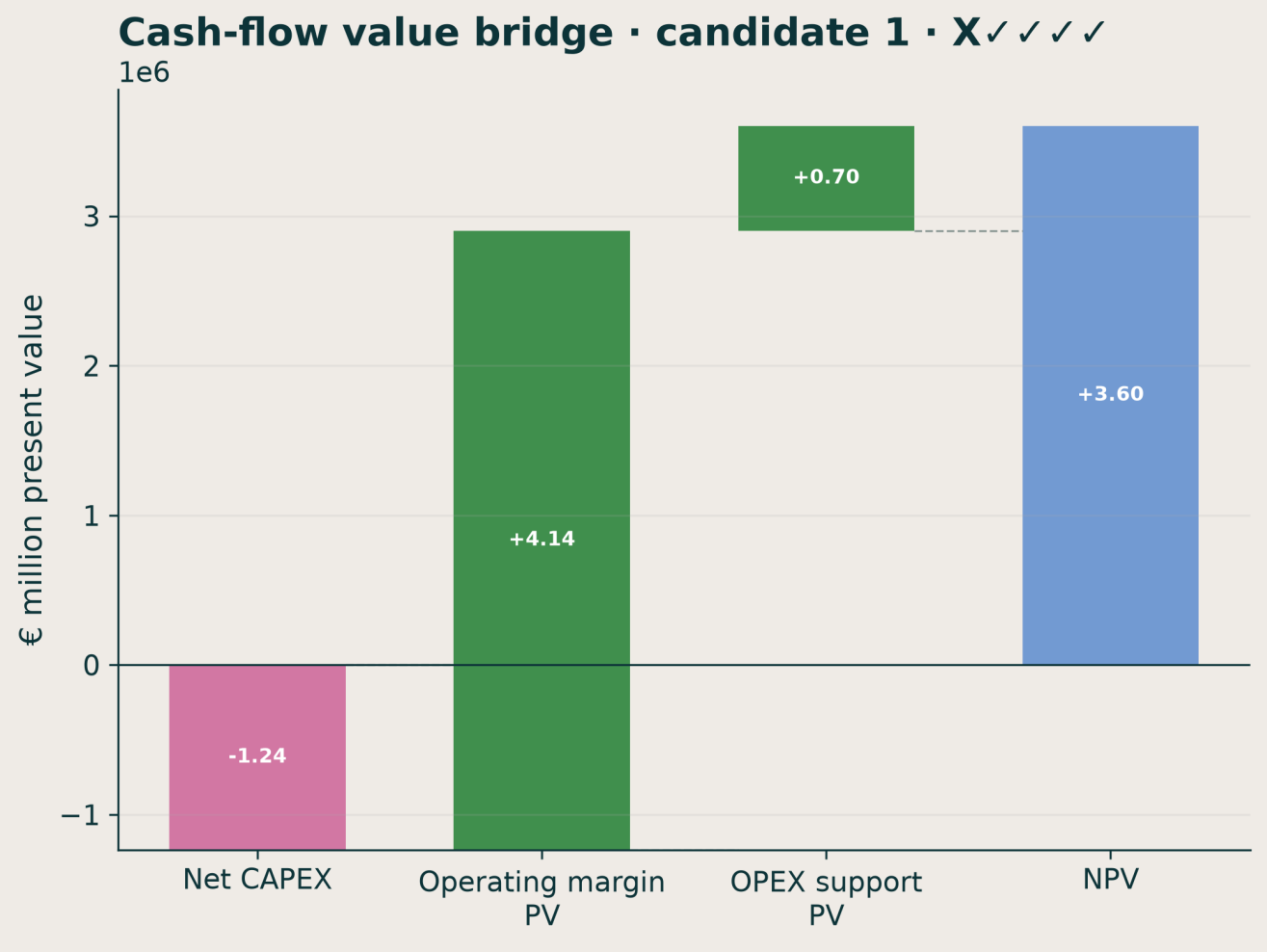


Figure 61. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 1; support C45 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

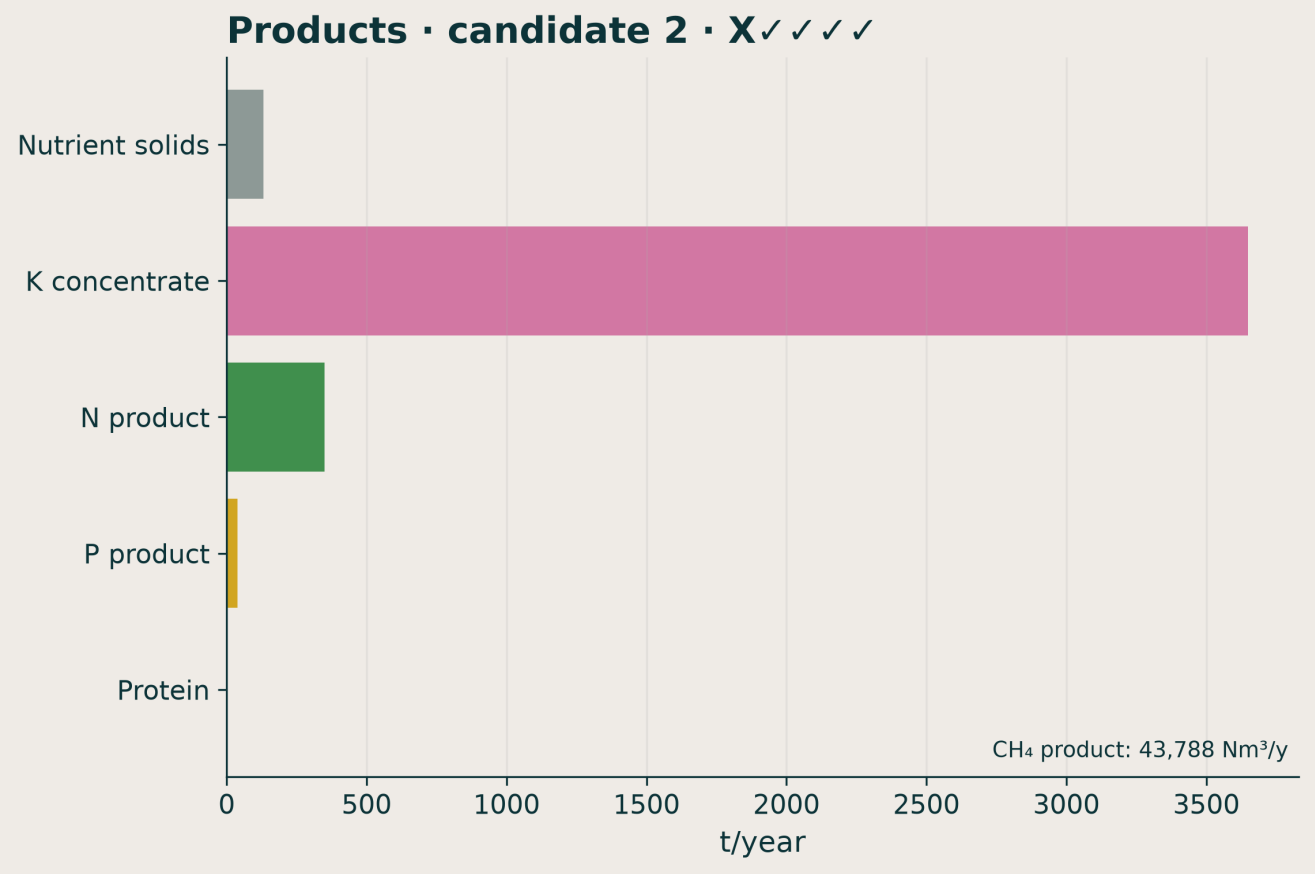


Figure 62. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

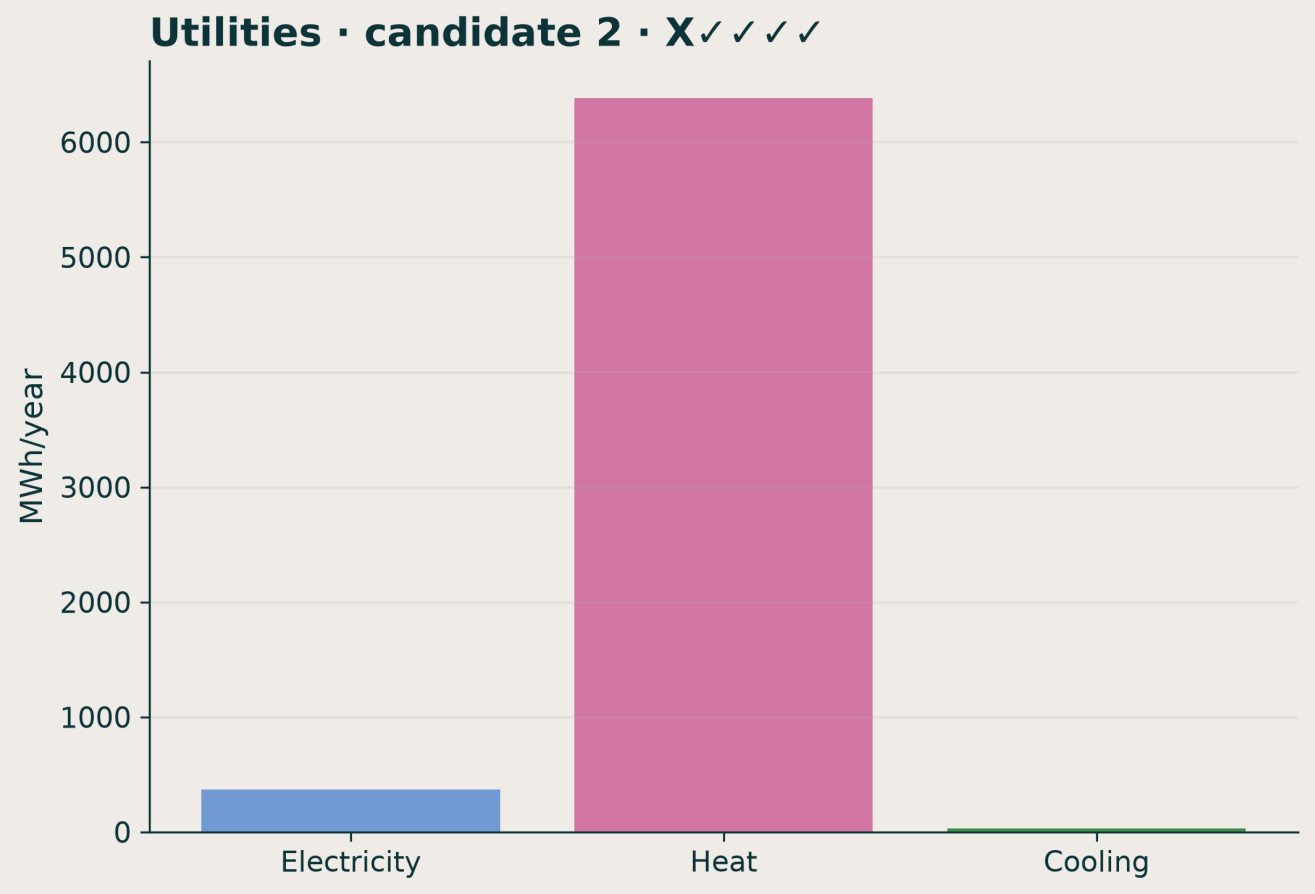


Figure 63. Annual electricity, heating and cooling duty

protein_bypass; rank 2; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

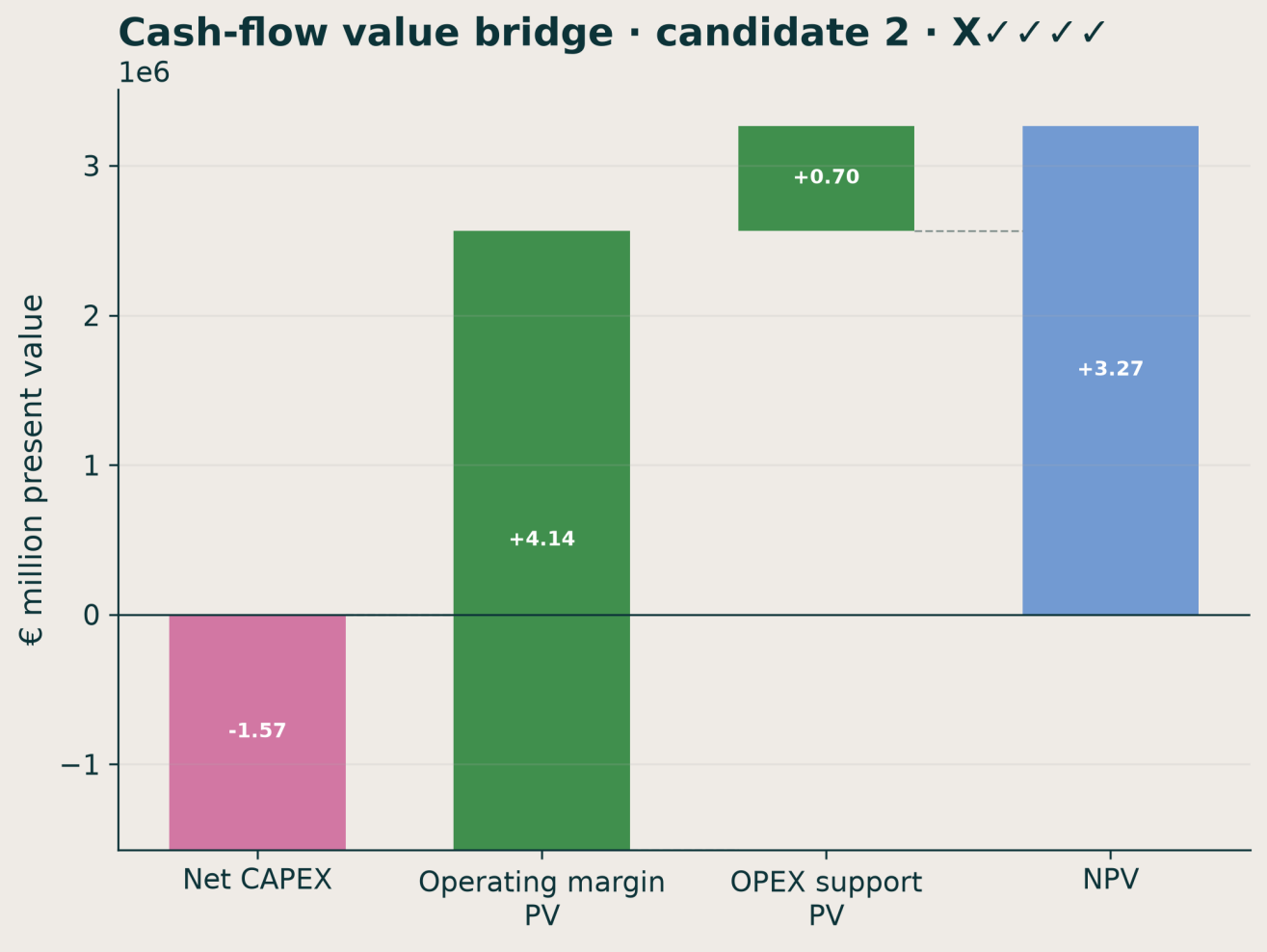


Figure 64. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 2; support C30 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

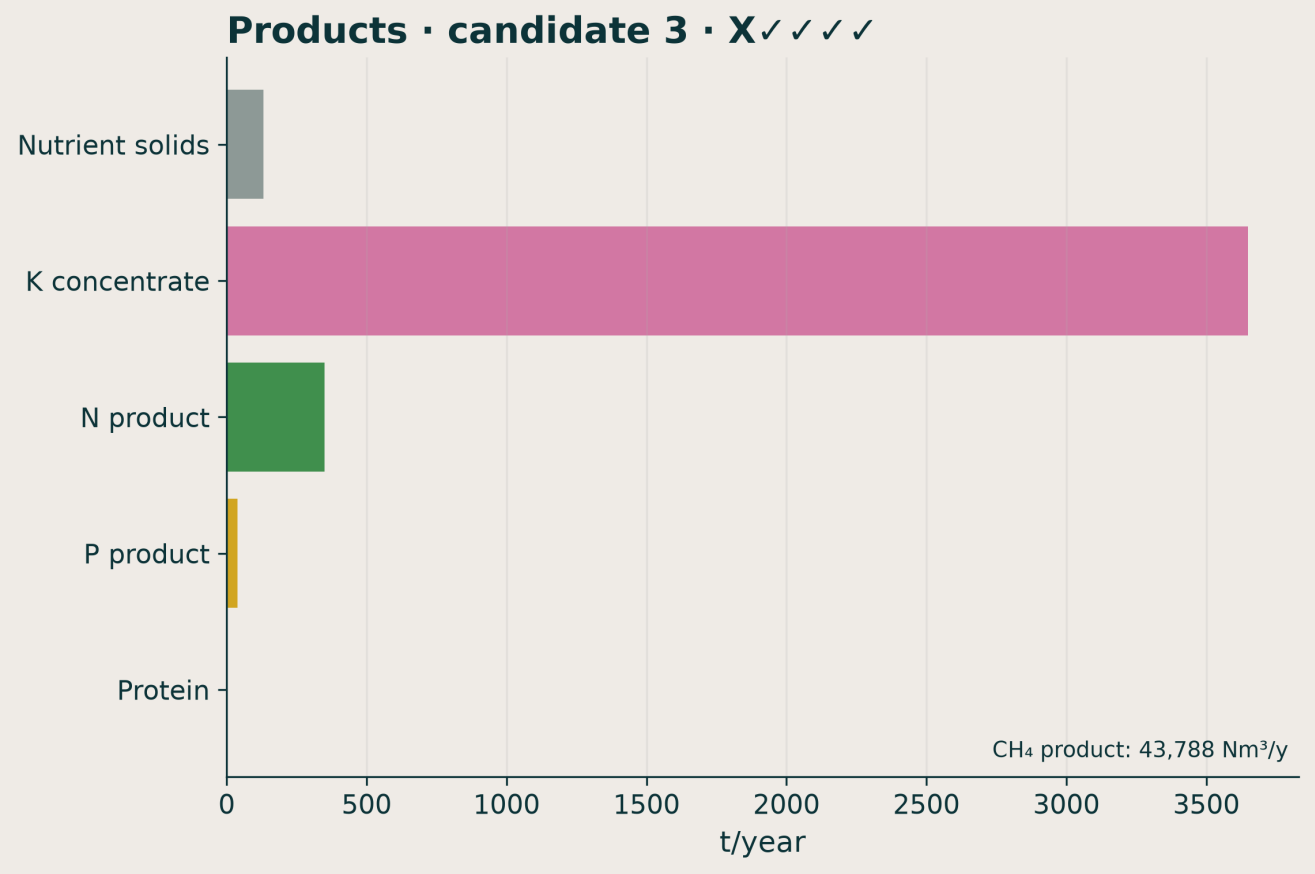


Figure 65. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

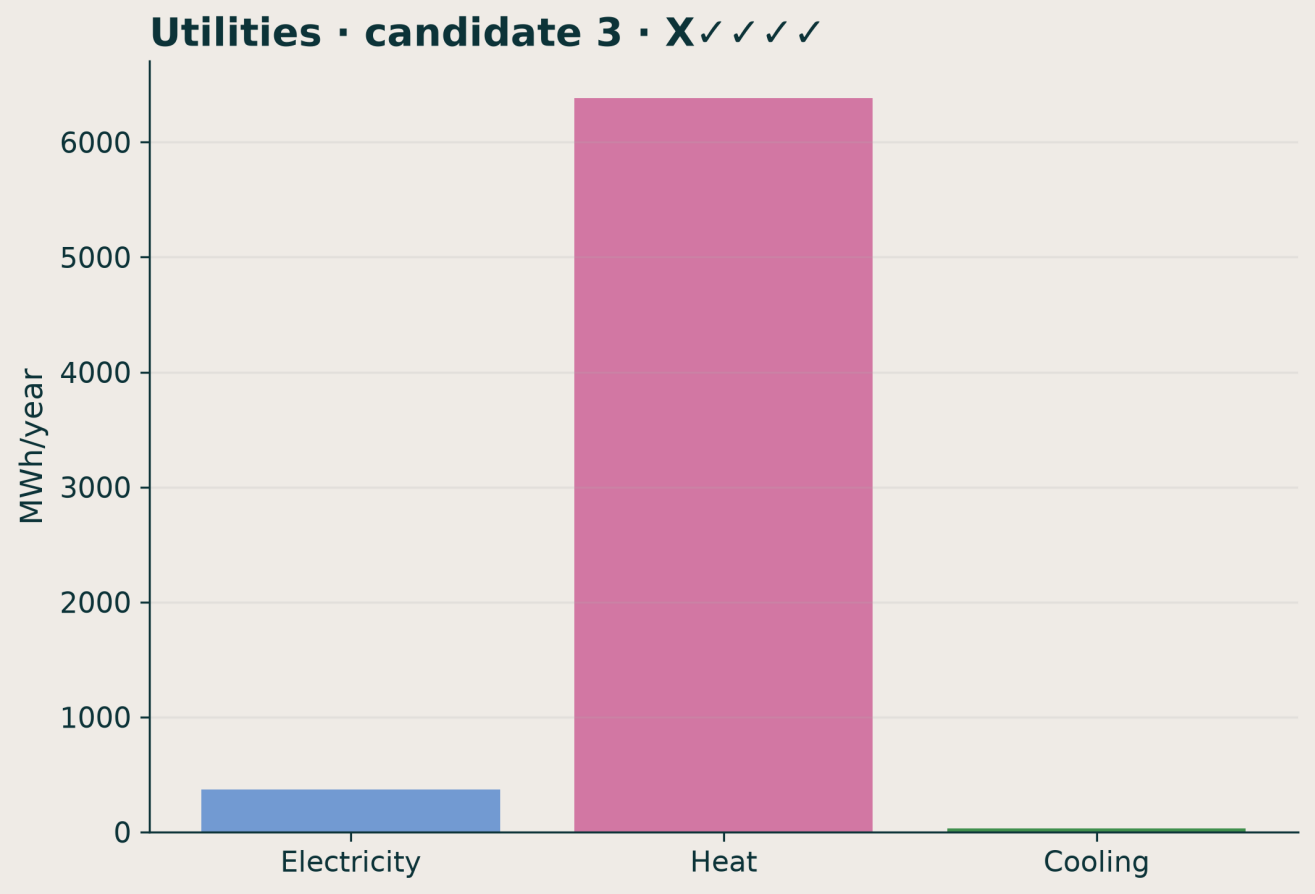


Figure 66. Annual electricity, heating and cooling duty

protein_bypass; rank 3; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR7', 'process_order': 'C'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

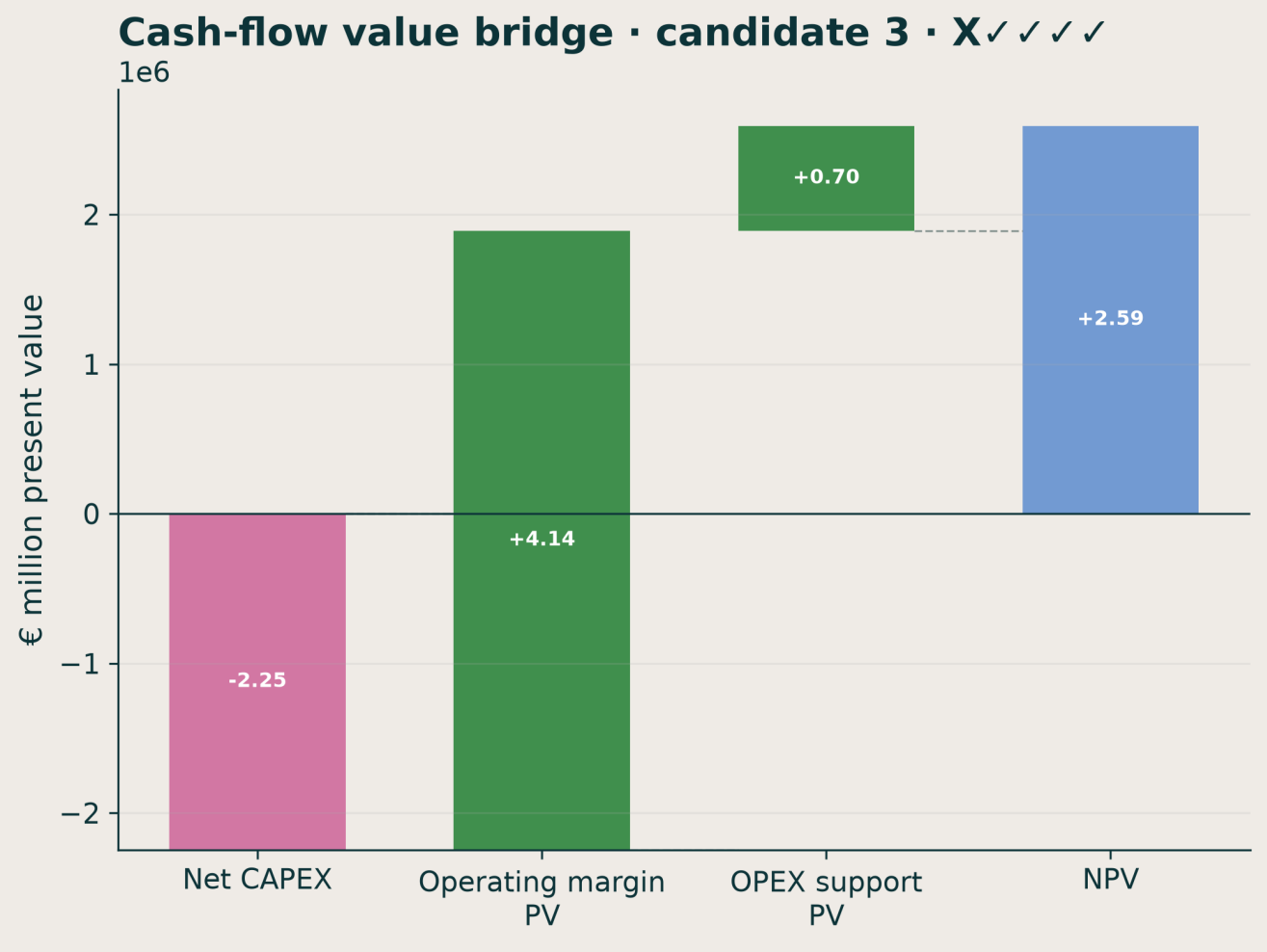


Figure 67. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 3; support C0 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

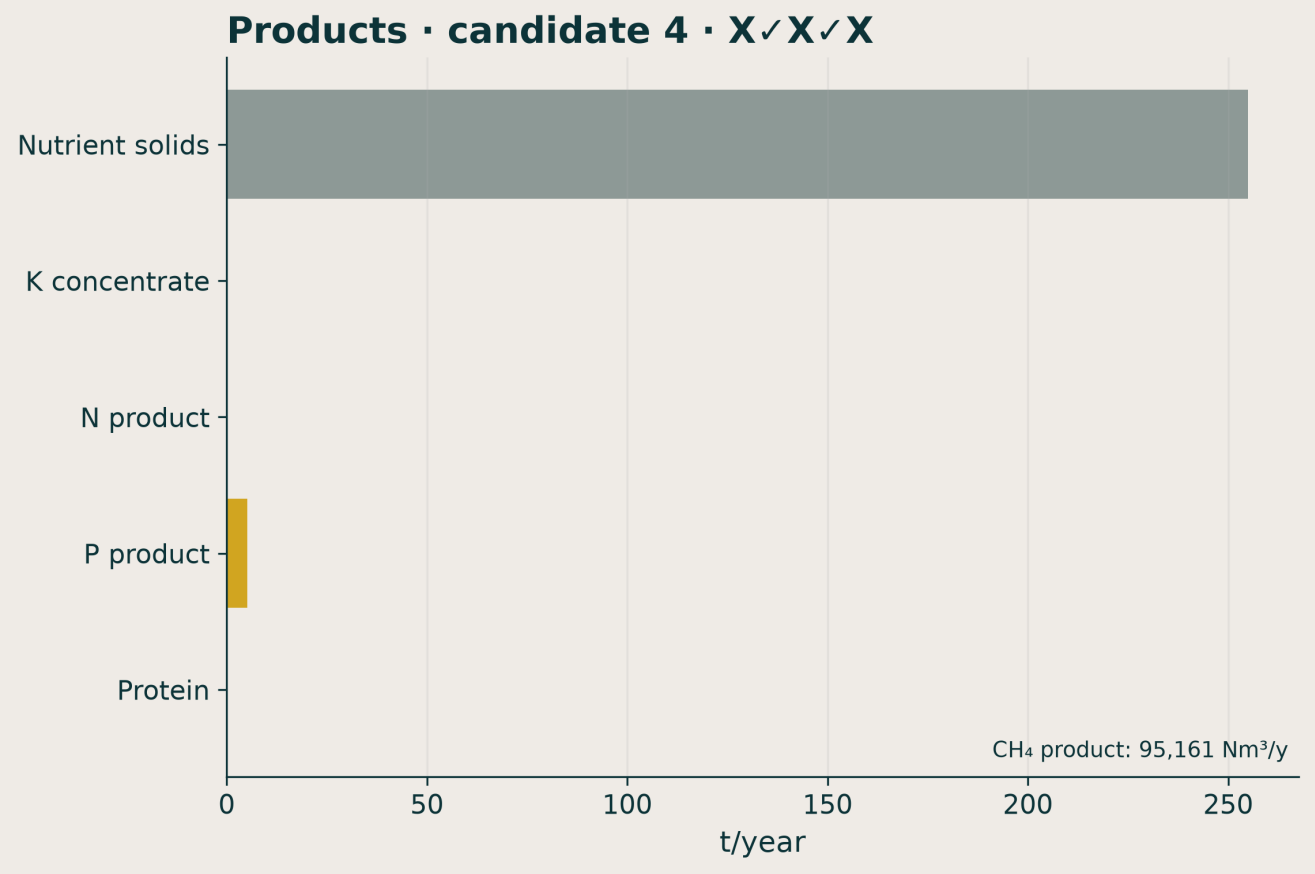


Figure 68. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

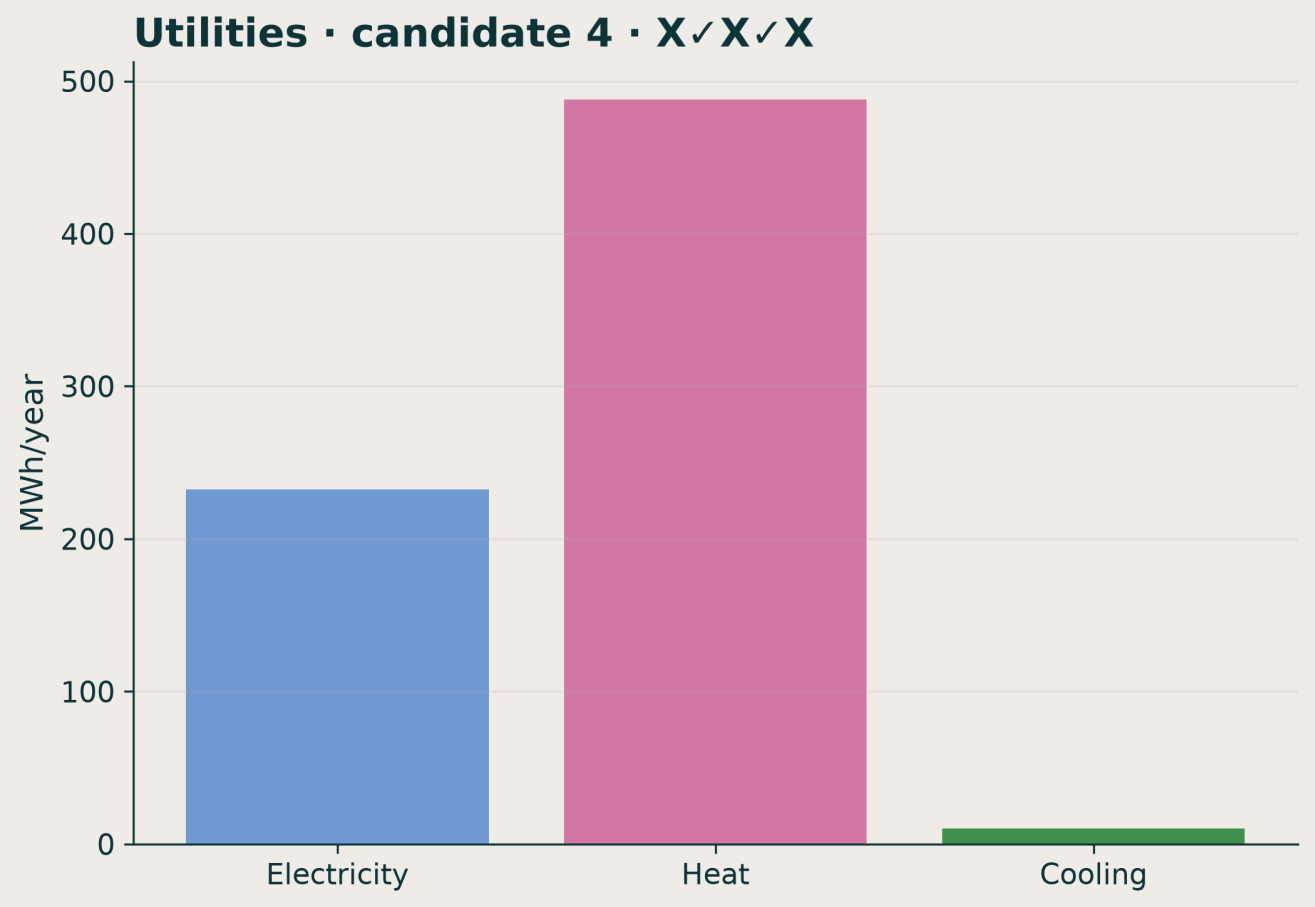


Figure 69. Annual electricity, heating and cooling duty

protein_bypass; rank 4; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

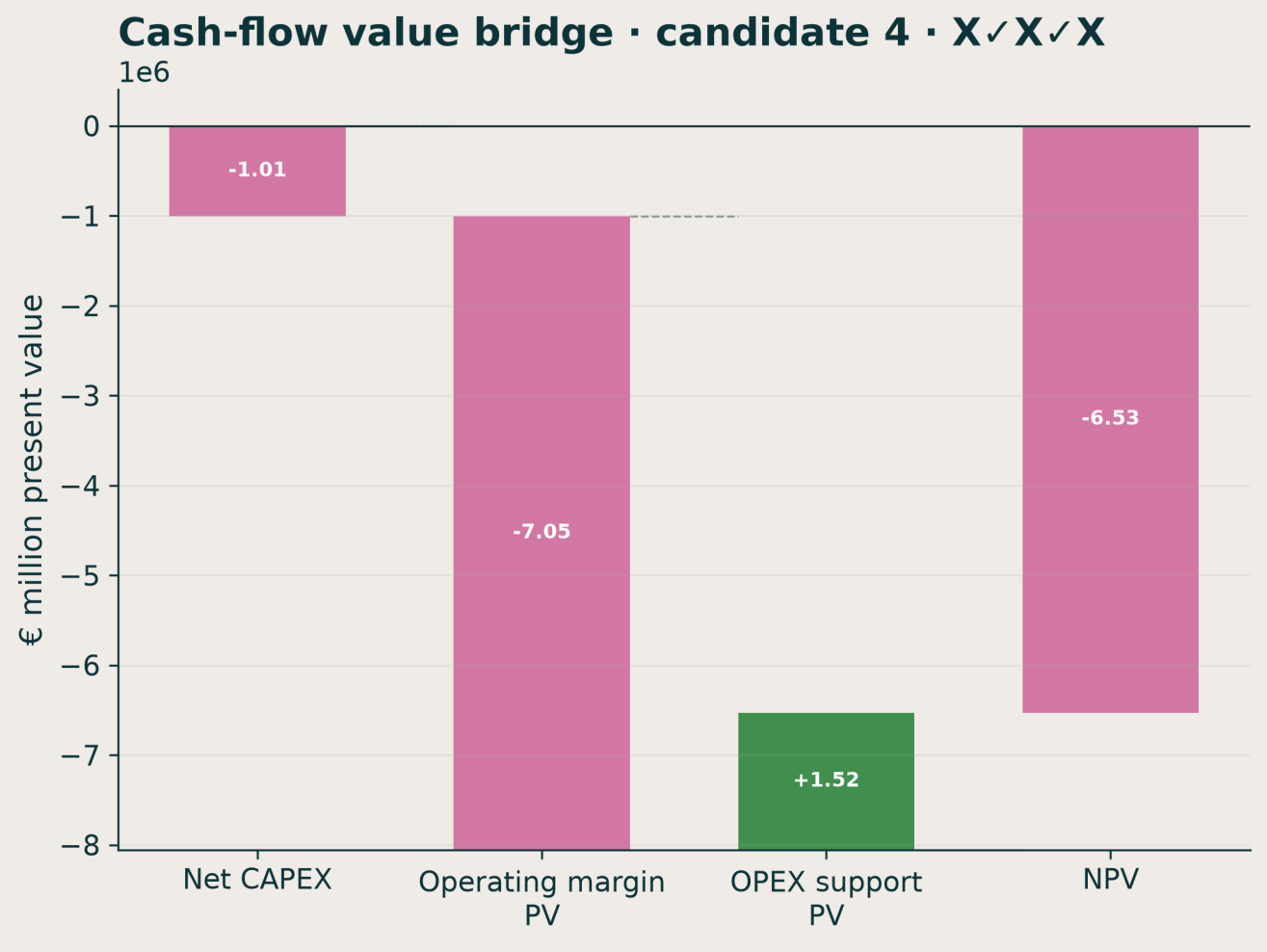


Figure 70. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 4; support C45 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

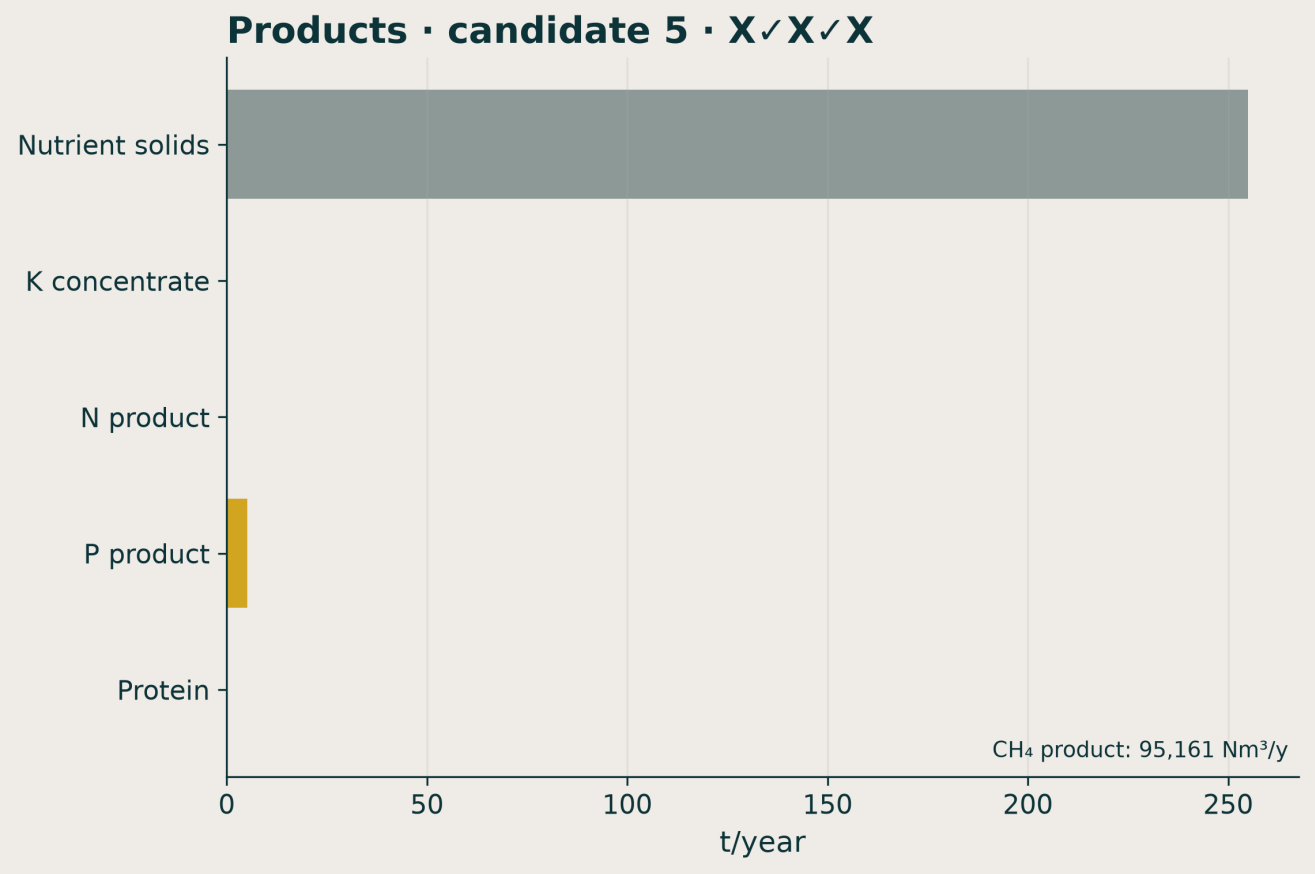


Figure 71. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

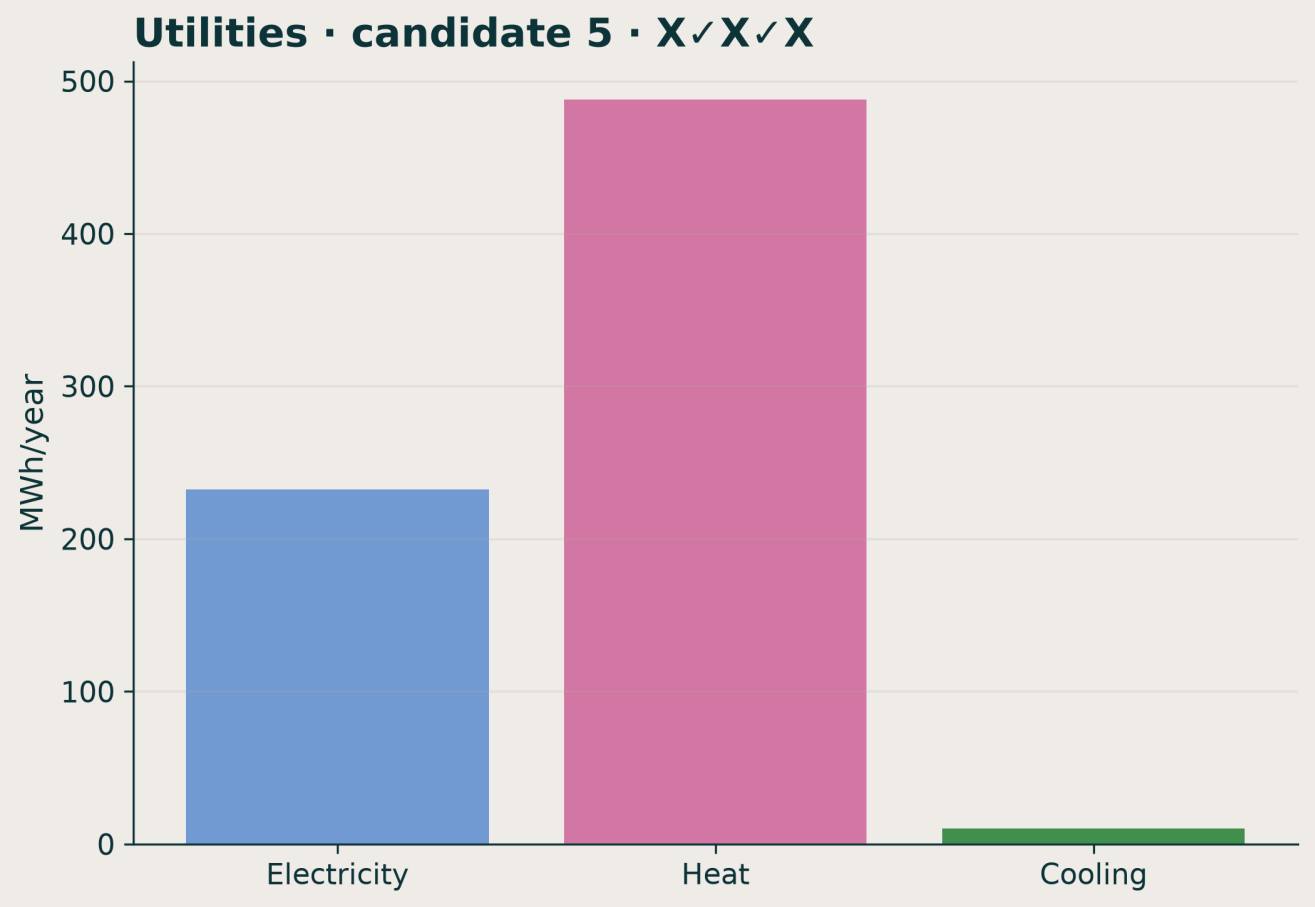


Figure 72. Annual electricity, heating and cooling duty

protein_bypass; rank 5; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

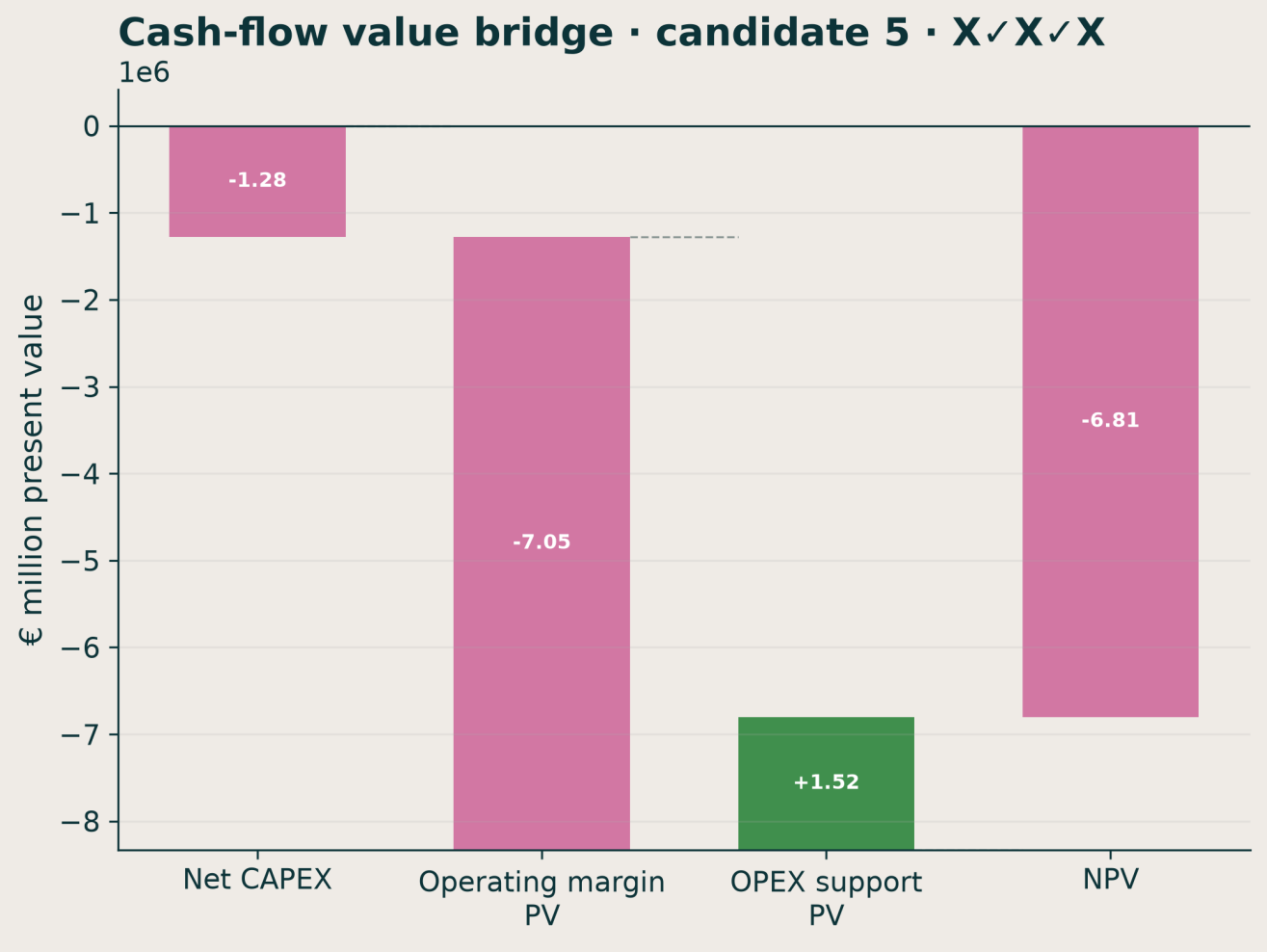


Figure 73. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 5; support C30 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

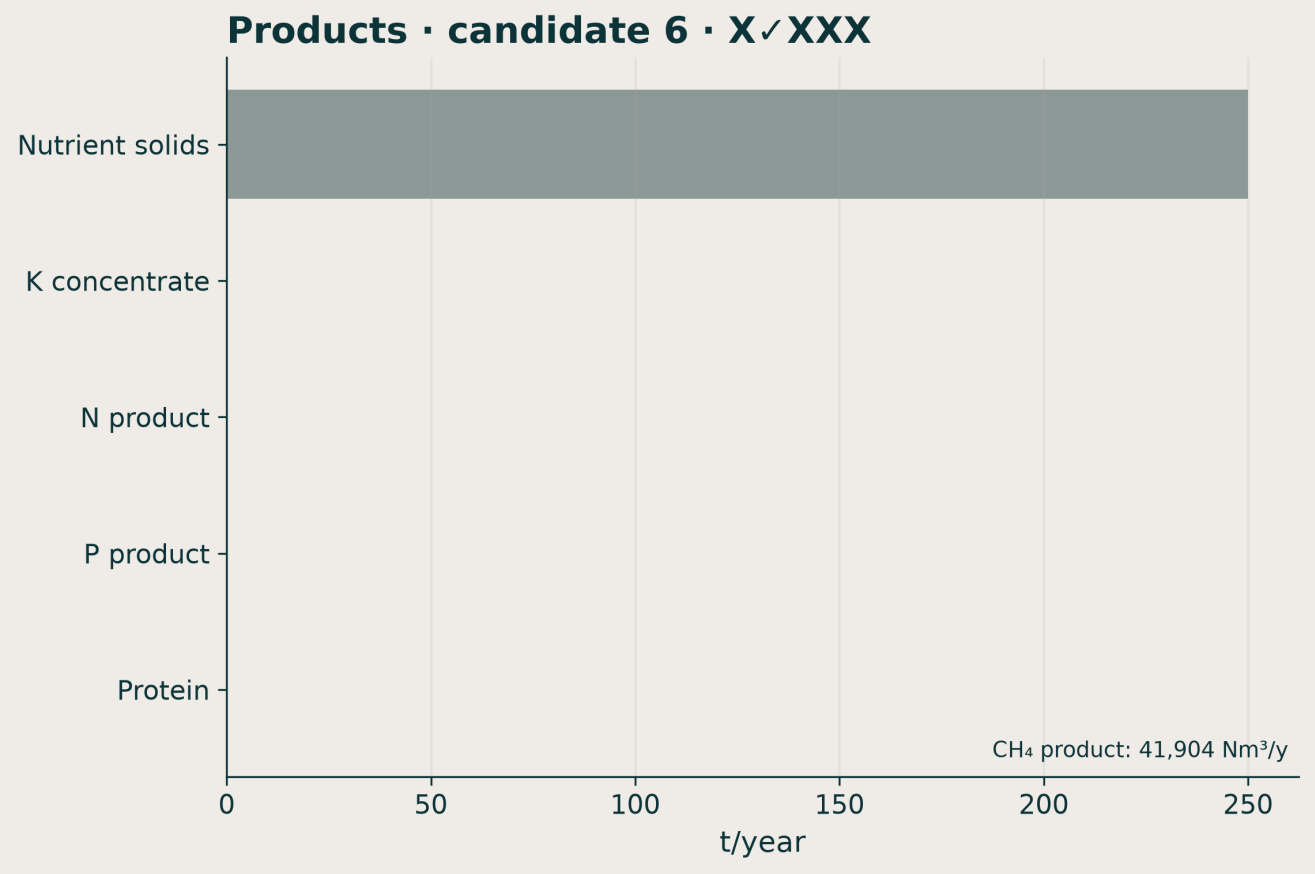


Figure 74. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

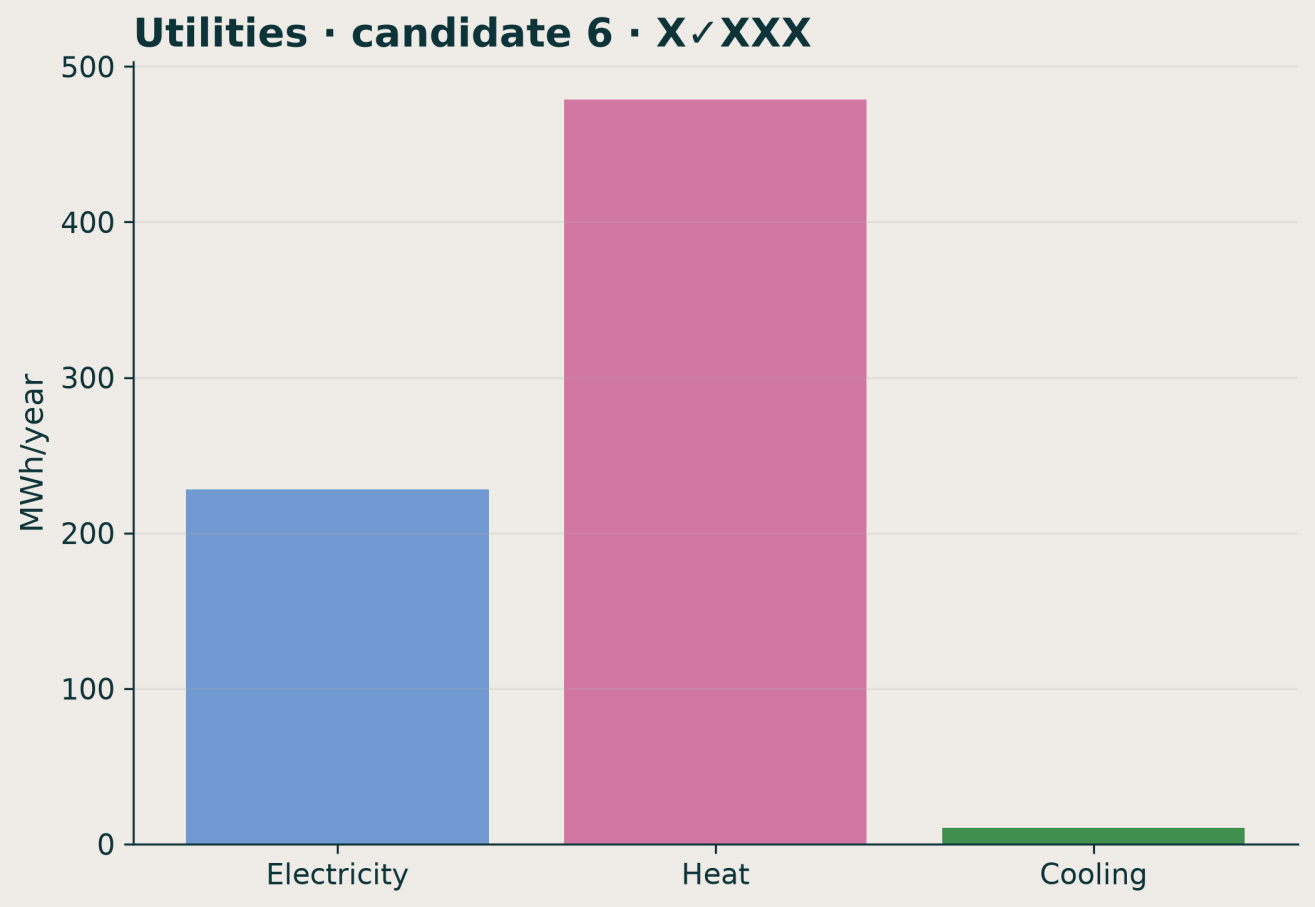


Figure 75. Annual electricity, heating and cooling duty

protein_bypass; rank 6; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

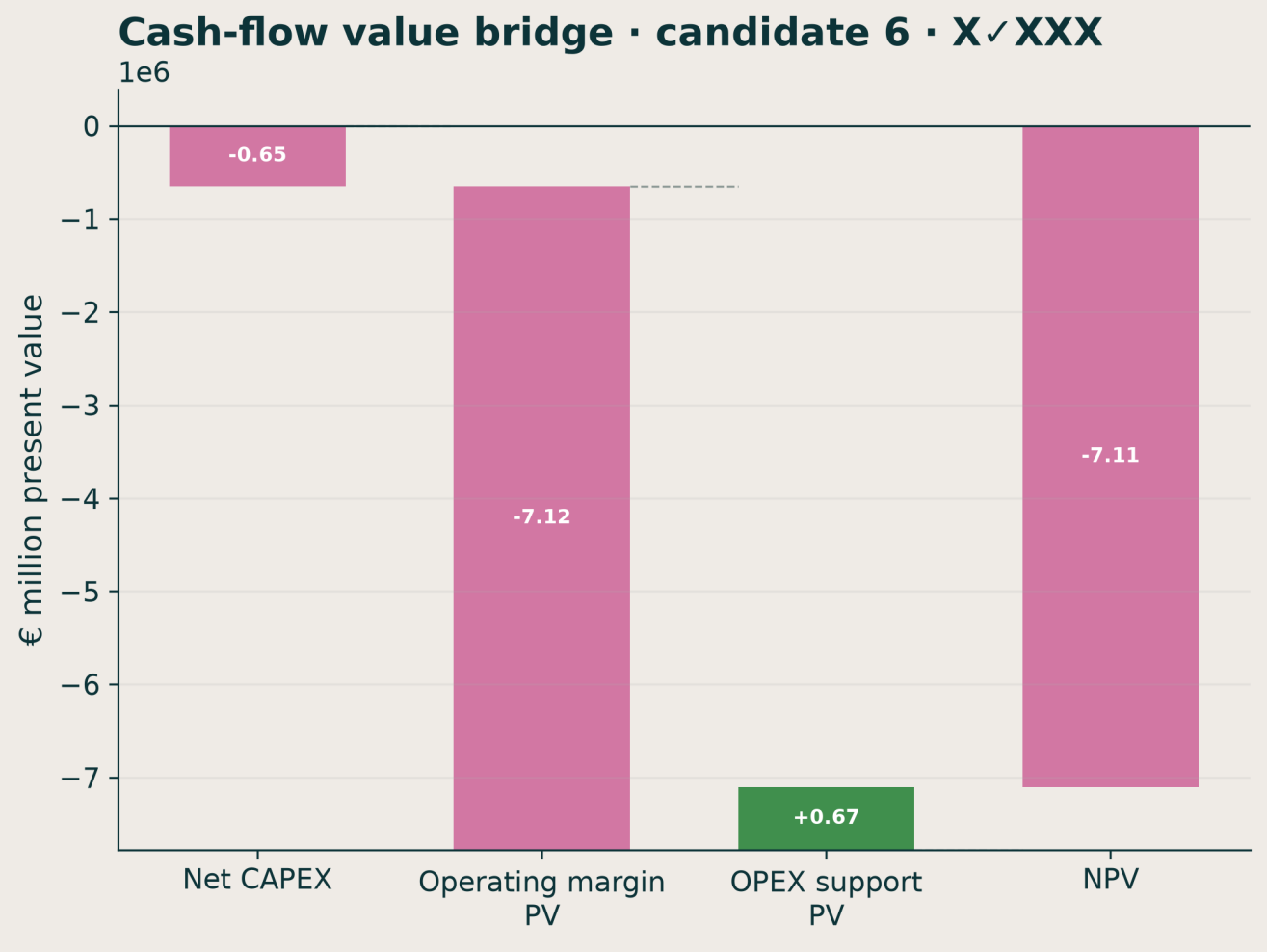


Figure 76. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 6; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

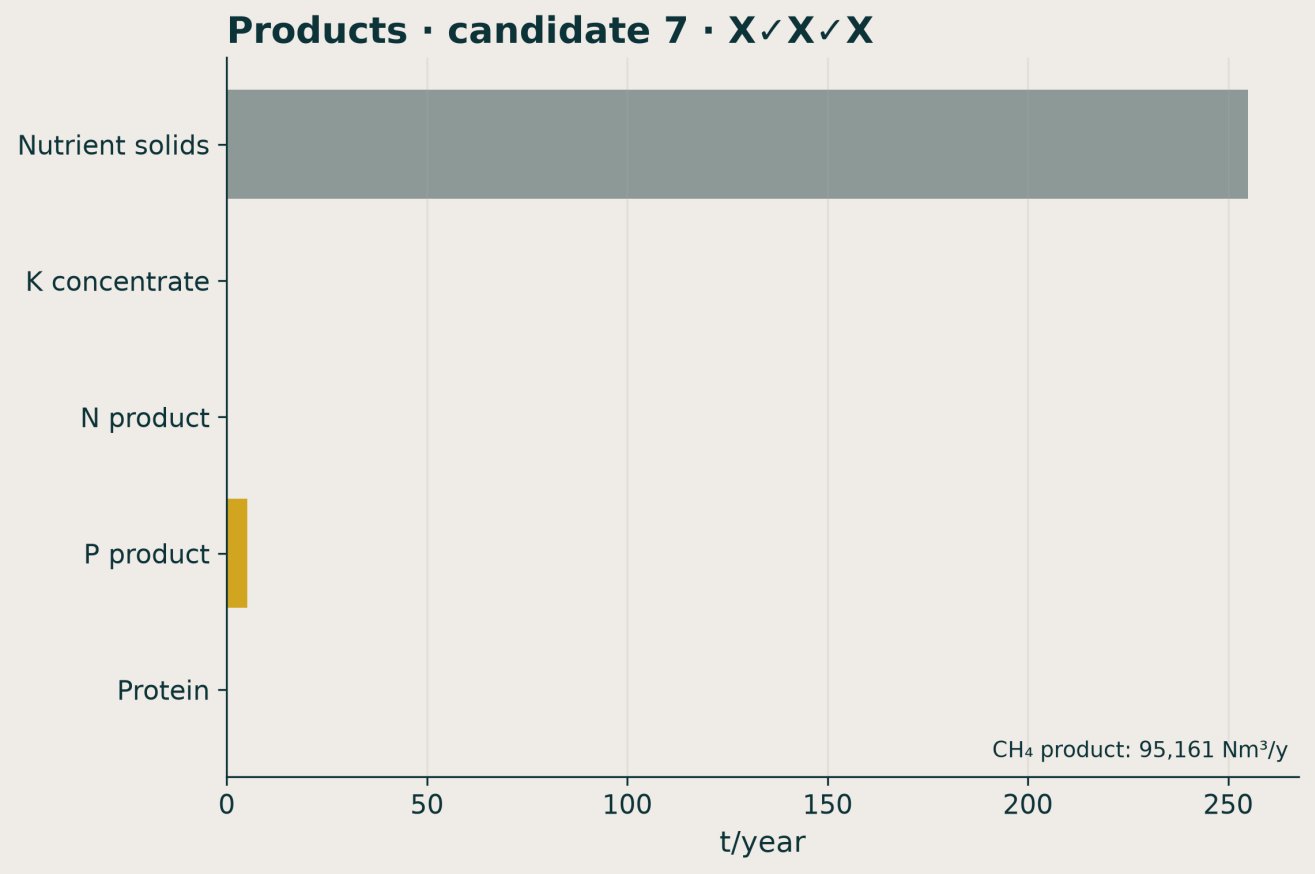


Figure 77. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

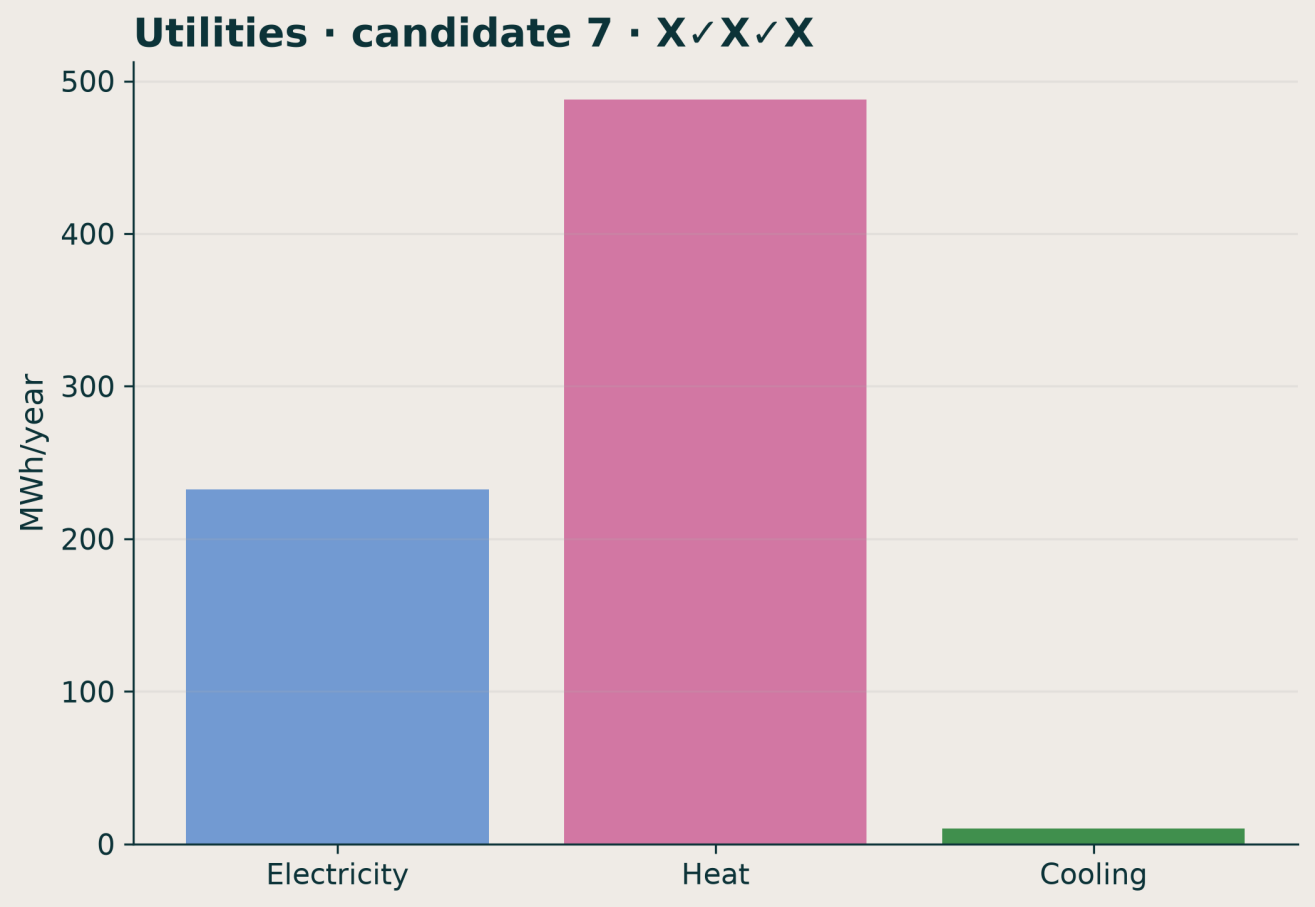


Figure 78. Annual electricity, heating and cooling duty

protein_bypass; rank 7; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

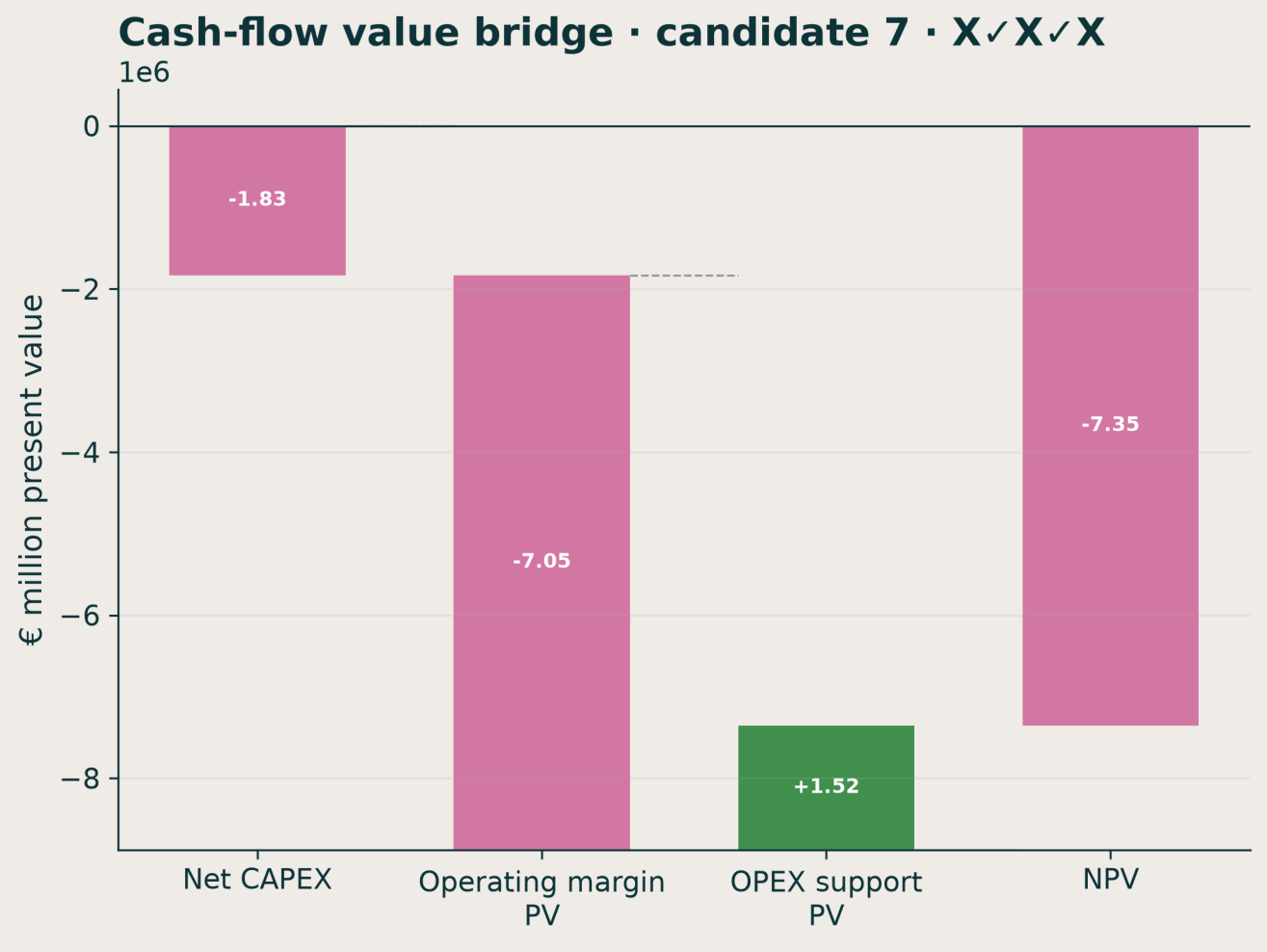


Figure 79. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 7; support C0 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

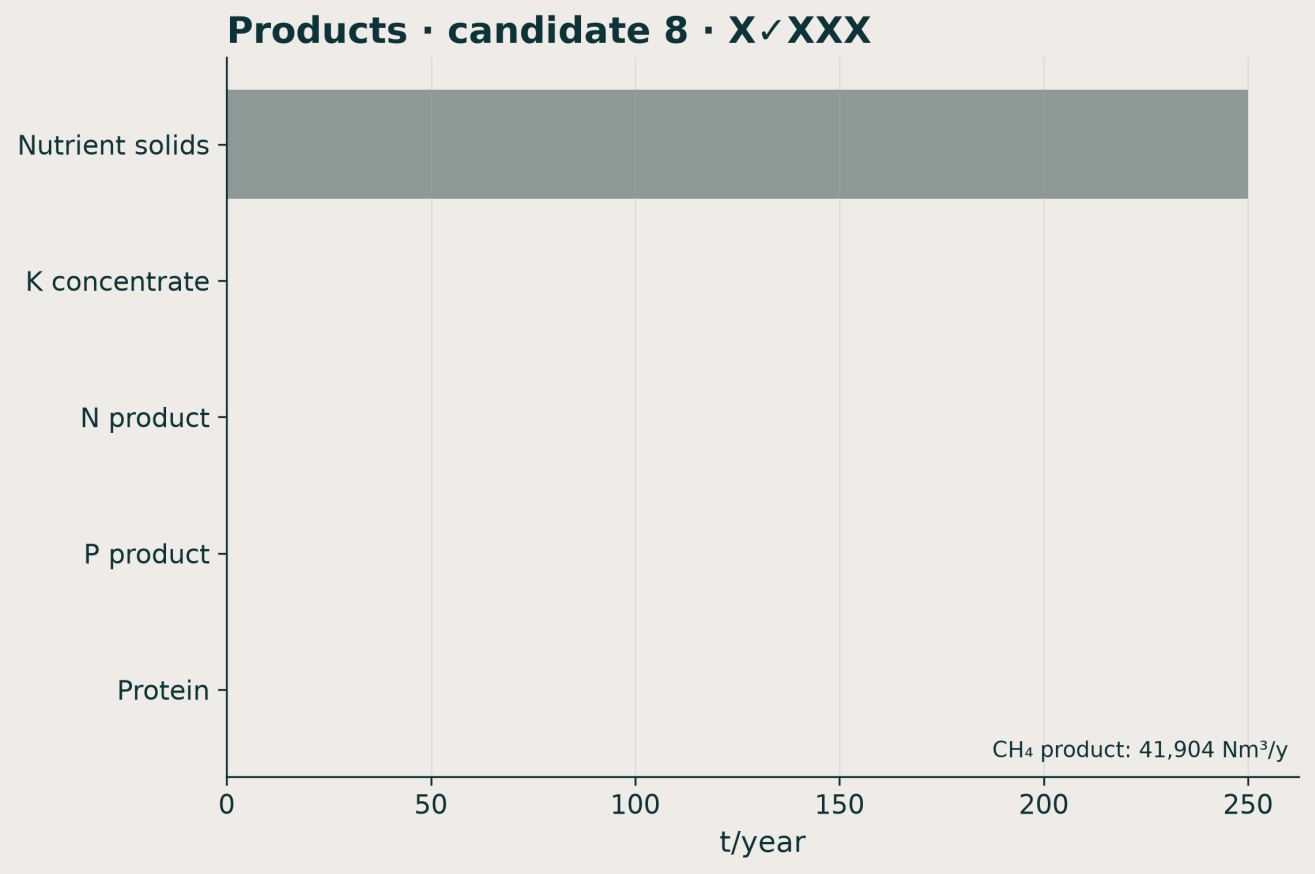


Figure 80. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

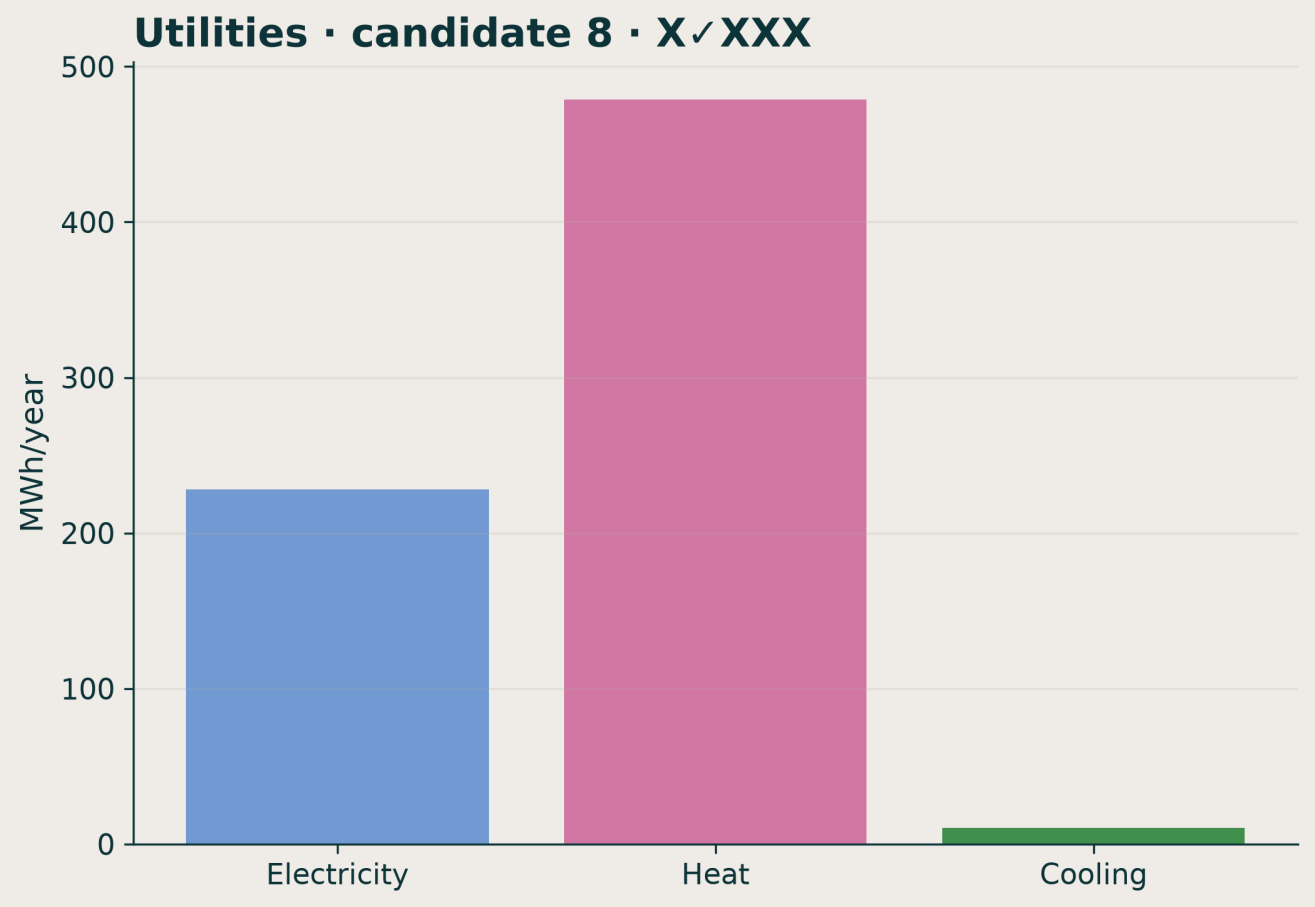


Figure 81. Annual electricity, heating and cooling duty

protein_bypass; rank 8; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

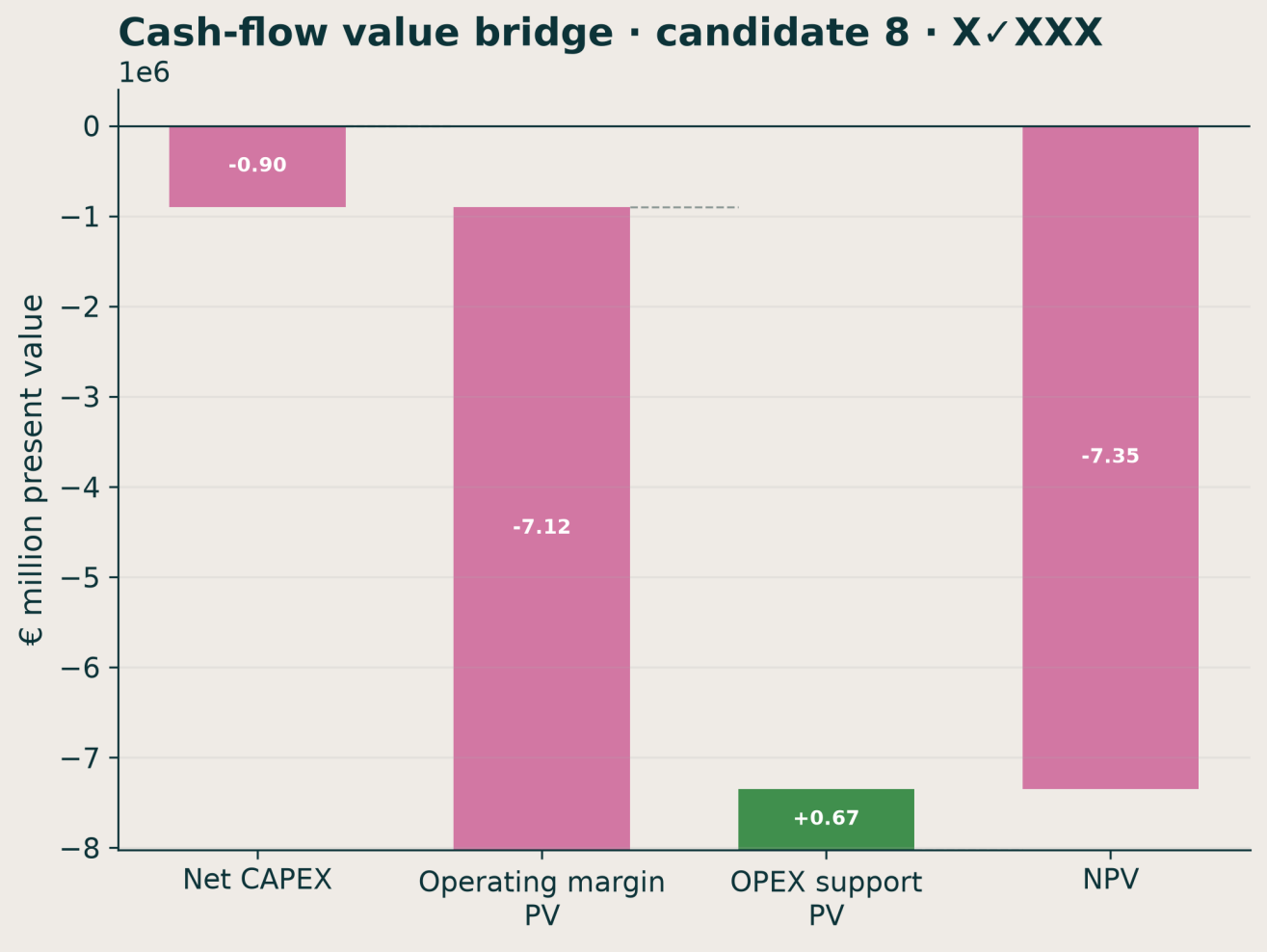


Figure 82. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 8; support C45 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

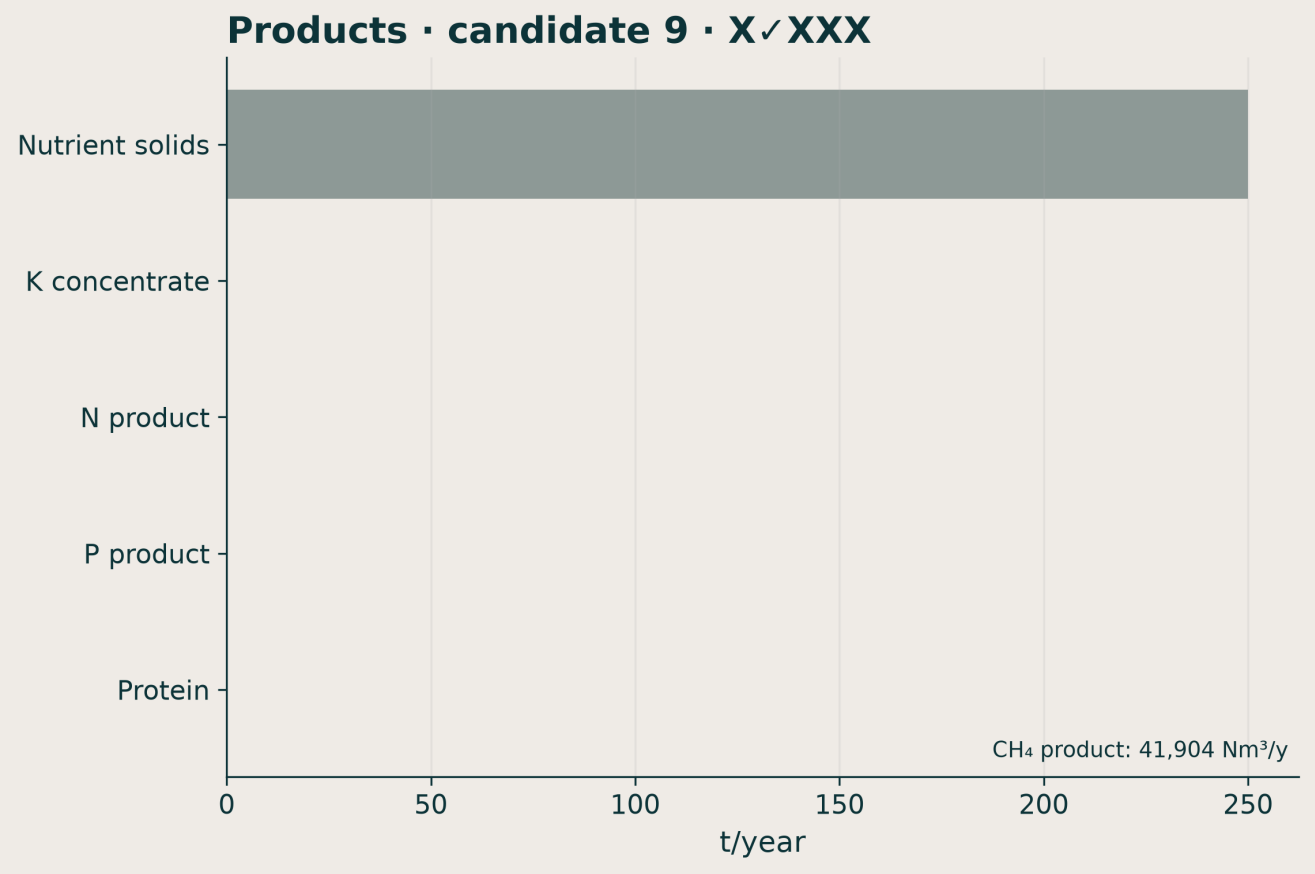


Figure 83. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions
protein_bypass; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

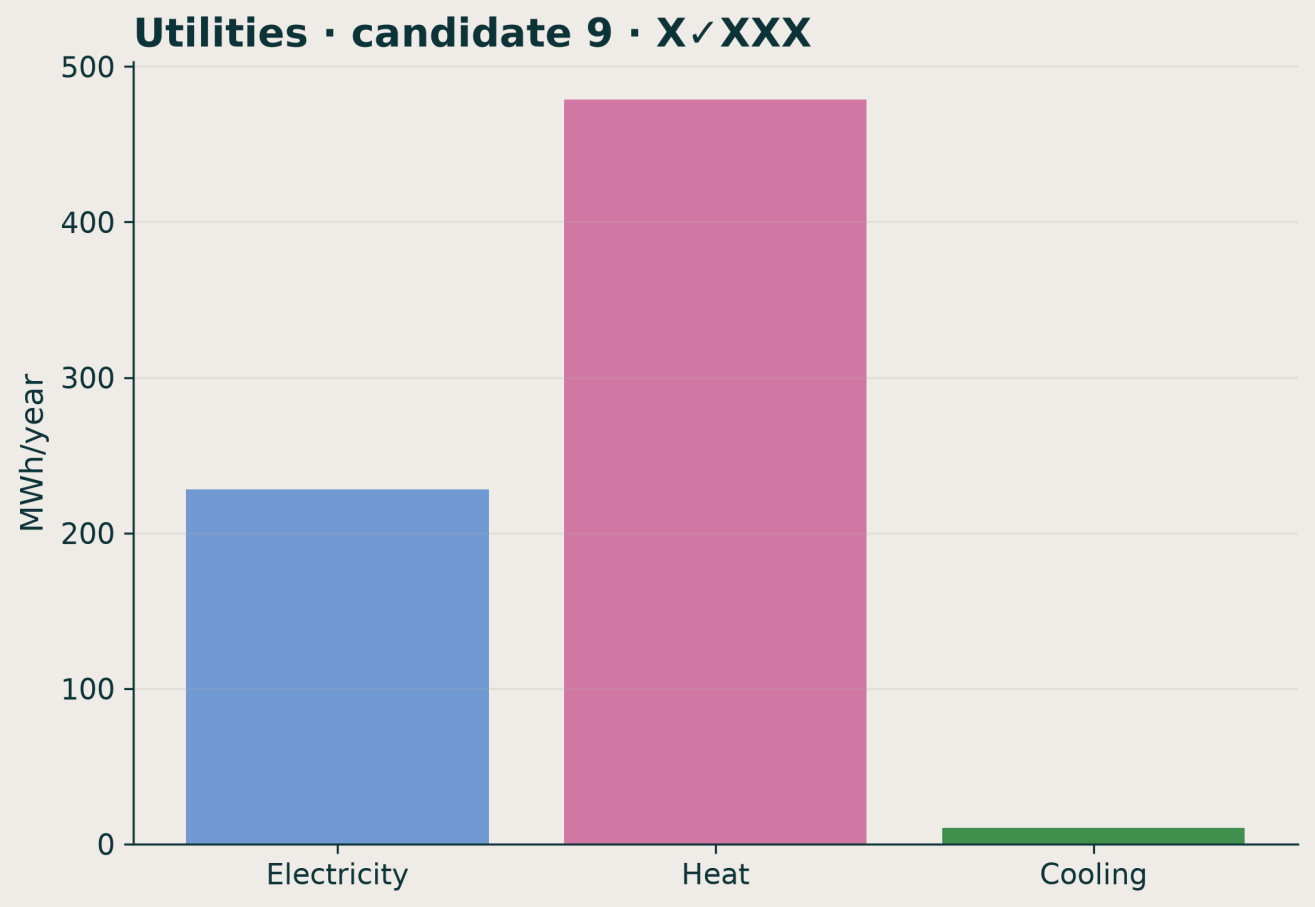


Figure 84. Annual electricity, heating and cooling duty

protein_bypass; rank 9; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG2', 'nutrient_recovery': 'NR1', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

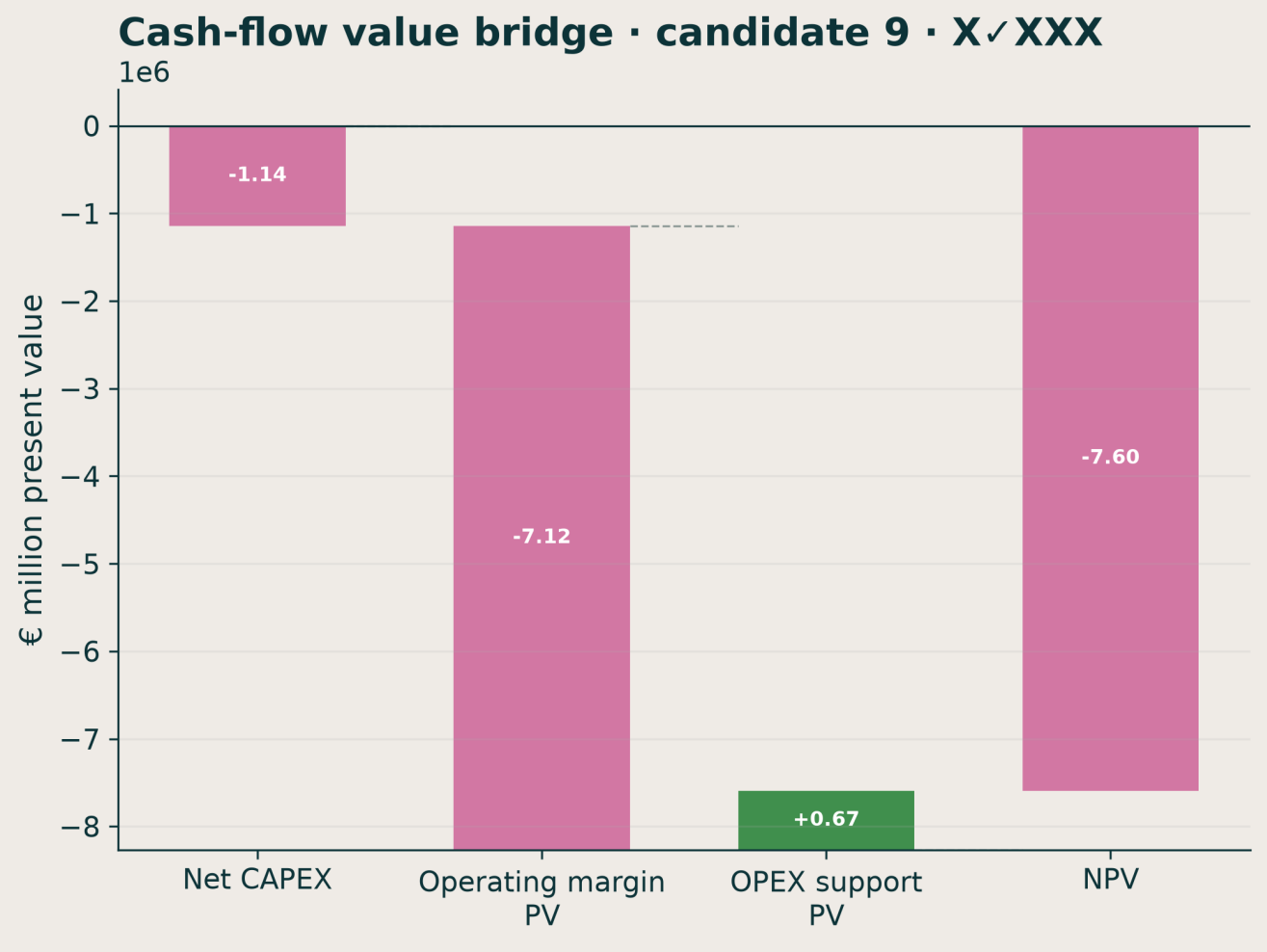


Figure 85. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 9; support C30 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate products

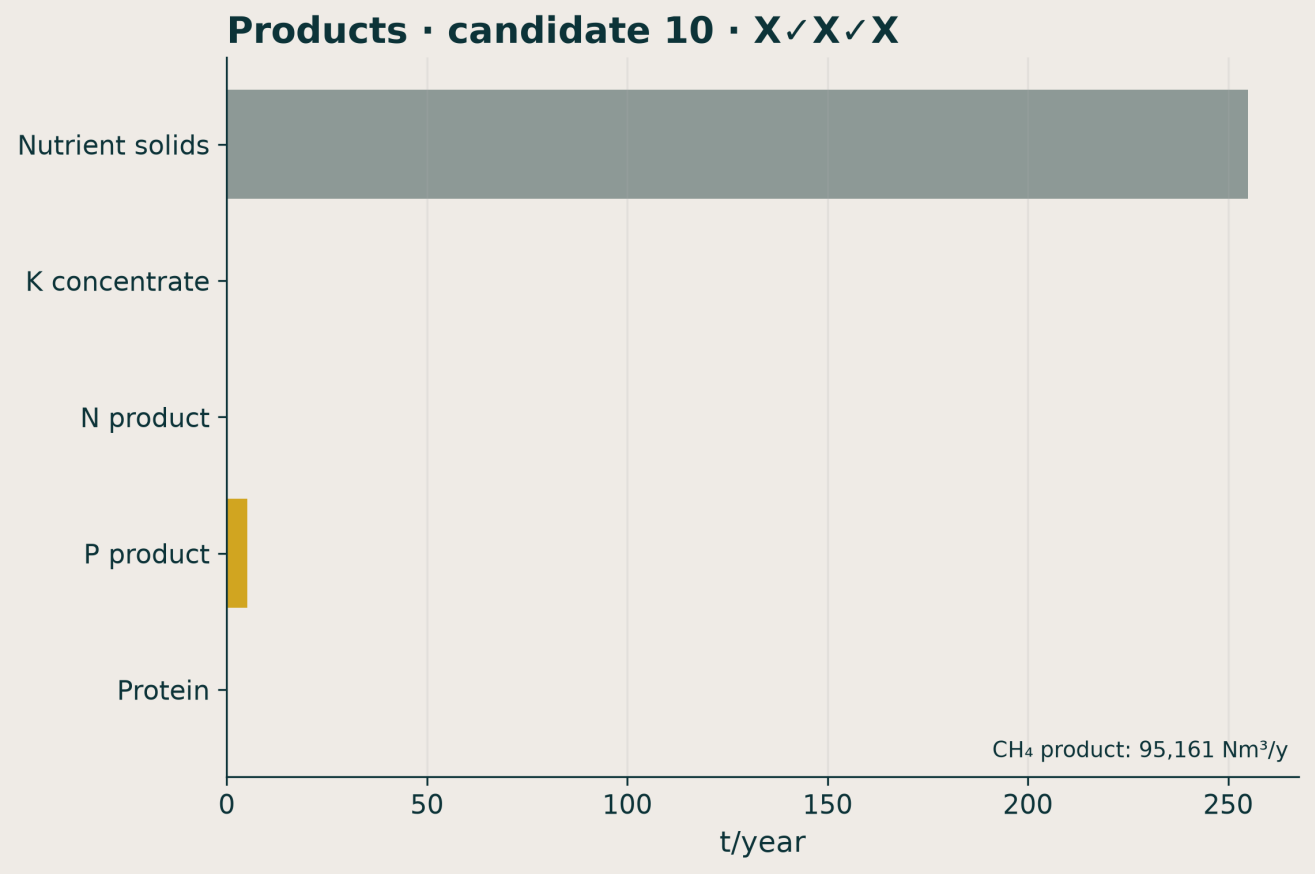


Figure 86. Annual solid products; methane separately shown as Nm3 using 0.716 kg/Nm3 at normal conditions

protein_bypass; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | t/year; Nm³/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate utilities

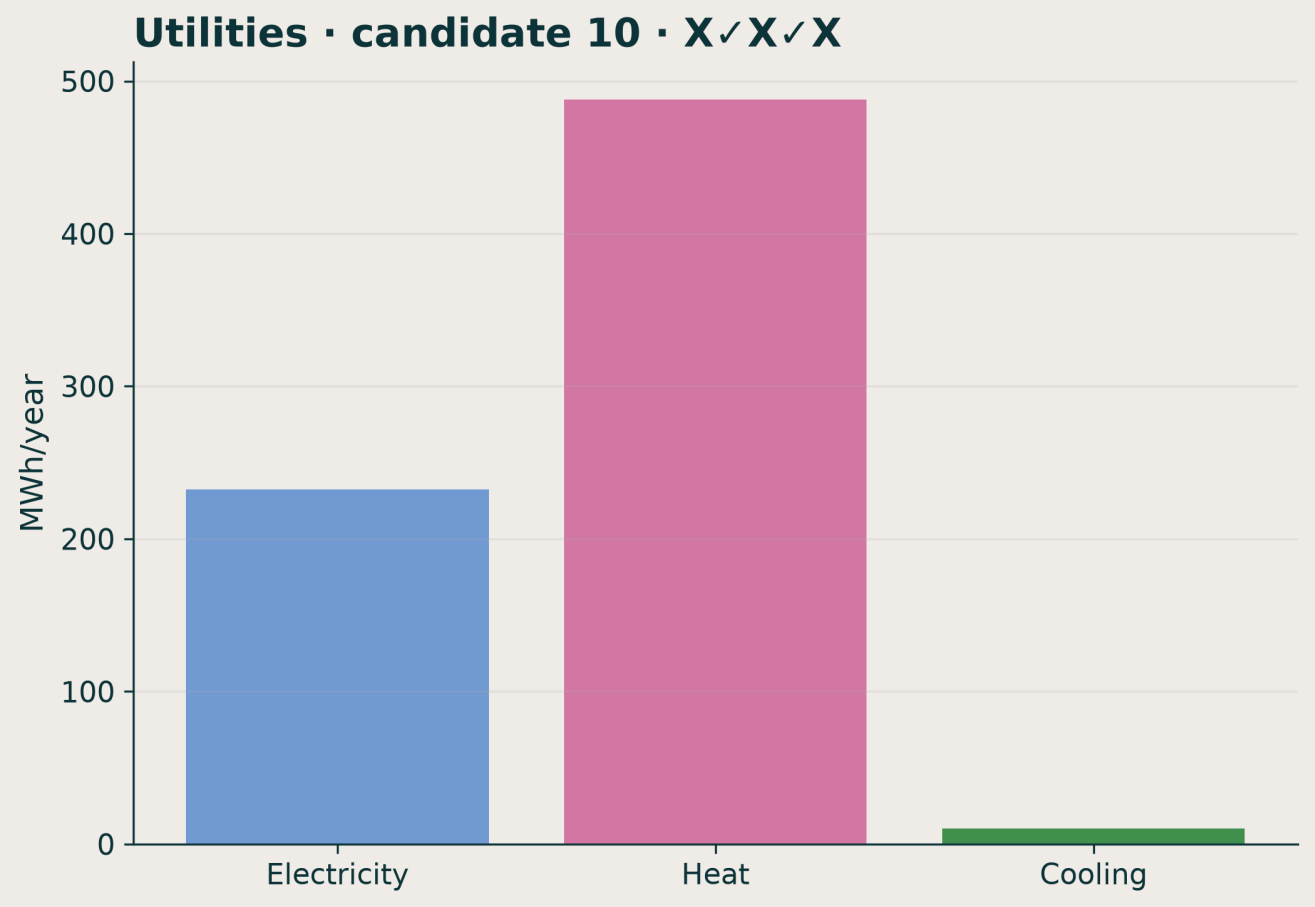


Figure 87. Annual electricity, heating and cooling duty

protein_bypass; rank 10; {'feed_preparation': 'F1', 'protein_recovery': 'PR0', 'anaerobic_digestion': 'AD3', 'biogas_upgrading': 'BG3', 'nutrient_recovery': 'NR2', 'process_order': 'A'} | MWh/year | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

candidate kpis

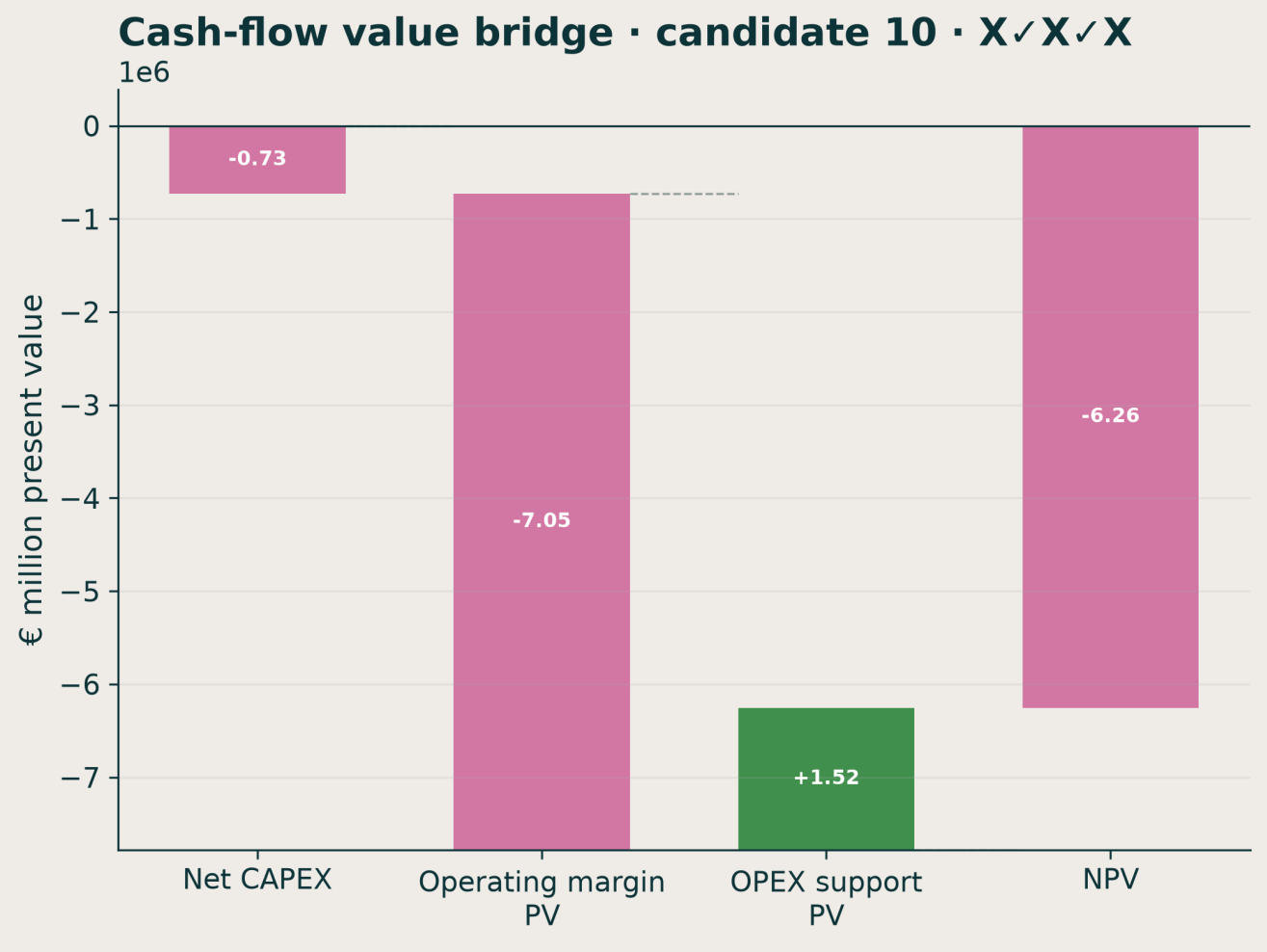


Figure 88. Present-value bridge: net direct CAPEX, discounted annual operating margin and operating-support PV to screening NPV

protein_bypass; rank 10; support C60 | € million present value | source: Greenhub/results/pareto_rvo_2026/protein_bypass_batch.json

baseline rvo support kpis

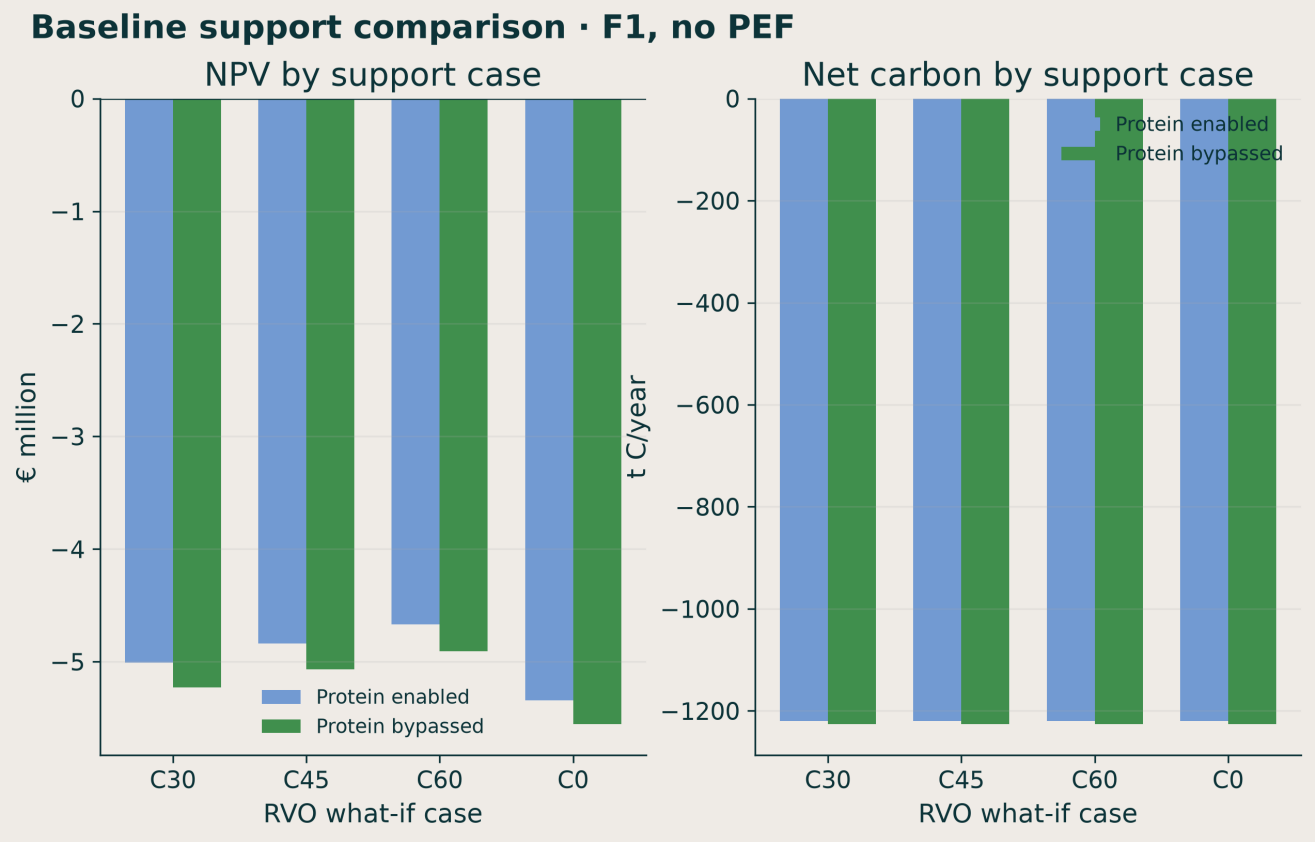


Figure 89. Baseline NPV and net carbon across paired RVO screening cases

No-PEF baseline; protein enabled and bypassed | € million; t C/year | source: Greenhub/results/pareto_rvo_2026

baseline utilities protein comparison

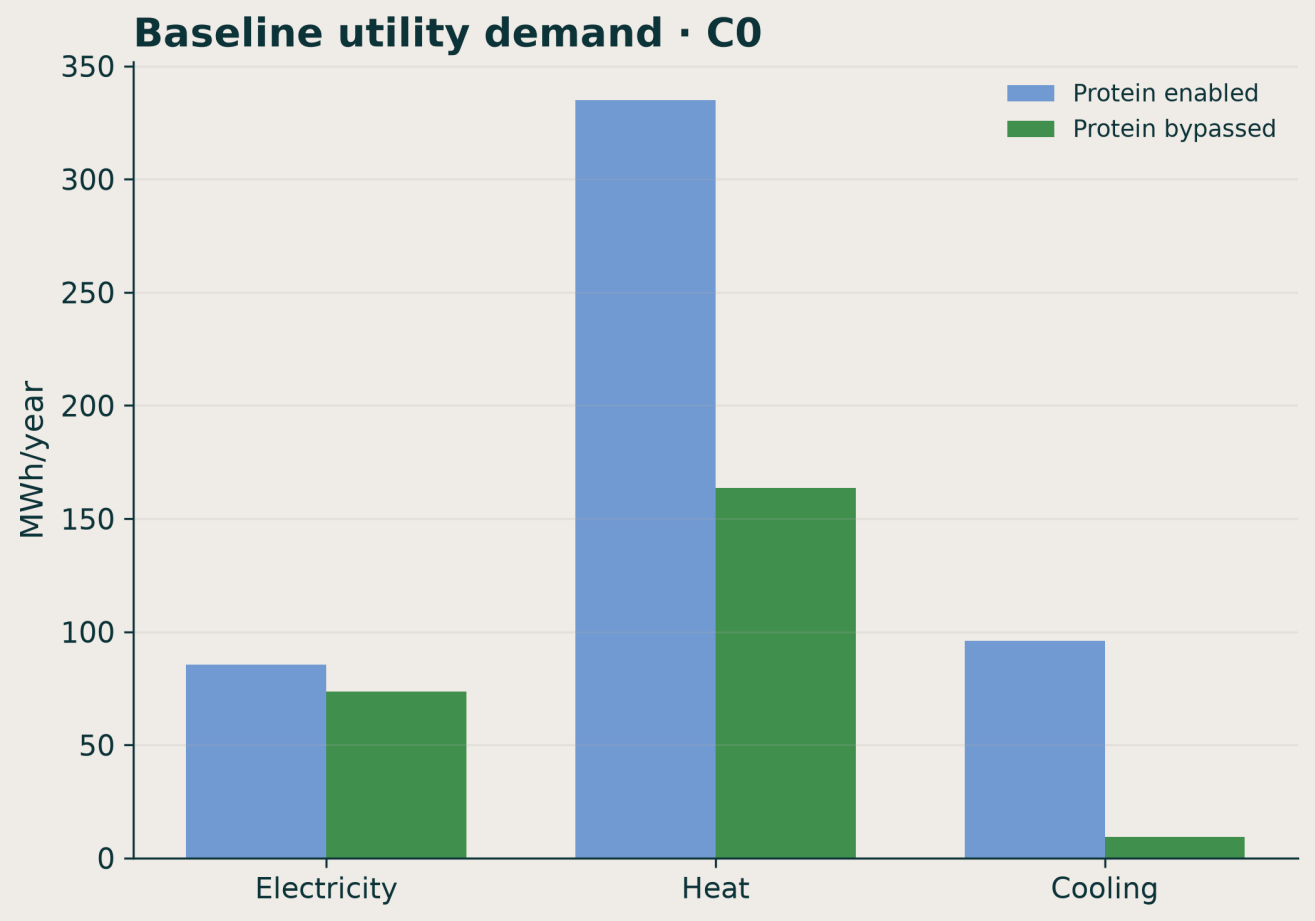


Figure 90. Baseline utility comparison for protein route and bypass

No-PEF baseline; C0 (utilities unaffected by support case) | MWh/year | source: Greenhub/results/pareto_rvo_2026

rvo support case npv comparison

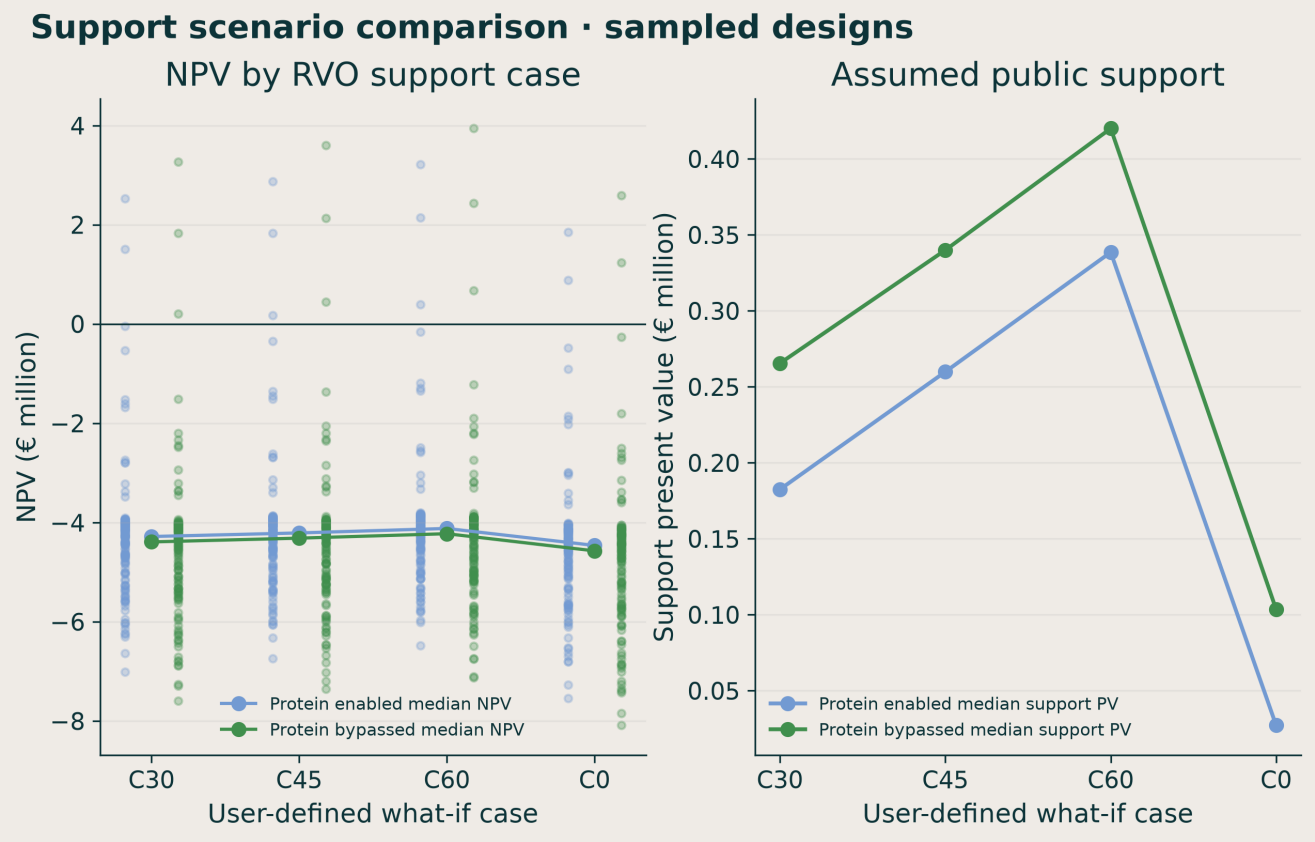


Figure 91. NPV samples and median support present value by support case

Protein enabled and bypassed; paired support cases | € million | source: Greenhub/results/pareto_rvo_2026

endpoint product yields

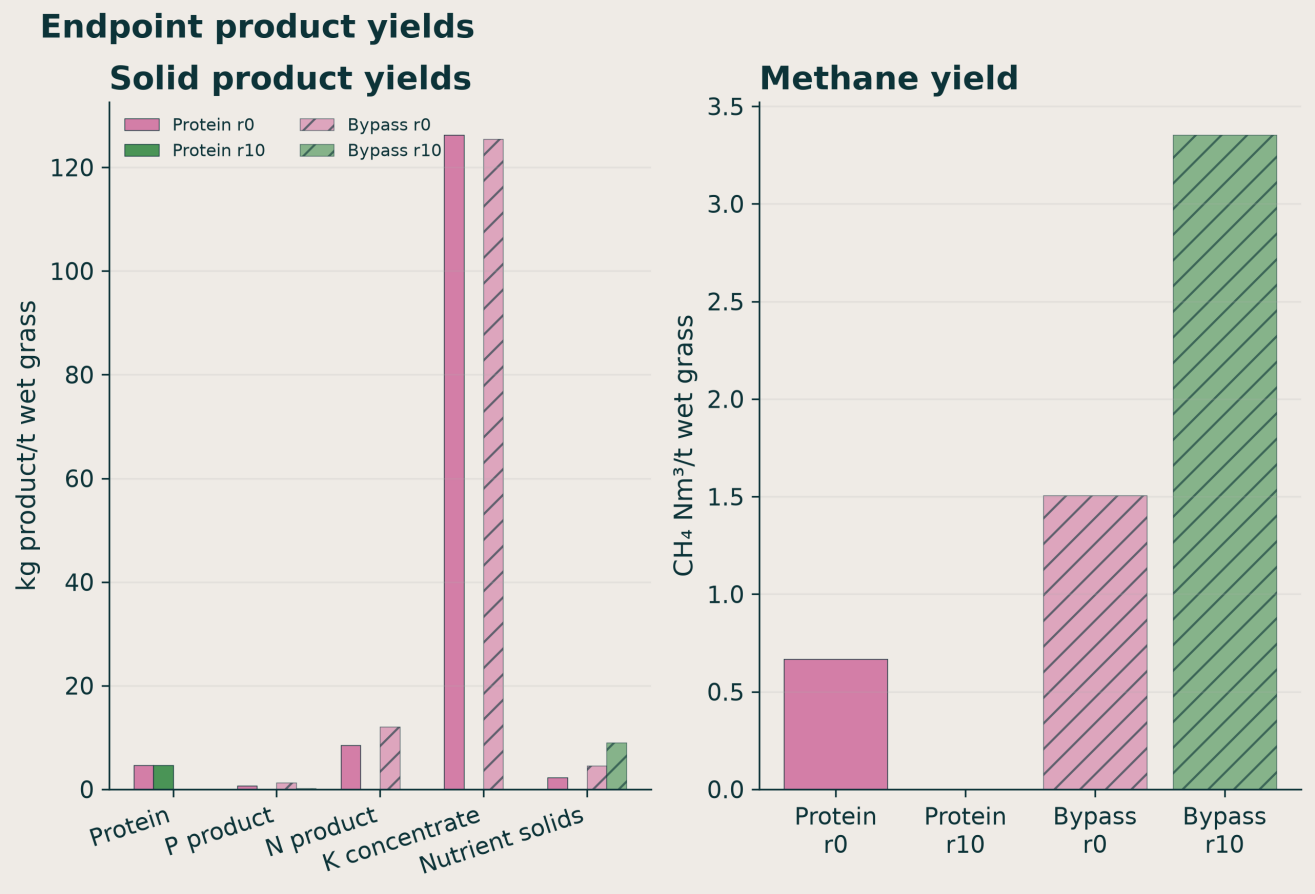


Figure 92. Solid product and methane yields at selected economic and carbon endpoints

Rank 0 and 10 from the three-objective front; no PEF | kg product/t wet grass; CH4 Nm3/t wet grass | source: Greenhub/results/pareto_rvo_2026

energy intensity by candidate

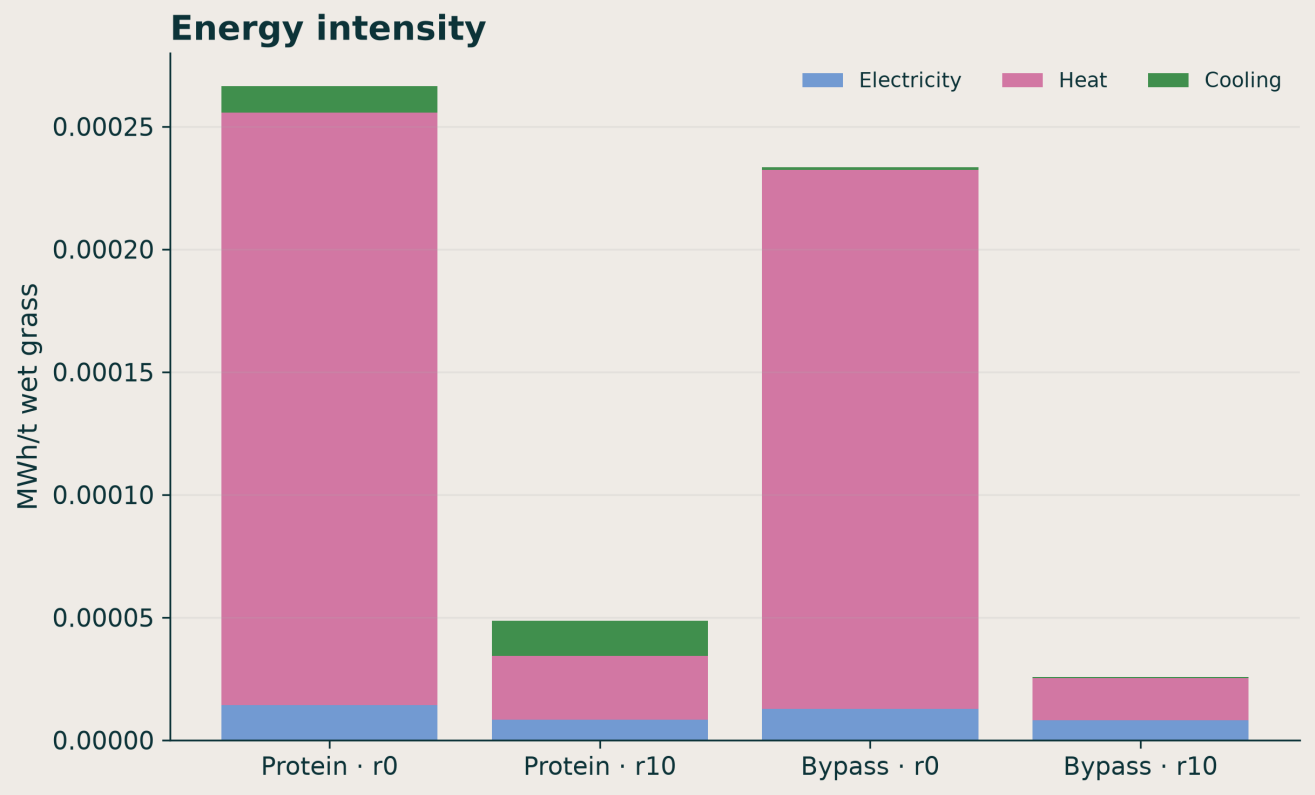


Figure 93. Electricity, heat and cooling intensity at selected endpoint candidates

Rank 0 and 10 from the three-objective front; no PEF | MWh/t wet grass | source: Greenhub/results/pareto_rvo_2026

npv profit by feed scale

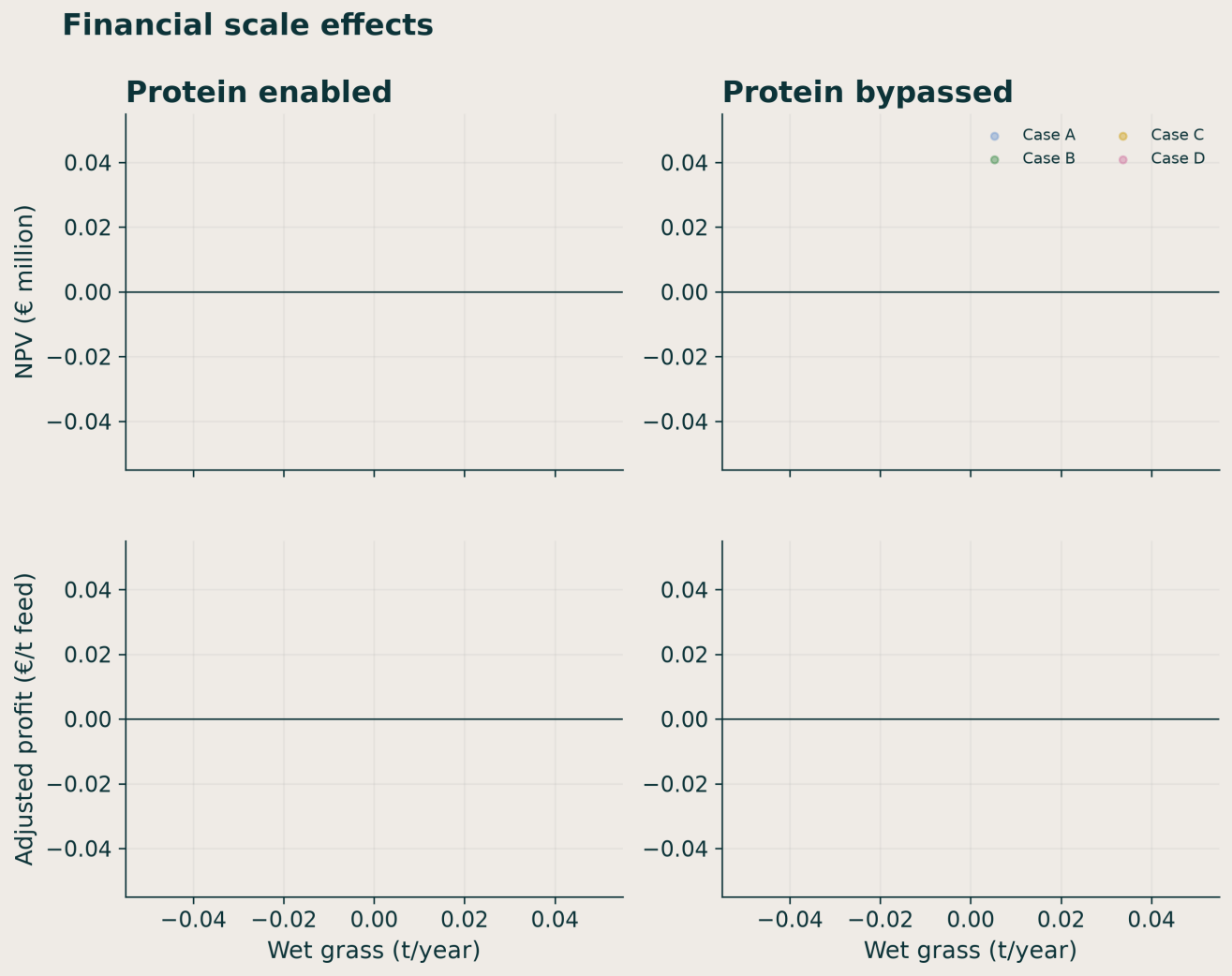


Figure 94. Screening NPV and original annual operating profit across evaluated feed scales

All evaluated designs; cases A-D; design-space scatter, not probabilistic uncertainty | NPV: € million; profit intensity: €/t wet grass | source: Greenhub/results/pareto_rvo_2026

rvo support composition

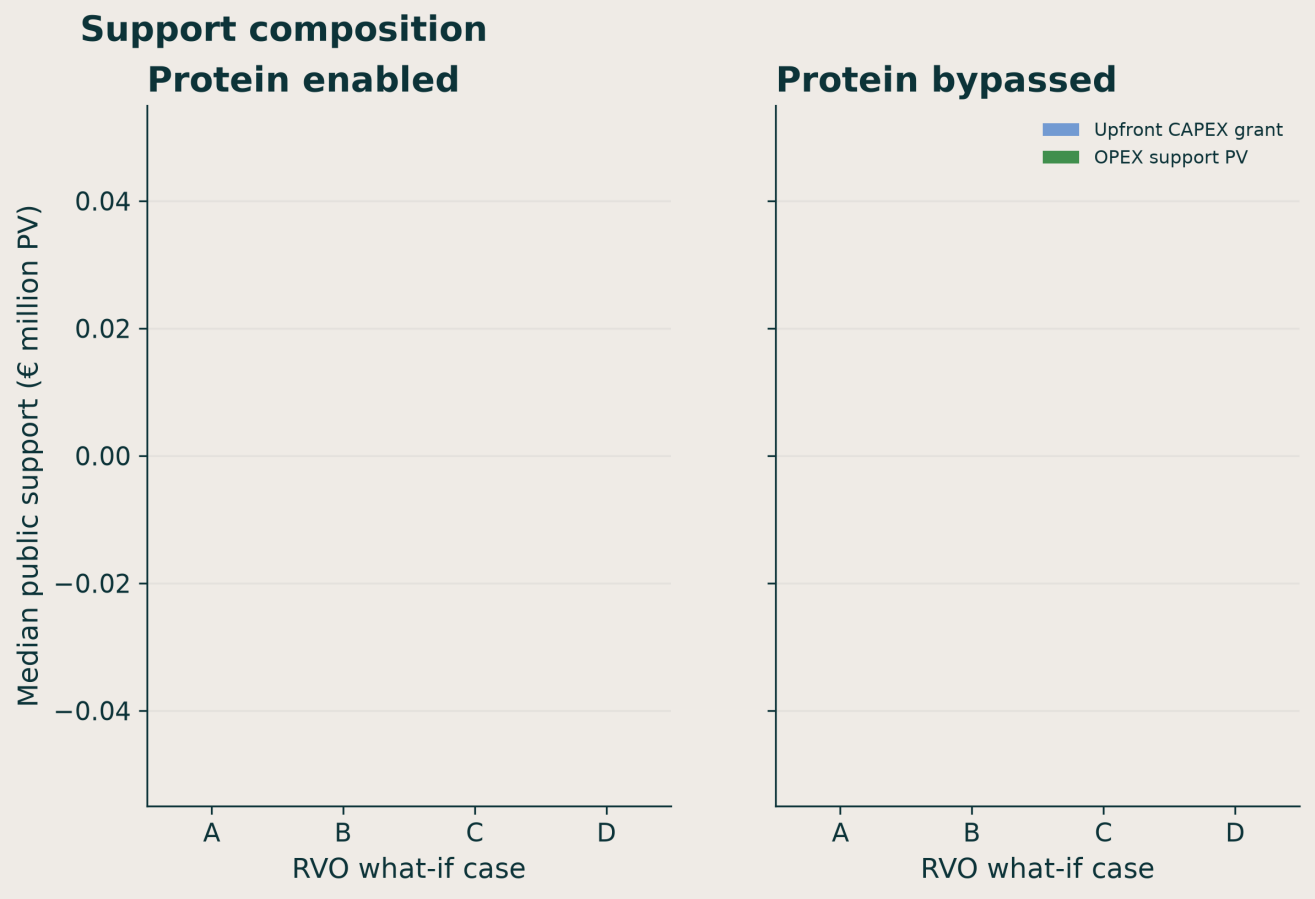


Figure 95. Median upfront grant and discounted operating-support components across paired support cases

Median across evaluated designs by protein group | € million present value | source: Greenhub/results/pareto_rvo_2026

operational climate by stage

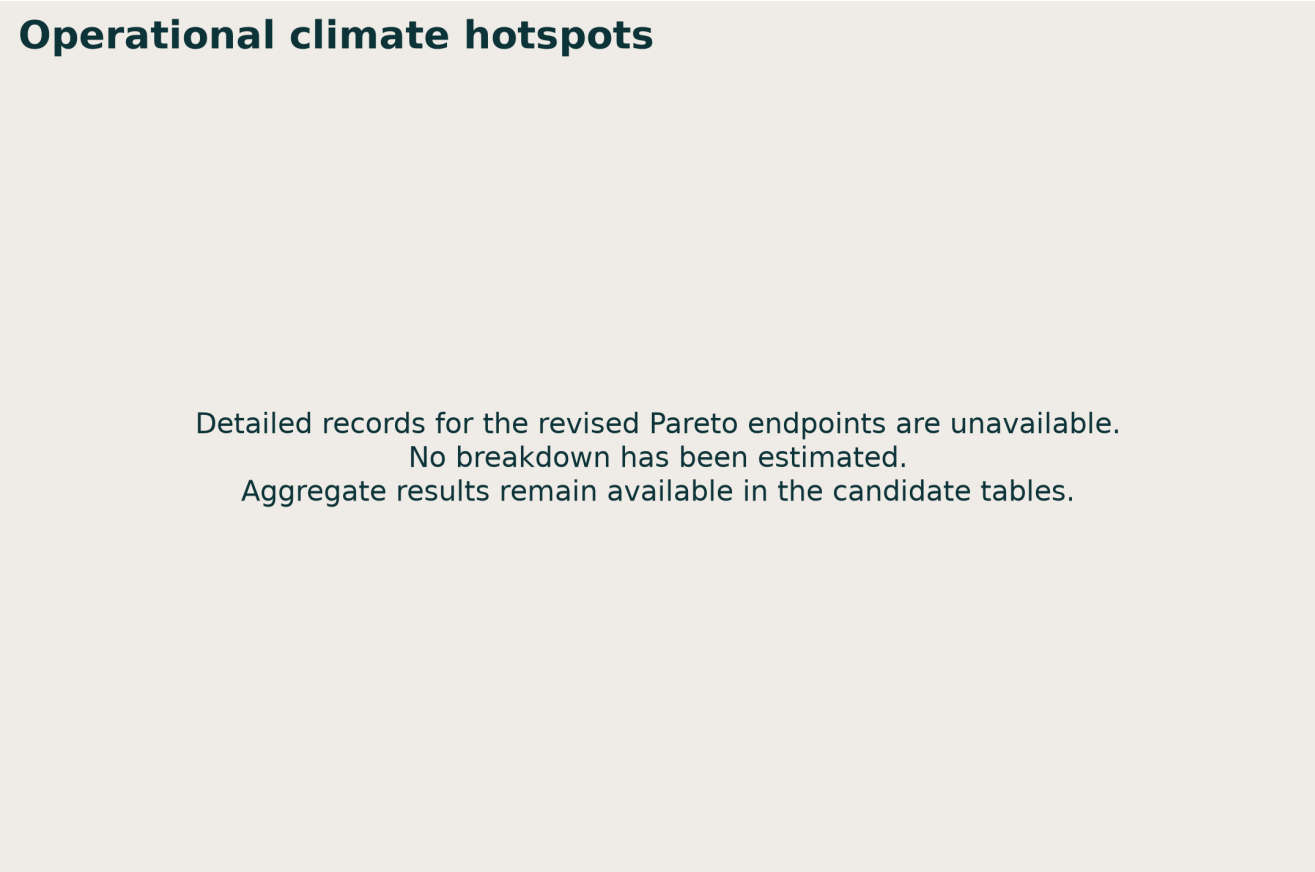


Figure 96. Unavailable: archived endpoint detail records missing; no breakdown estimated

Recalculated endpoints | Unavailable | source: Greenhub/results/pareto_rvo_2026/latest.json

opex categories by endpoint

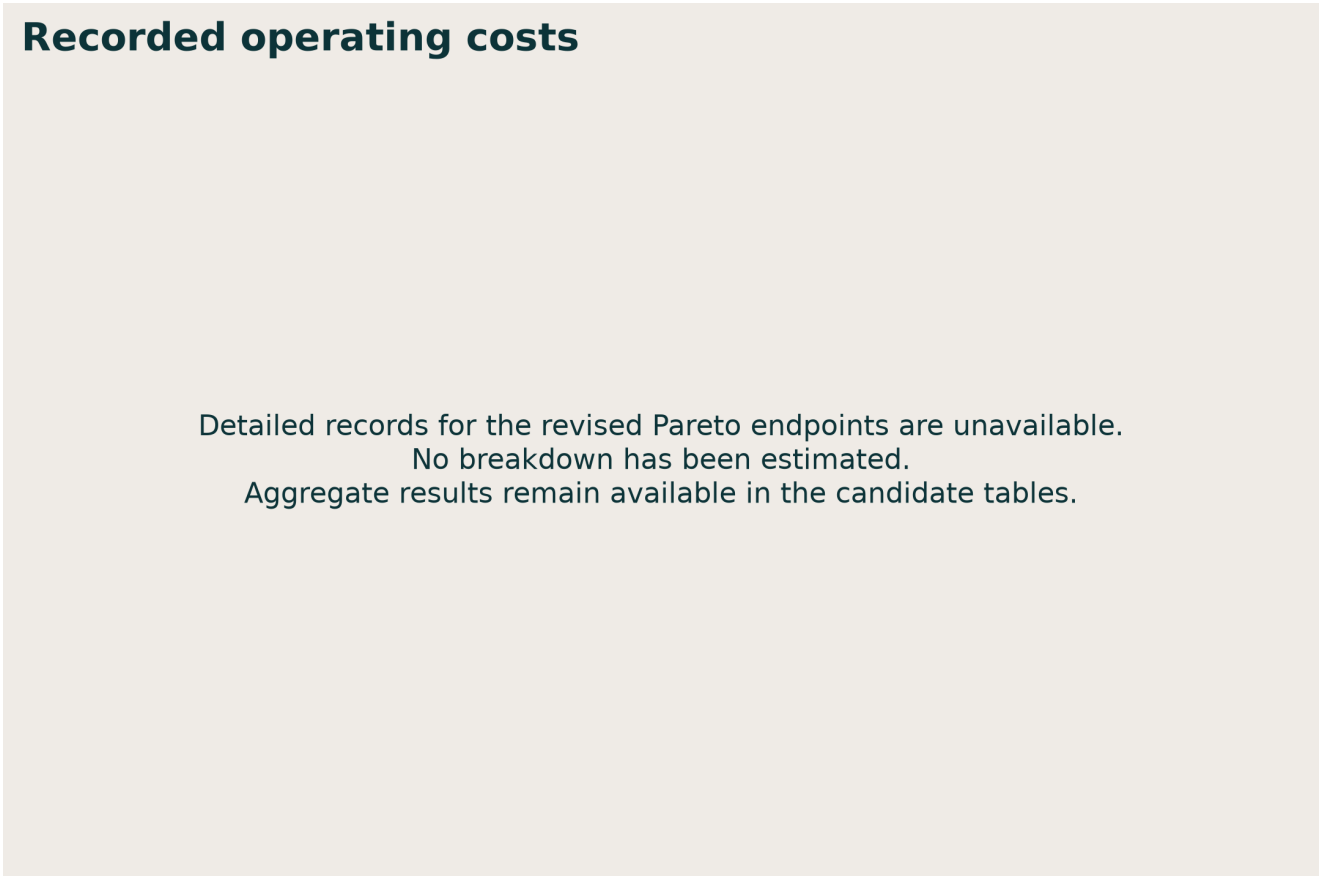


Figure 97. Unavailable: archived endpoint detail records missing; no breakdown estimated
Recalculated endpoints | Unavailable | source: Greenhub/results/pareto_rvo_2026/latest.json

equipment energy by endpoint

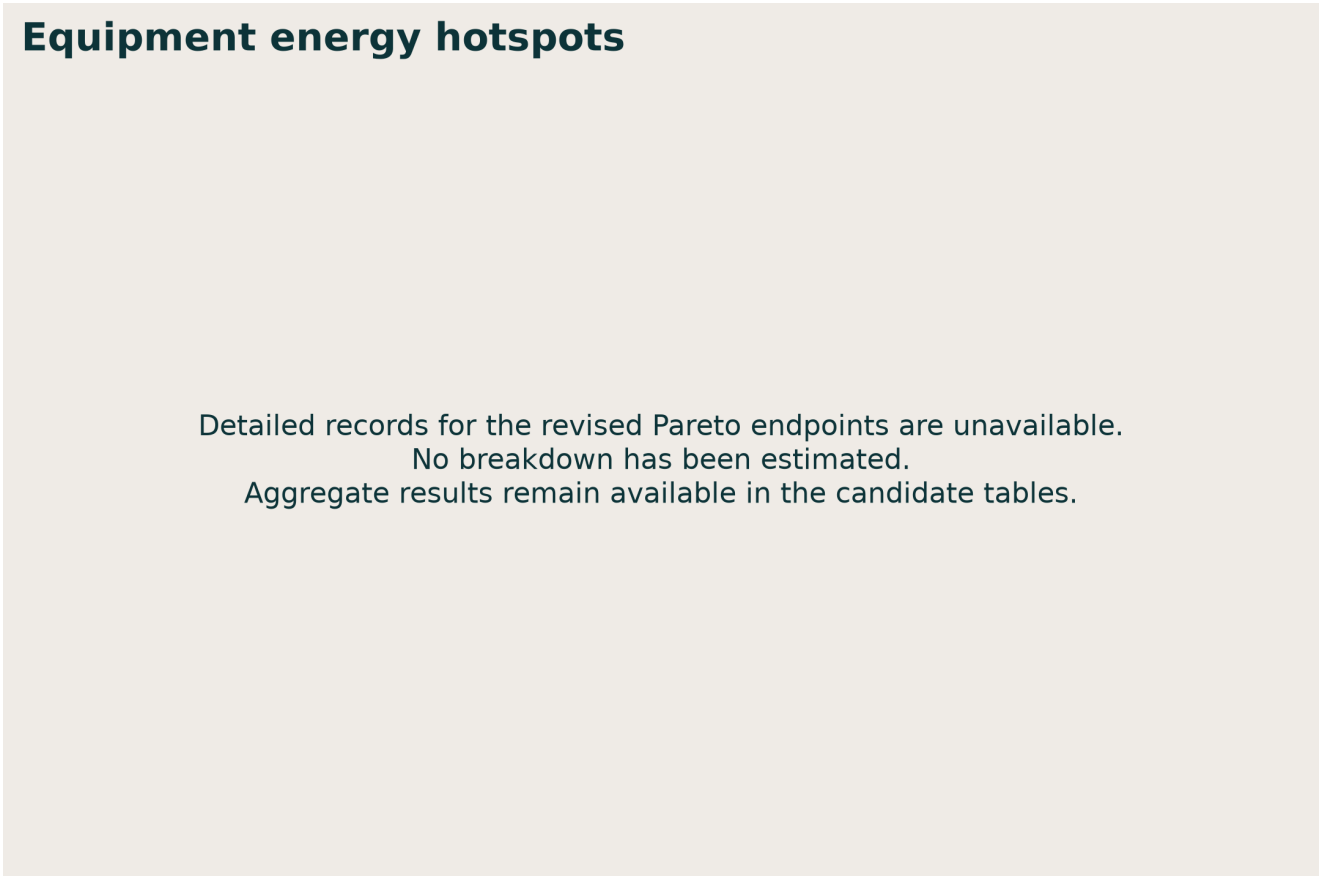


Figure 98. Unavailable: archived endpoint detail records missing; no breakdown estimated
Recalculated endpoints | Unavailable | source: Greenhub/results/pareto_rvo_2026/latest.json